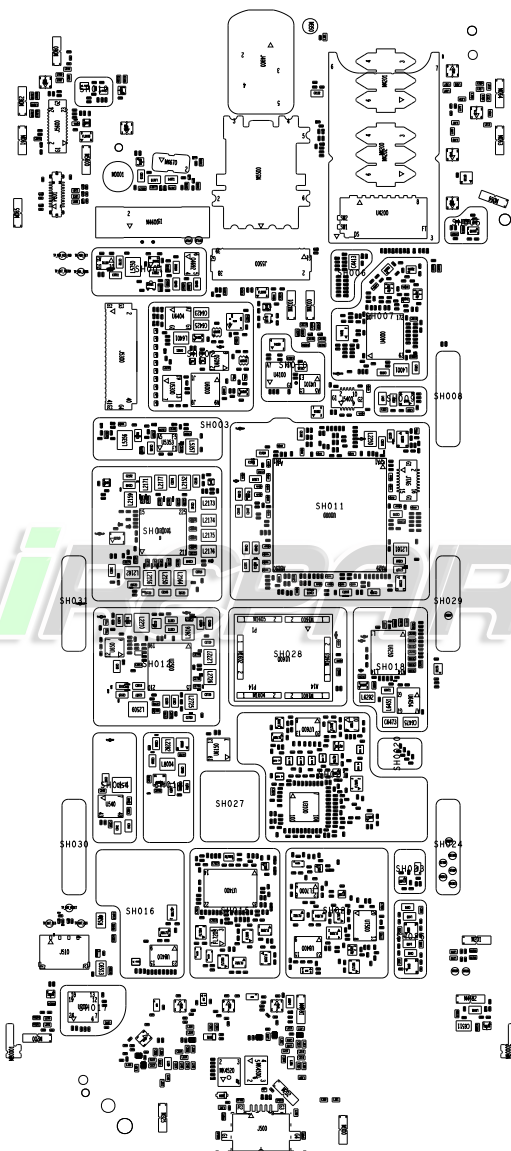






X+2
SYSTEM 5.0

D

C

B

A

IMAGERS

AP_PMIC

DISPLAY

AUDIO_FE

XMCS

SLICE_IF

USER_INPUT

POWER

BLK_WTR3925

RFFE_CA+DSDA

ANTENNAS

CONNECTIVITY

REVISIONS

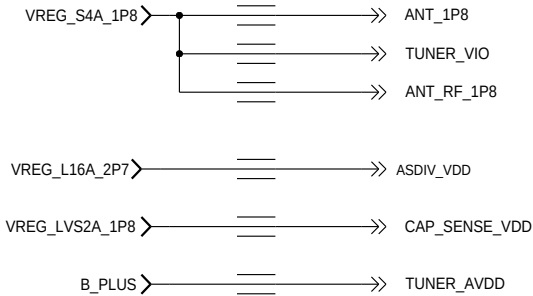
ZONE	LTR	DESCRIPTION	DATE	APPROVED

APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA MOBILITY		84016440001_b	
LAST MODIFIED BY: NATE CONNELL Wed Apr 08 11:47:13 2015		PROJECT NAME 84016440001_b.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED		REV 01	
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ANT: PORTS

REFDES: 000 - 499

SUPPLIES



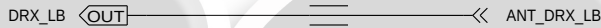
TRX ULTRA LOW/MID BAND ANTENNA



TRX LOW/HIGH BAND ANTENNA



DRX LOW BAND ANTENNA



DRX MID BAND ANTENNA



DRX HIGH BAND ANTENNA



GPS ANTENNA



BT/WLAN ANTENNA



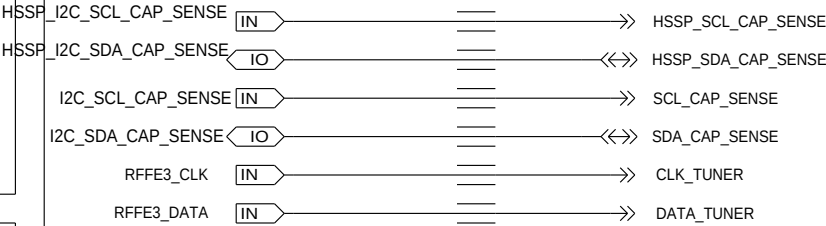
WLAN ANTENNA 2 (FOR 2X2 MIMO SUPPORT)



COUPLER PORTS



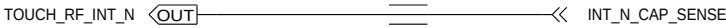
INTERFACES



GPIO - INPUT



GPIO - OUTPUT



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: antennas	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:57 2015		PROJECT NAME 84016440001_b.cpm	
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		PAGE 3 OF 78	

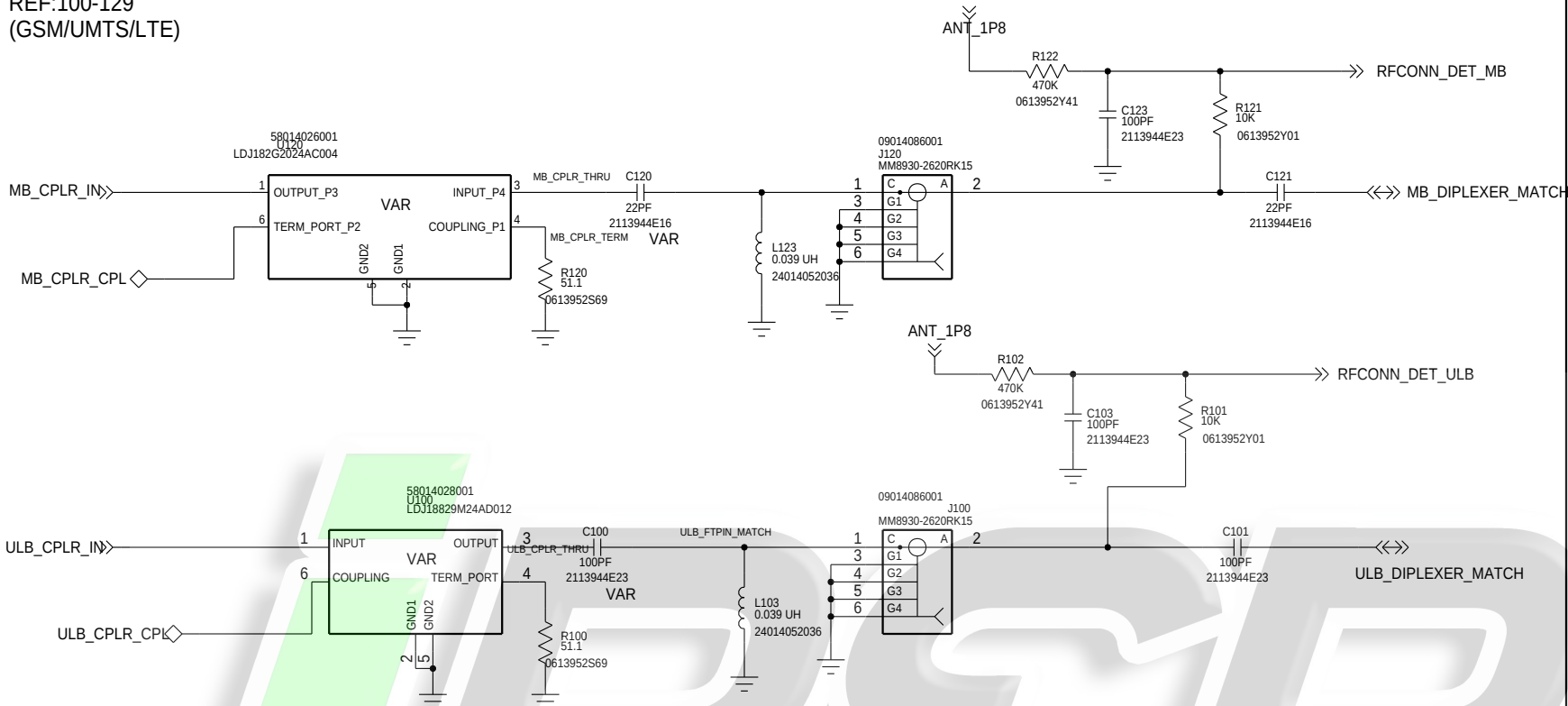
ANT: ANTENNAS

REFDES: 100 - 199

TRX ULTRA LOW/ MID BAND ANTENNA - ASM2

REF:100-129

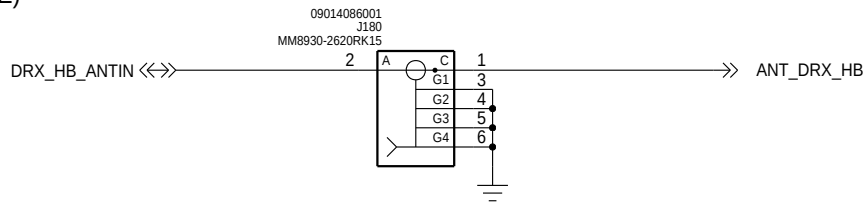
(GSM/UMTS/LTE)



DRX HIGH BAND ANTENNA

REF:180-189

(GSM/UMTS/LTE)

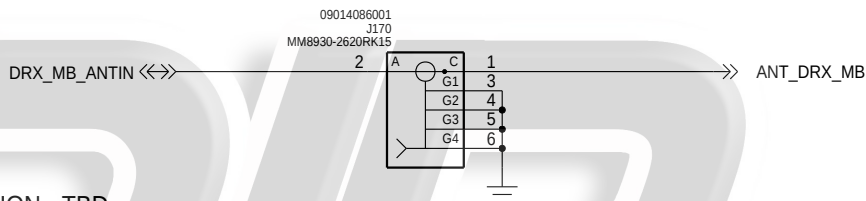


DEVICE LOCATION - TBD

DRX MID BAND ANTENNA

REF:170-179

(GSM/UMTS/LTE)

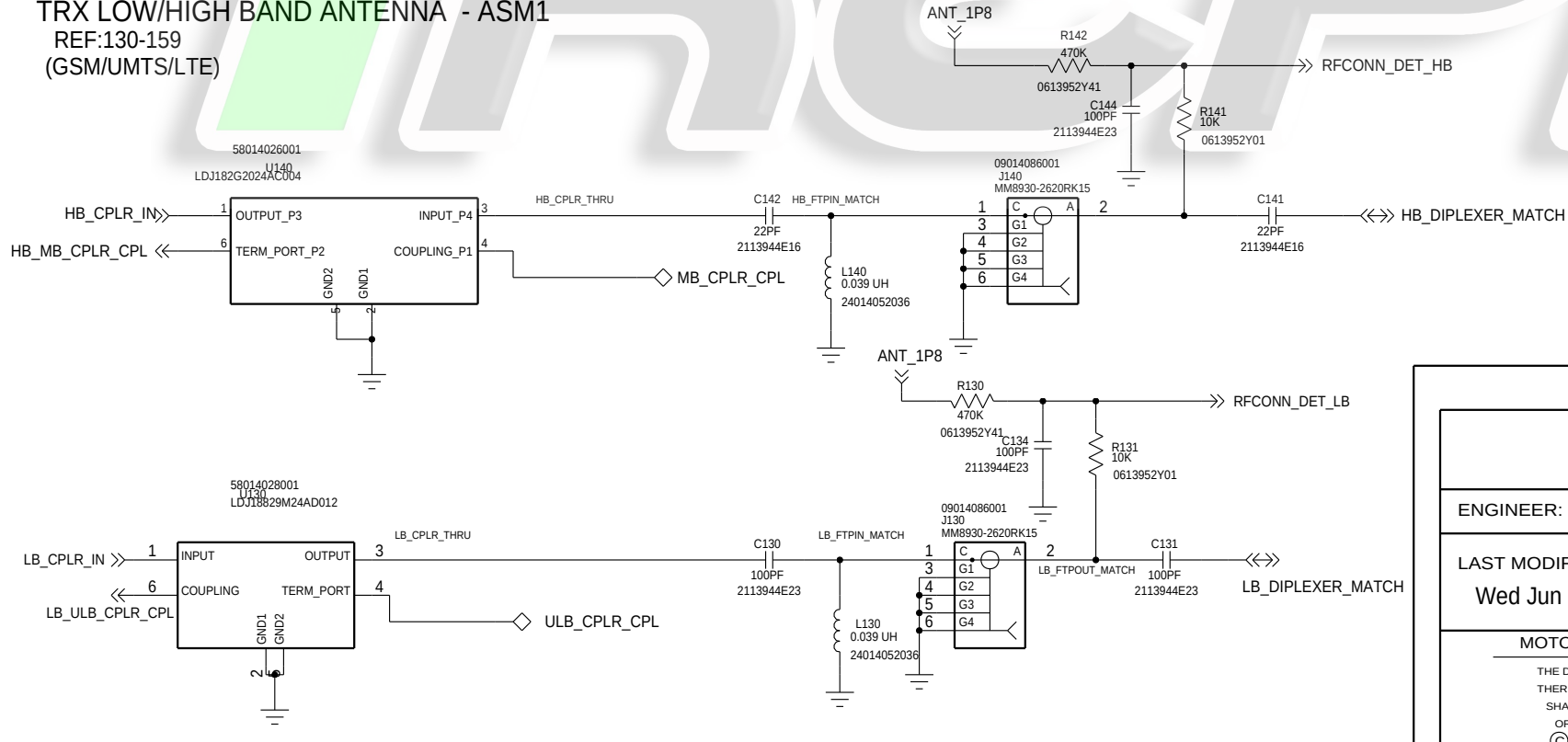


DEVICE LOCATION - TBD

TRX LOW/HIGH BAND ANTENNA - ASM1

REF:130-159

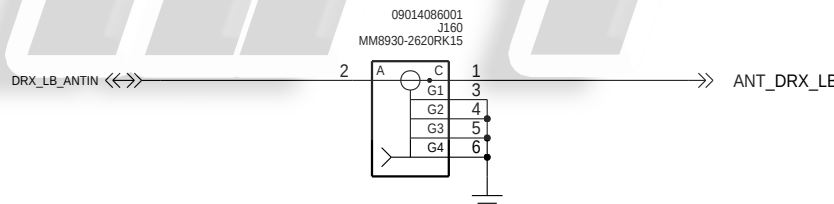
(GSM/UMTS/LTE)



DRX LOW BAND ANTENNA

REF:160-169

(GSM/UMTS/LTE)



DEVICE LOCATION - TBD

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Wed Jun 10 14:32:58 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

antennas

PROJECT NAME

B 84016440001_b.cpm

CAD FILE NAME:

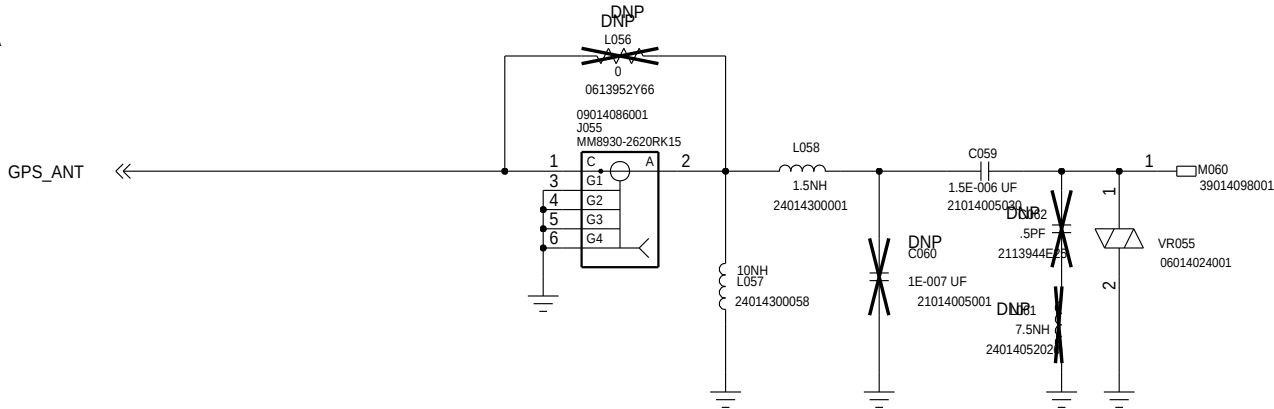
REV

01

PAGE 4 OF 78

ANT: GPS/WLAN ANTENNAS

GPS ANTENNA
REF: 055-069



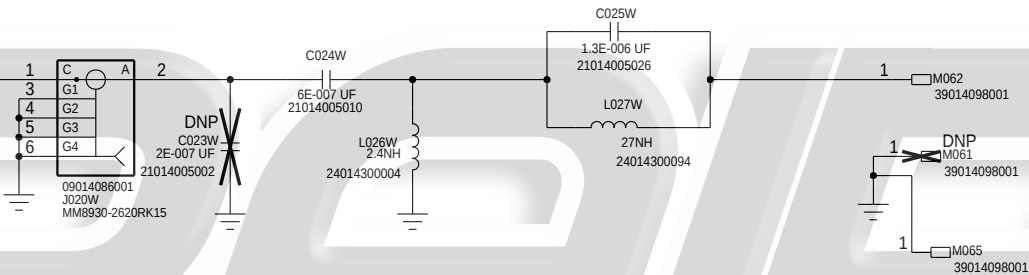
LOCATION - TBD

WLAN CHAIN1 ANTENNA
REF: 060-85

MIMO AND MAIN PATHS ARE SWAPPED BEFORE THE RF CONNECTORS J010 AND J020W

REF:020W-029W

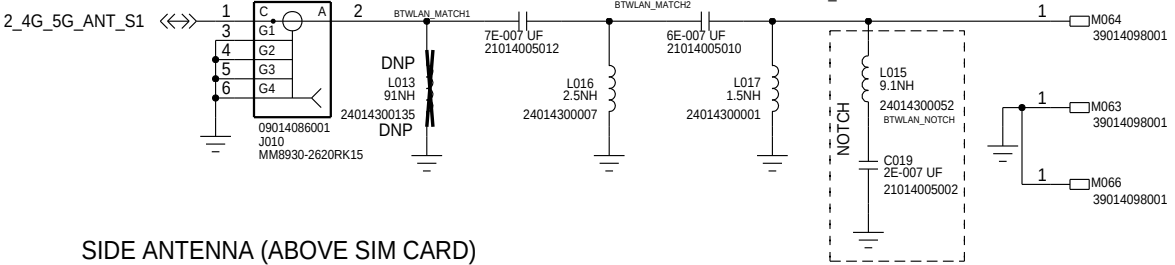
2_4G_5G_ANT_S0 <<>>



DEVICE LOCATION - TBD, APPLIES ONLY IF QCA6174 IS USED
ALL CHAIN1 ANTENNA COMPONENTS HAVE W SUFFIX FOR NOW.

BT/WLAN CHAIN0 ANTENNA
REF:010-019

MIMO AND MAIN PATHS ARE SWAPPED BEFORE THE RF CONNECTORS J010 AND J020W



SIDE ANTENNA (ABOVE SIM CARD)

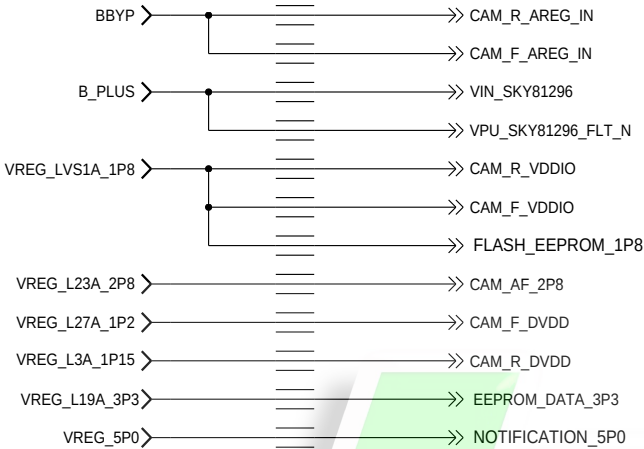
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Wed Jun 10 14:33:00 2015		antennas	
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		REV 01	
		PAGE 5 OF 78	

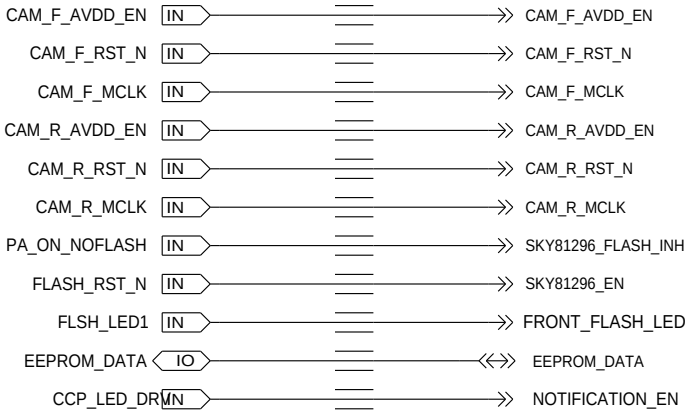
CAM: PORTS

REF 5500-5999

SUPPLIES



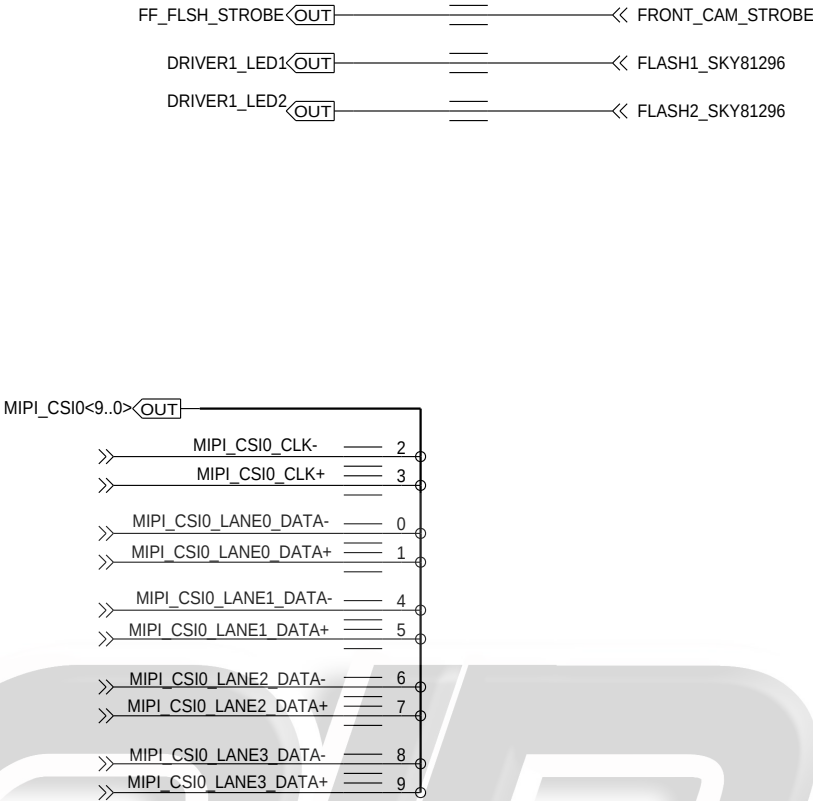
INPUTS



INTERFACES



OUTPUTS



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Wed Jun 10 14:32:53 2015

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MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

imagers

PROJECT NAME

B 84016440001_b.cpm

REV

01

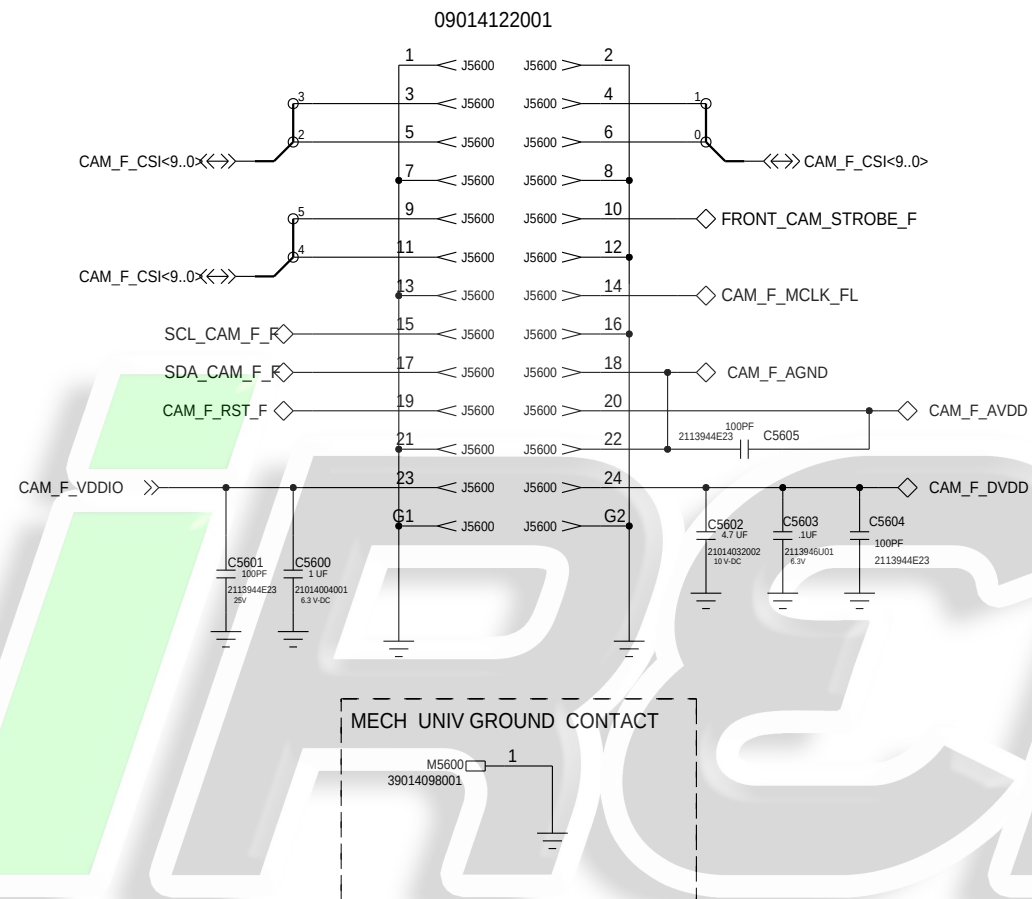
CAD FILE NAME:

PAGE 10 OF 78

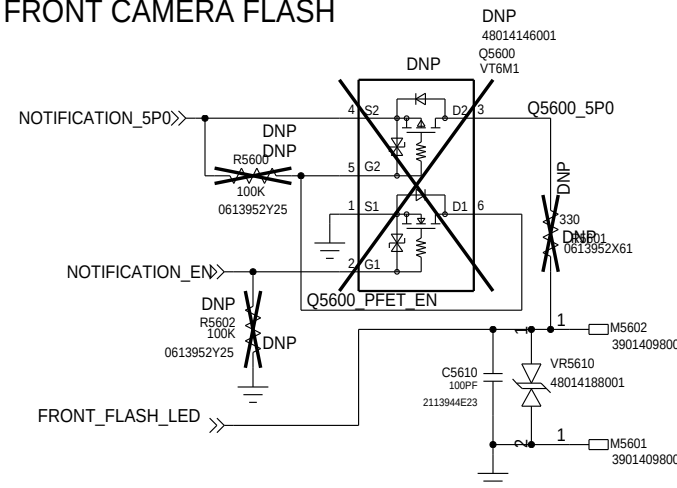
CAM: FRONT
REF 5600-5699

*** IMAGE SENSOR I2C ADD. 0X6C

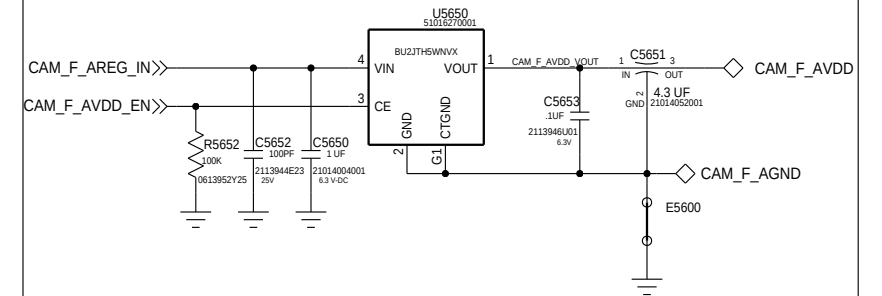
MATES WITH 2871491E05



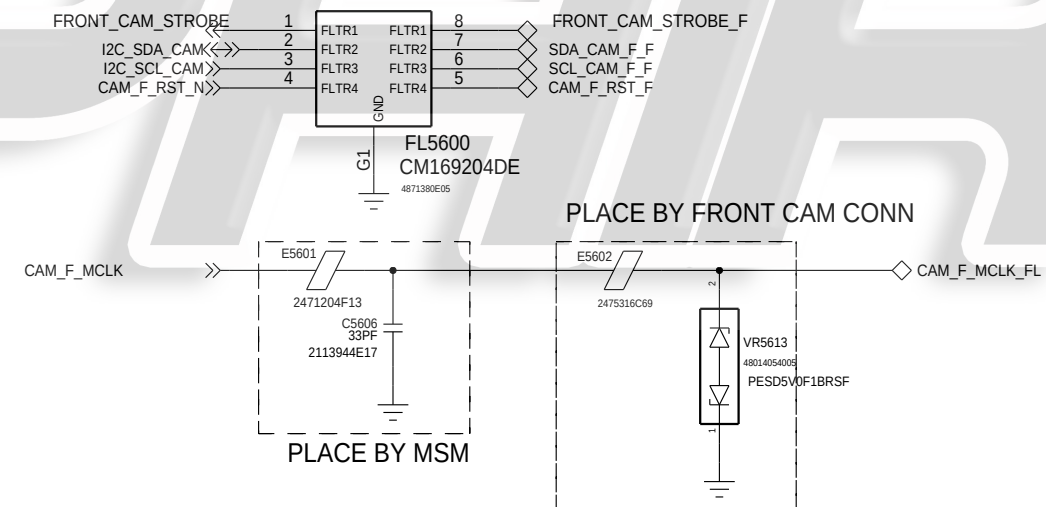
FRONT CAMERA FLASH



ANALOG REGULATOR



EMI/ESD FILTER



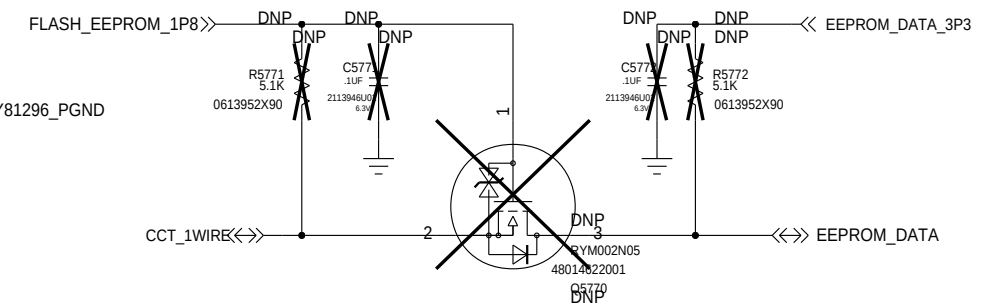
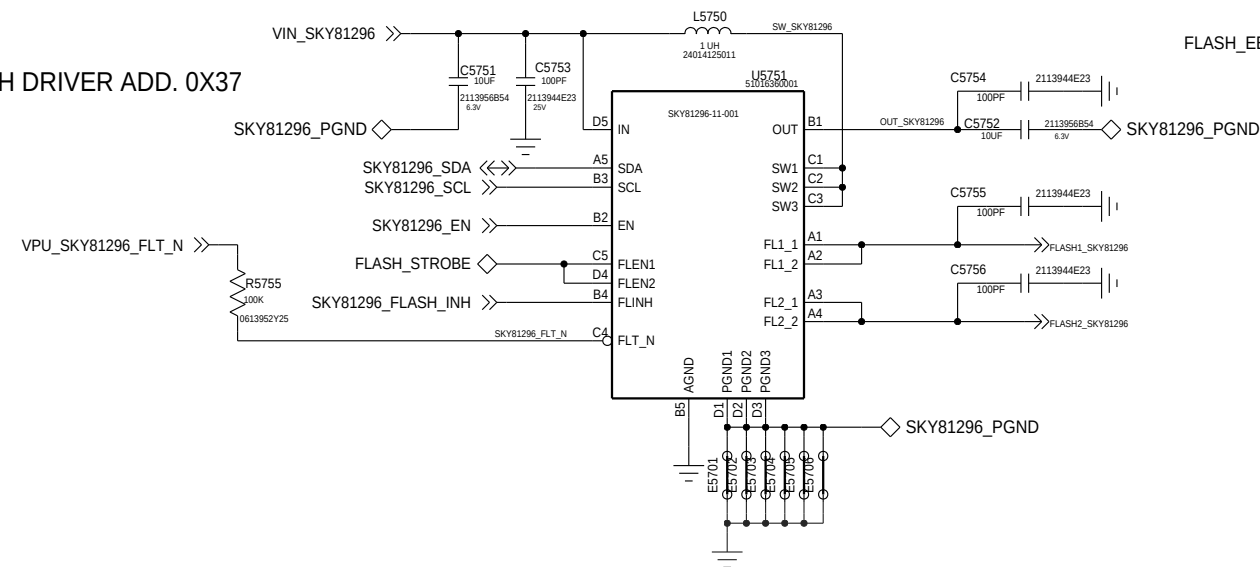
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER	BLOCK TITLE: imagers		
LAST MODIFIED BY: UID ELRAN Wed Jun 10 14:32:54 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME B 84016440001_b.cpm	REV 01
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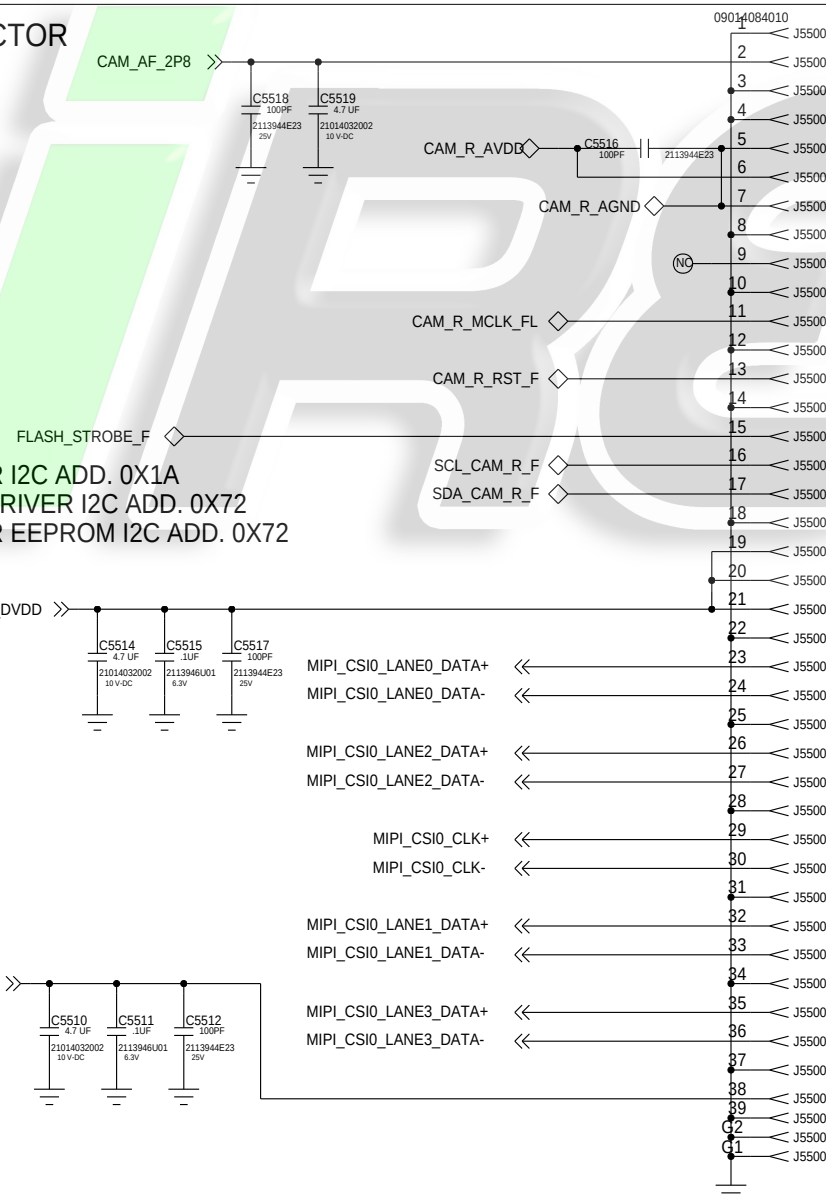
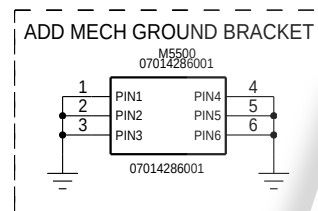
CAM: REAR
REF:5500-5599

LED FLASH DRIVER
REF:5750-5799

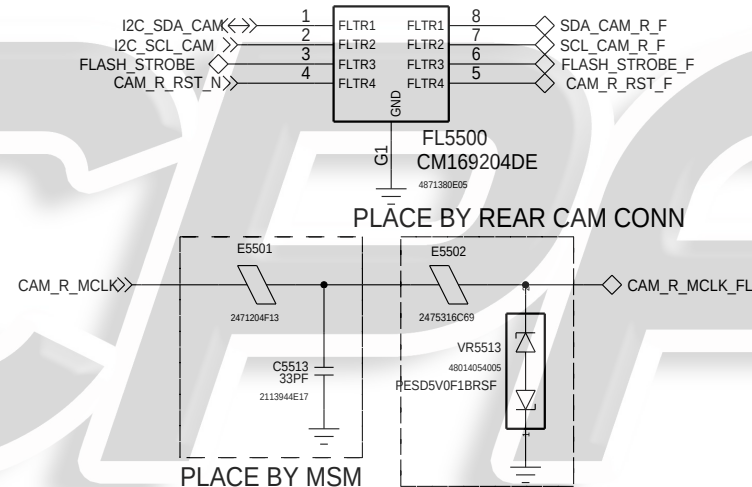
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*** FLASH DRIVER ADD. 0X37
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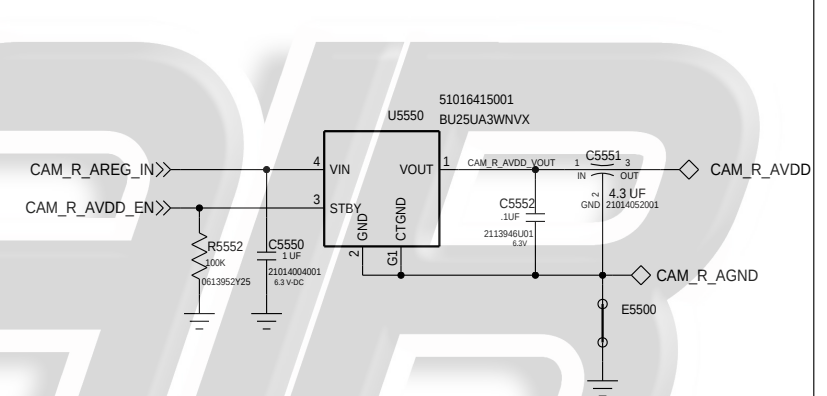
IMAGER FLEX CONNECTOR



EMI/ESD FILTER



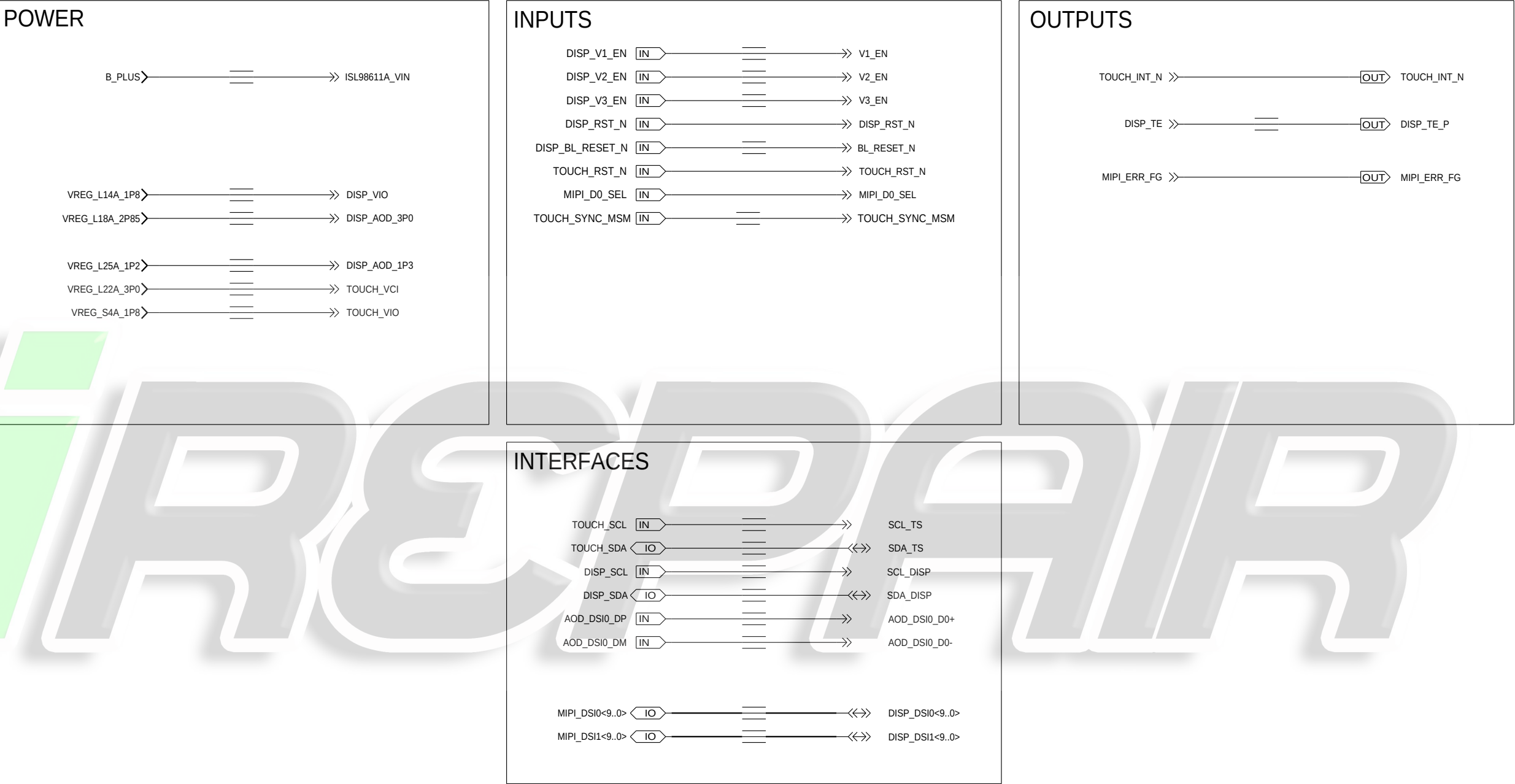
ANALOG REGULATOR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: imagers		
LAST MODIFIED BY: MOTOROLA Wed Jun 10 14:32:56 2015			
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		CAD FILE NAME:	PAGE 12 OF 78

DISP: PORTS

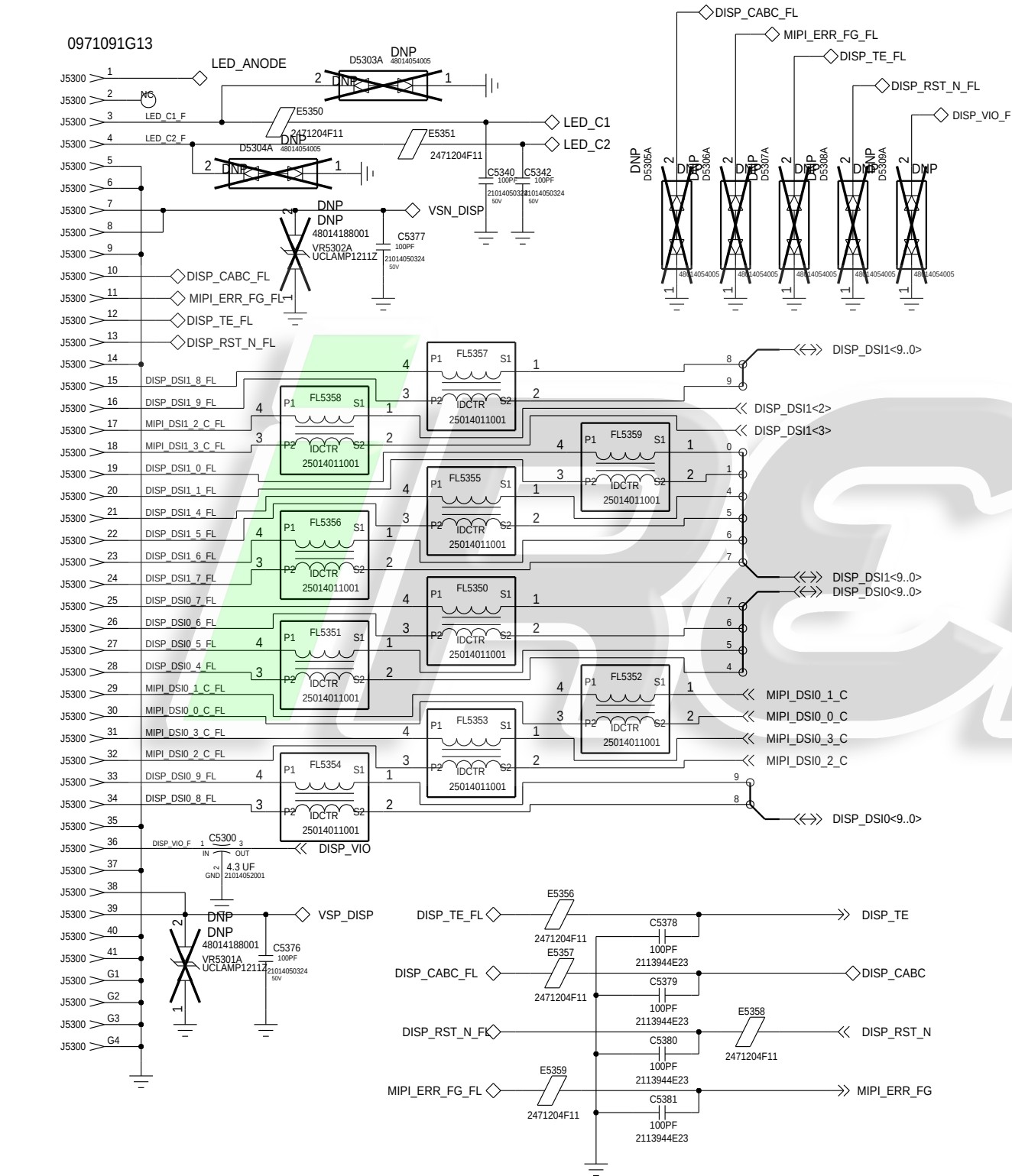
REF 5000-5499



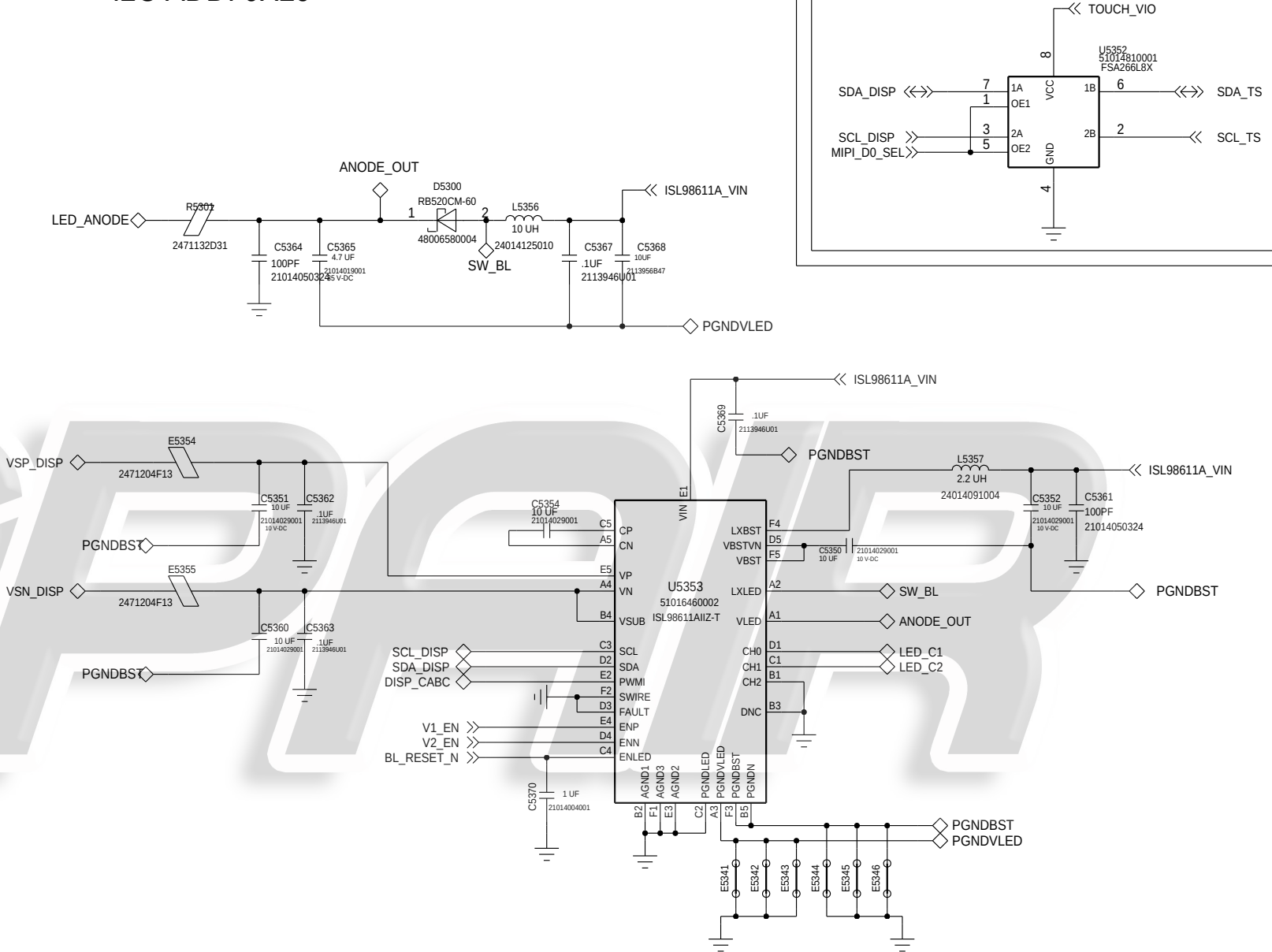
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: display	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:48 2015			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME 84016440001_b.cpm	REV 01
		CAD FILE NAME: PAGE 13 OF 78	

DISP: CONNECTOR
REF 5300-5349

DISPLAY CONNECTOR



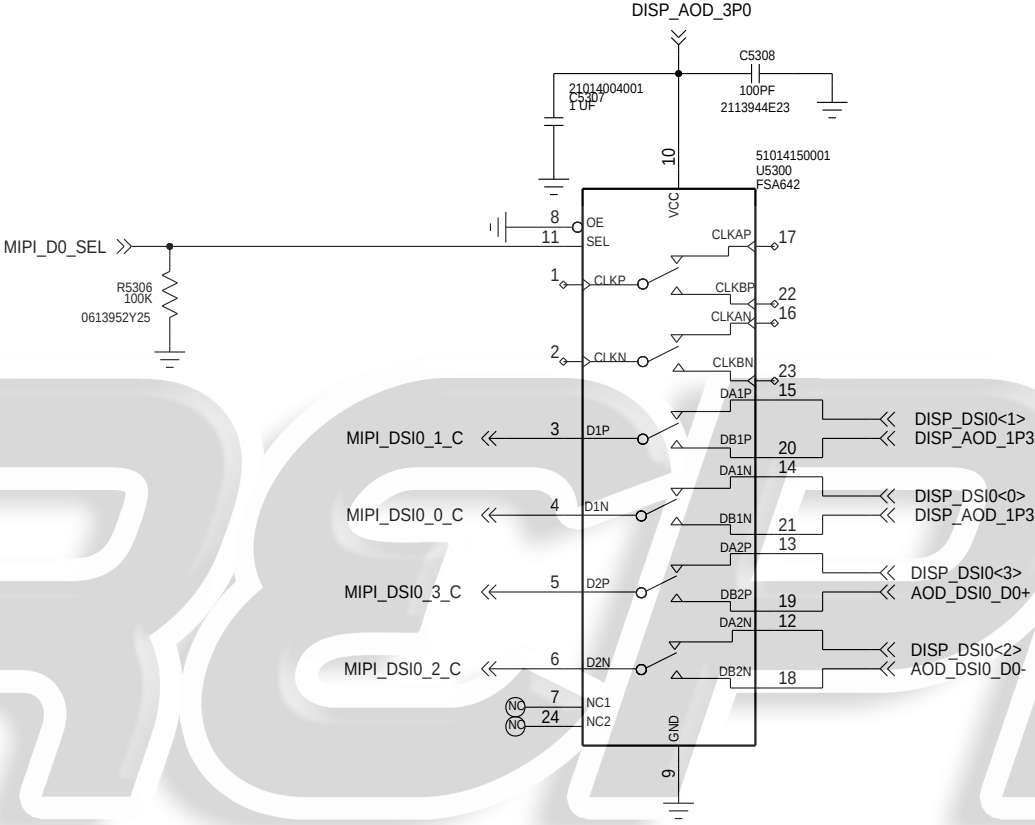
DISPLAY DUAL OUTPUT POWER SUPPLY
*** I2C ADD. 0X29



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:50 2015		display	
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CAD FILE NAME:		PAGE 14 OF 78	

DISP: AOD

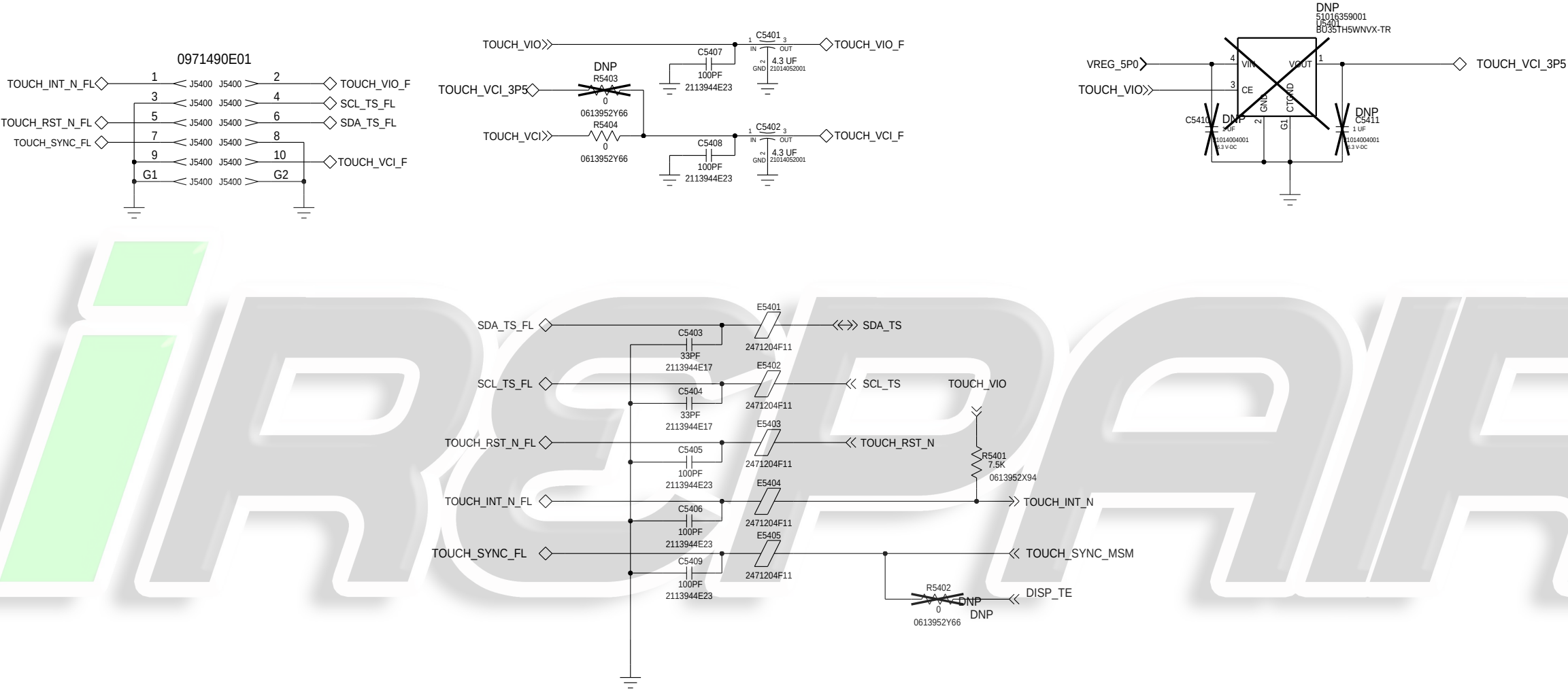


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Wed Jun 10 14:32:50 2015		display	
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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 15 OF 78	

TOUCH: CONNECTOR

REF 5400-5439

*** TOUCH SENSOR I2C ADD. 0X20



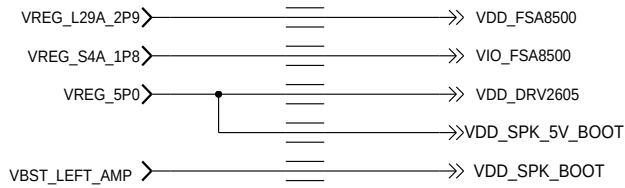
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: display	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:51 2015			
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		CAD FILE NAME: PAGE 16 OF 78	

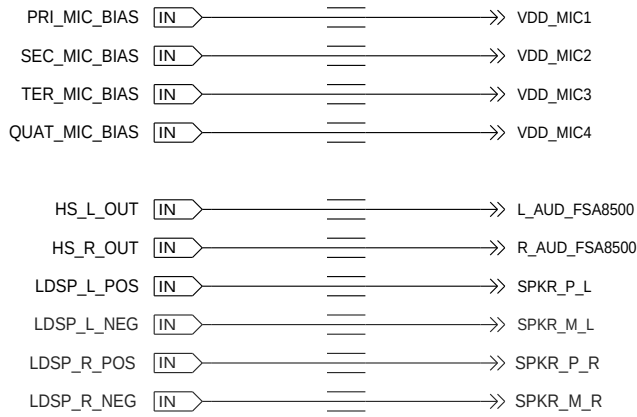
AFE: PORTS

REF 4500-4999

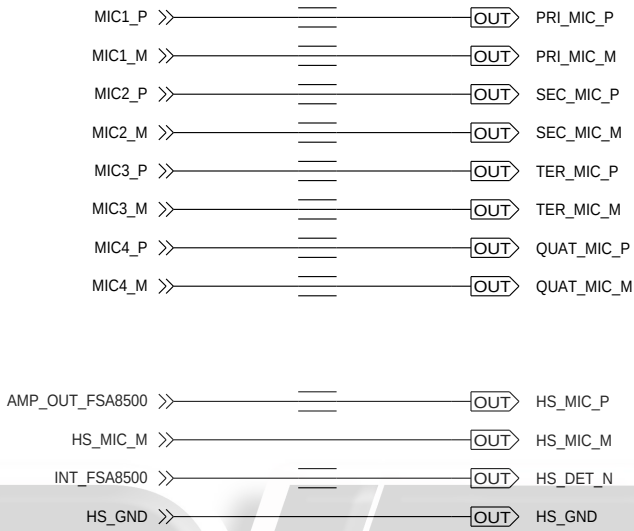
POWER



INPUTS



OUTPUTS

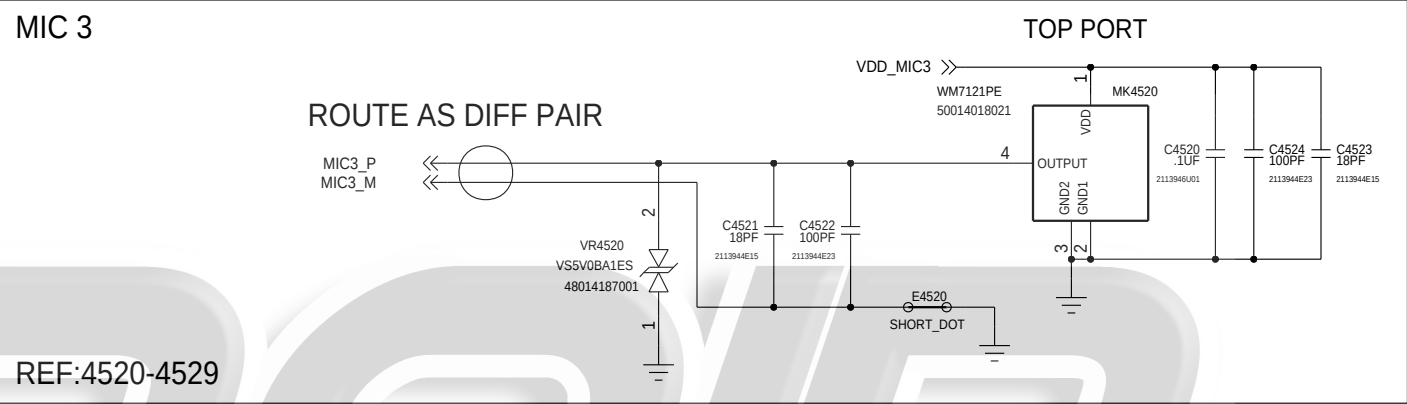
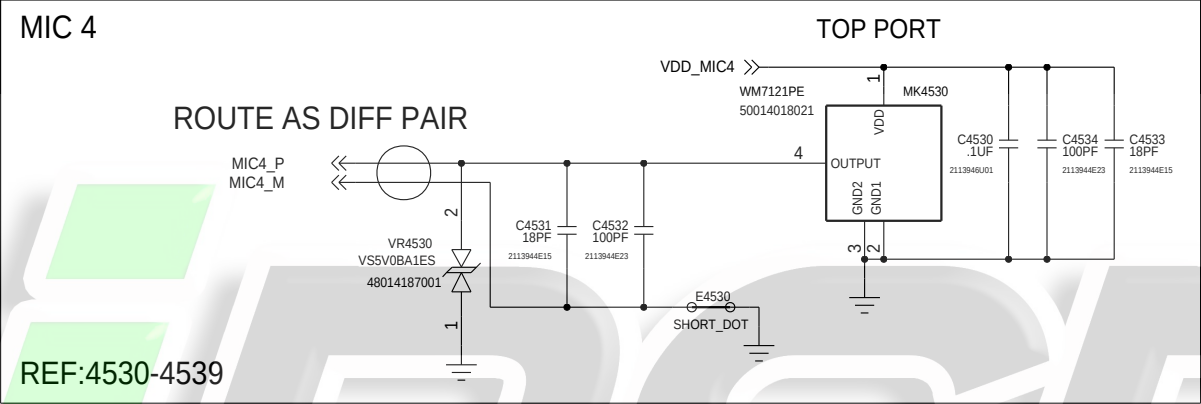
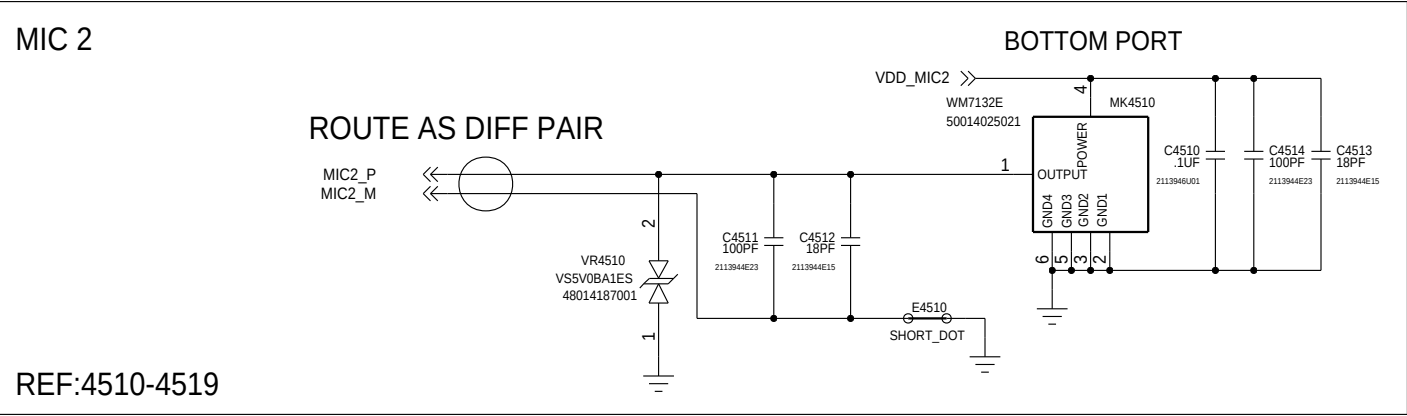
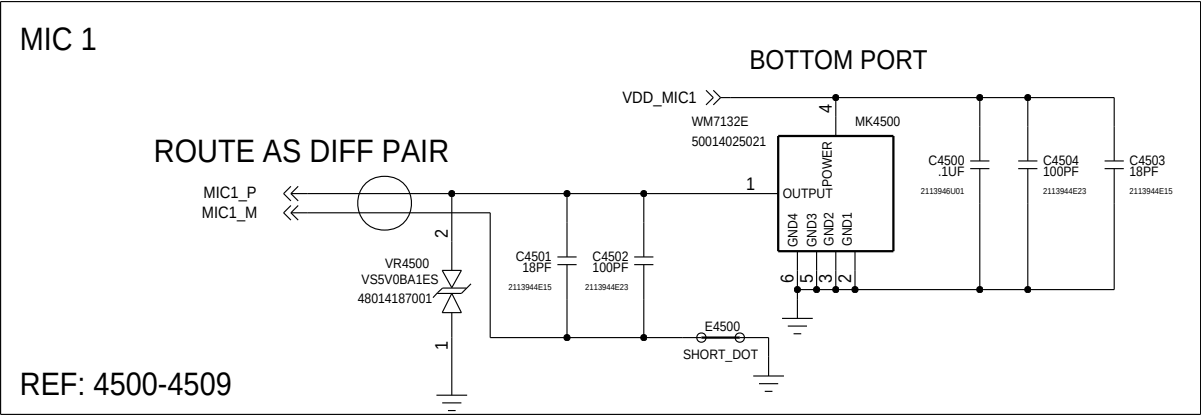


INTERFACES

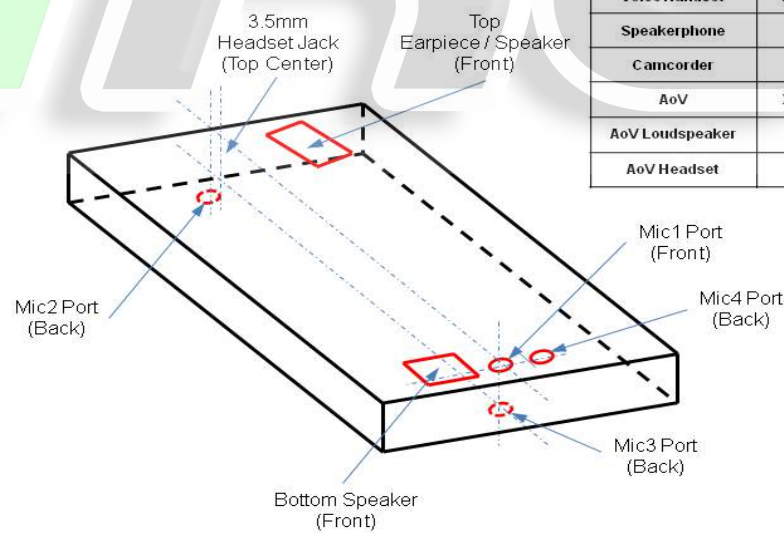


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:39 2015			
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		CAD FILE NAME:	PAGE 17 OF 78

AFE: MICS
REF: 4500-4599



Clark (Front View)

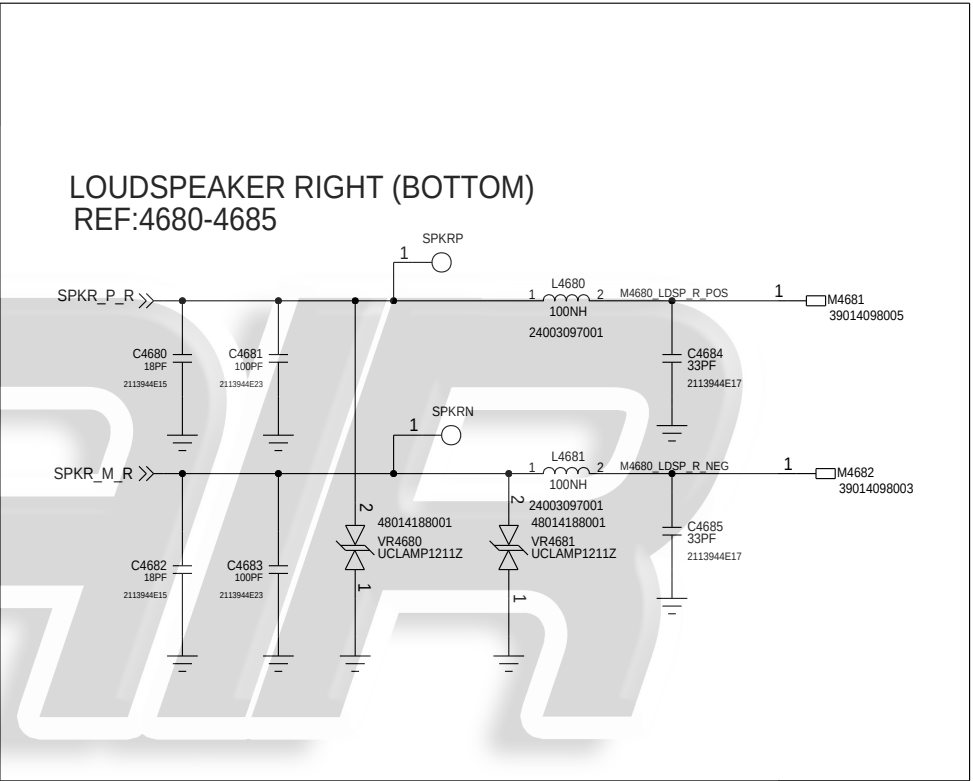
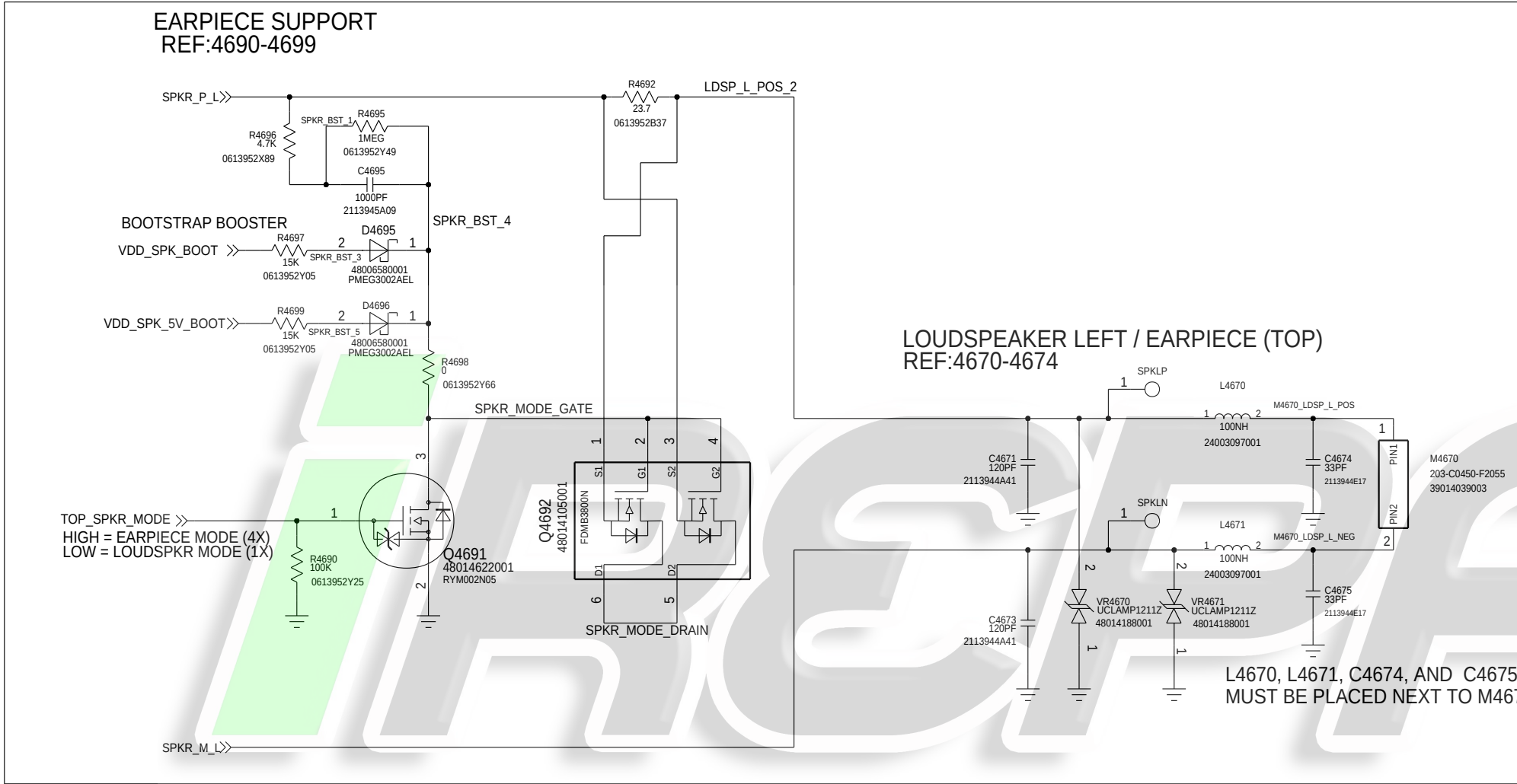


	Mic1	Mic2	Mic3	Mic4	HS Mic
Voice Handset	X	X			
Speakerphone		X			
Camcorder		X	X		
AoV	X		X	X	
AoV Loudspeaker		X			
AoV Headset					X

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MOTOROLA Wed Jun 10 14:32:42 2015		PROJECT NAME B 84016440001_b.cpm	
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		REV 01	

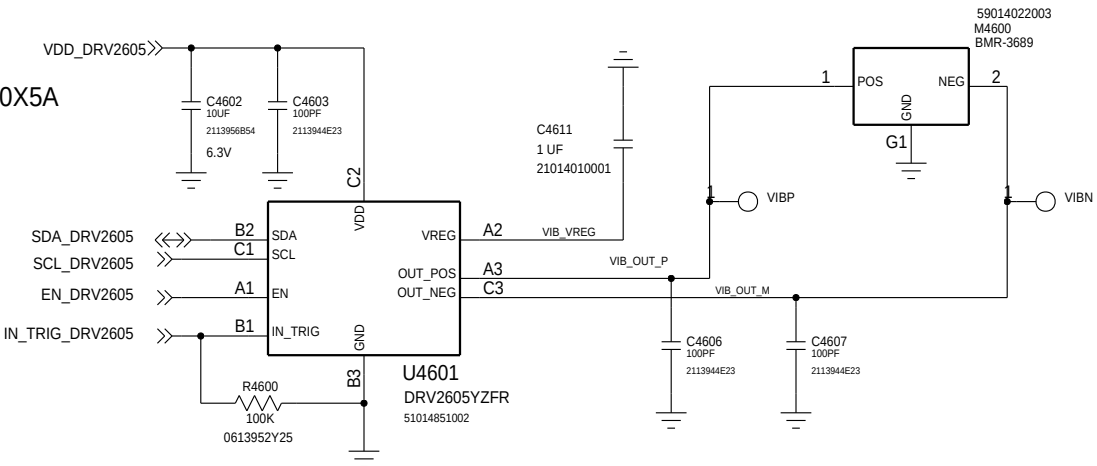
AFE: SPEAKERS

REF: 4600-4699



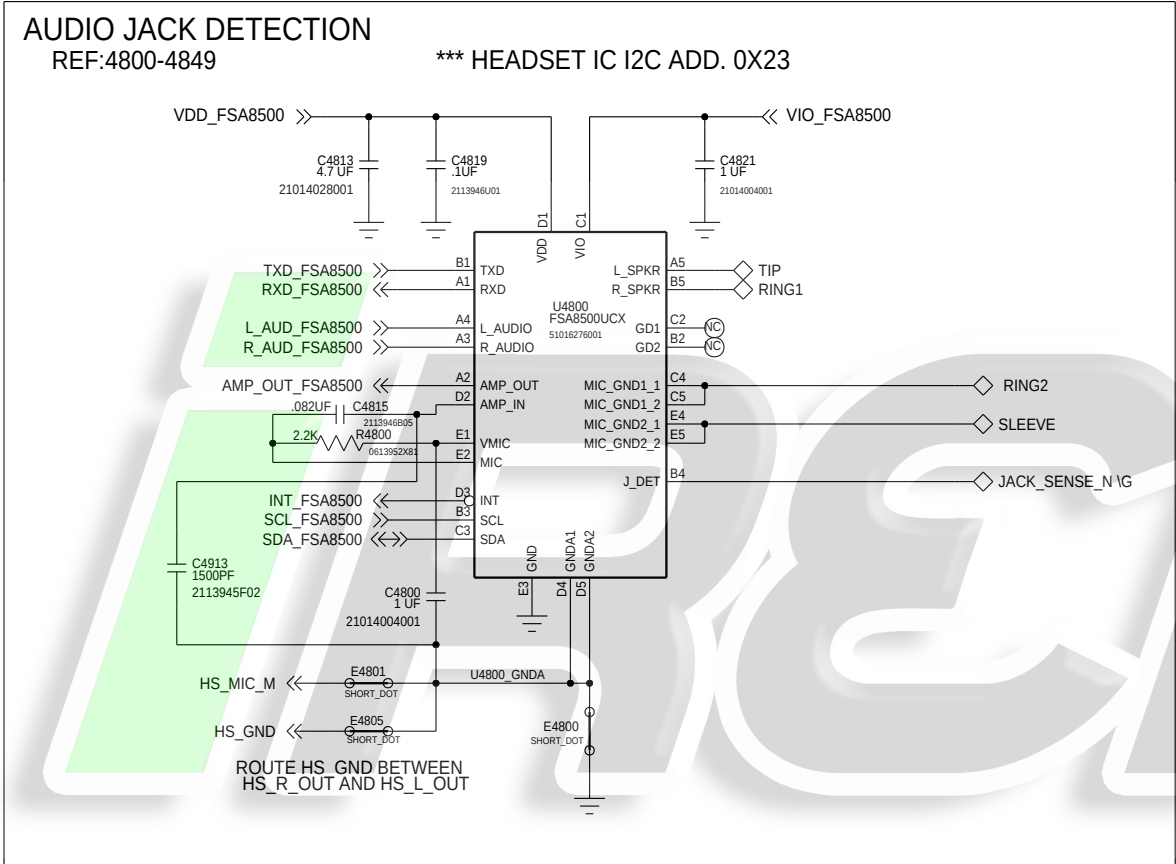
ROTARY VIBRATOR
REF DES 4600-4619

*** VIBRATOR DRIVER I2C ADD. 0X5A

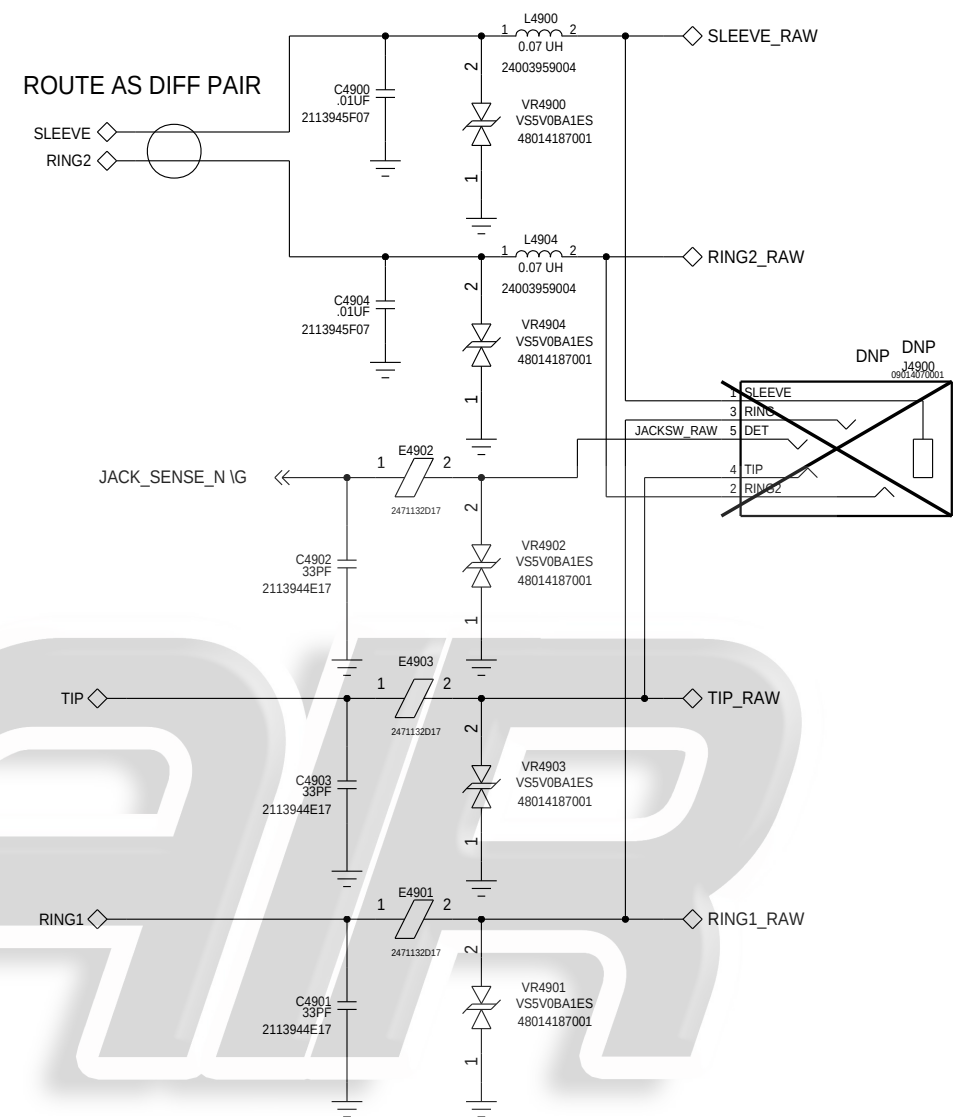


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MOTOROLA Wed Jun 10 14:32:44 2015		PROJECT NAME 84016440001_b.cpm	
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		PAGE 19 OF 78	

AFE: 3.5MM HEADSET



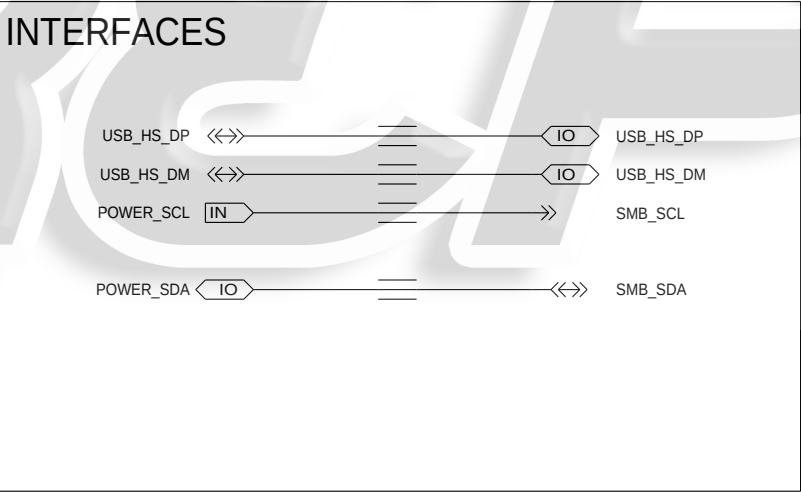
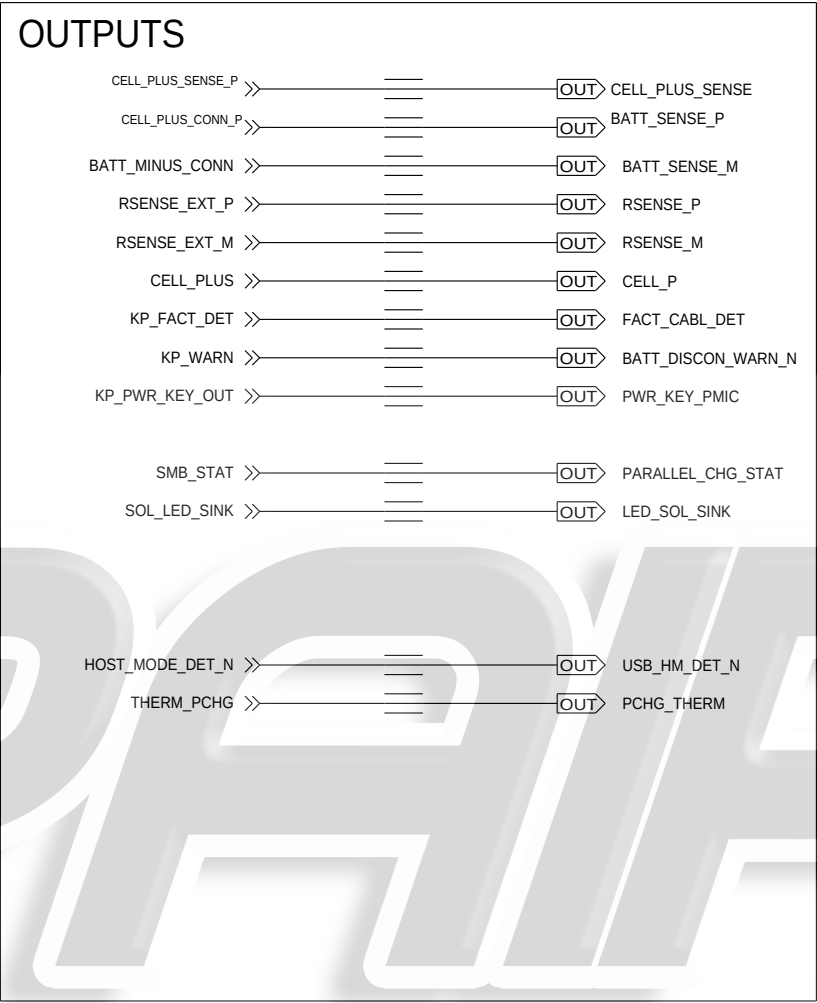
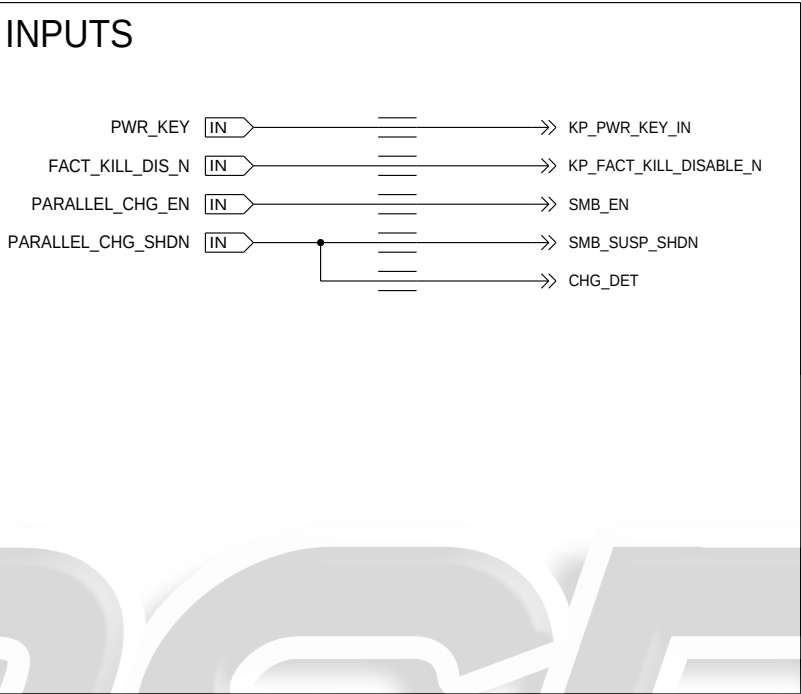
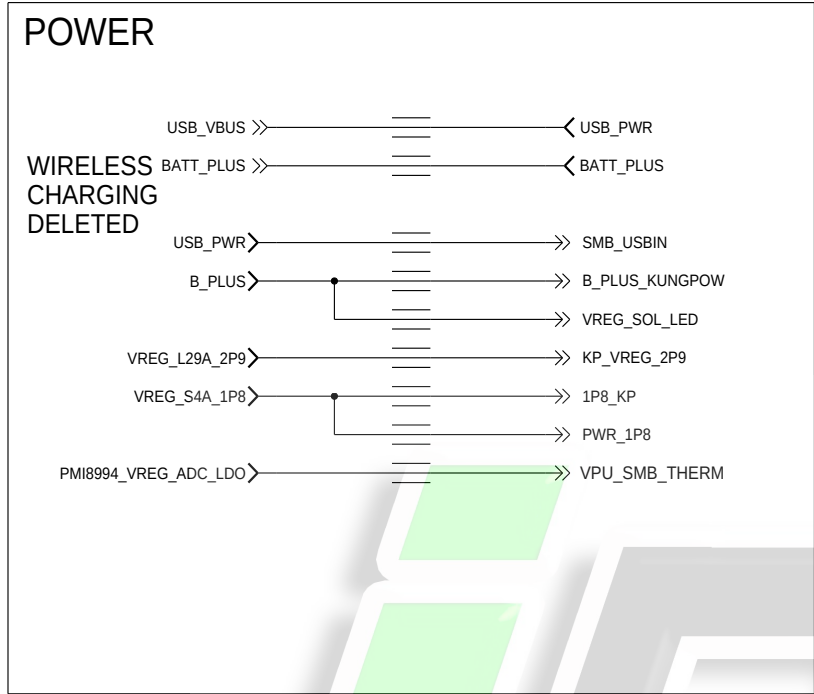
3.5MM HSJ
REF:4900-4949



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MATT MOTTIER		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:45 2015		PROJECT NAME 84016440001_b.cpm	
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		REV 01	
		PAGE 20 OF 78	

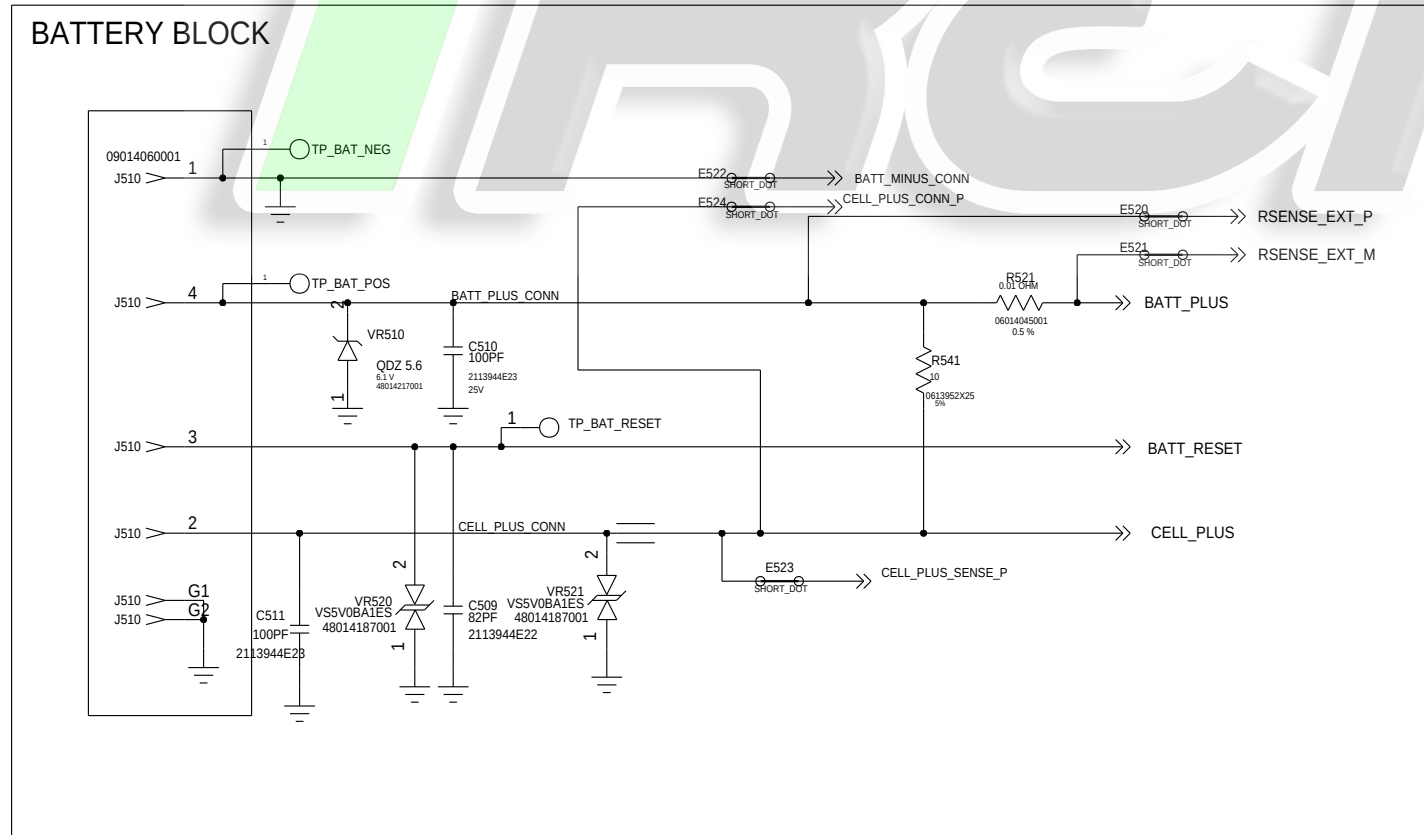
PWR: PORTS

REF 500-799



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:35 2015		power	
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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 21 OF 78	

REF: 500-530



KUNG ROW BATTERY DISCONNECT

U650 SLG4K4583 51016448001

Components:

- R650: 249K
- R651: 249K
- R652: 470K
- R653: 1MEG
- R654: 249K
- R655: 1MEG
- R656: 100K
- R657: 100K
- R662: 2.2K
- C655: 0.1uF
- C660: 22uF
- C661: 210.406001

Connections:

- EMU_ID_RAW to R652
- HOST_MODE_DET_N to R655
- KP_VREG_2P9 to R650 and R651
- KP_PWR_KEY_IN to R650
- CHG_DET to R653
- KP_FACT_KILL_DISABLE_N to R654
- B_PLUS_KUNGPOW to R656 and R657
- BPLUS_IN to R656
- PWR_KEY_IN_N to R650
- PWR_KY_OUT_N to R657
- FACT_DET to R656
- BATT_DIS to R657
- WARN_N to R662
- KP_PWR_KEY_OUT to R656
- KP_FACT_DET to R657
- BATT_RESET to R662
- KP_WARN to R662

Notes:

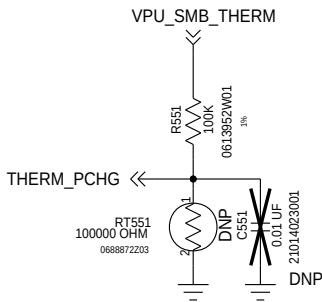
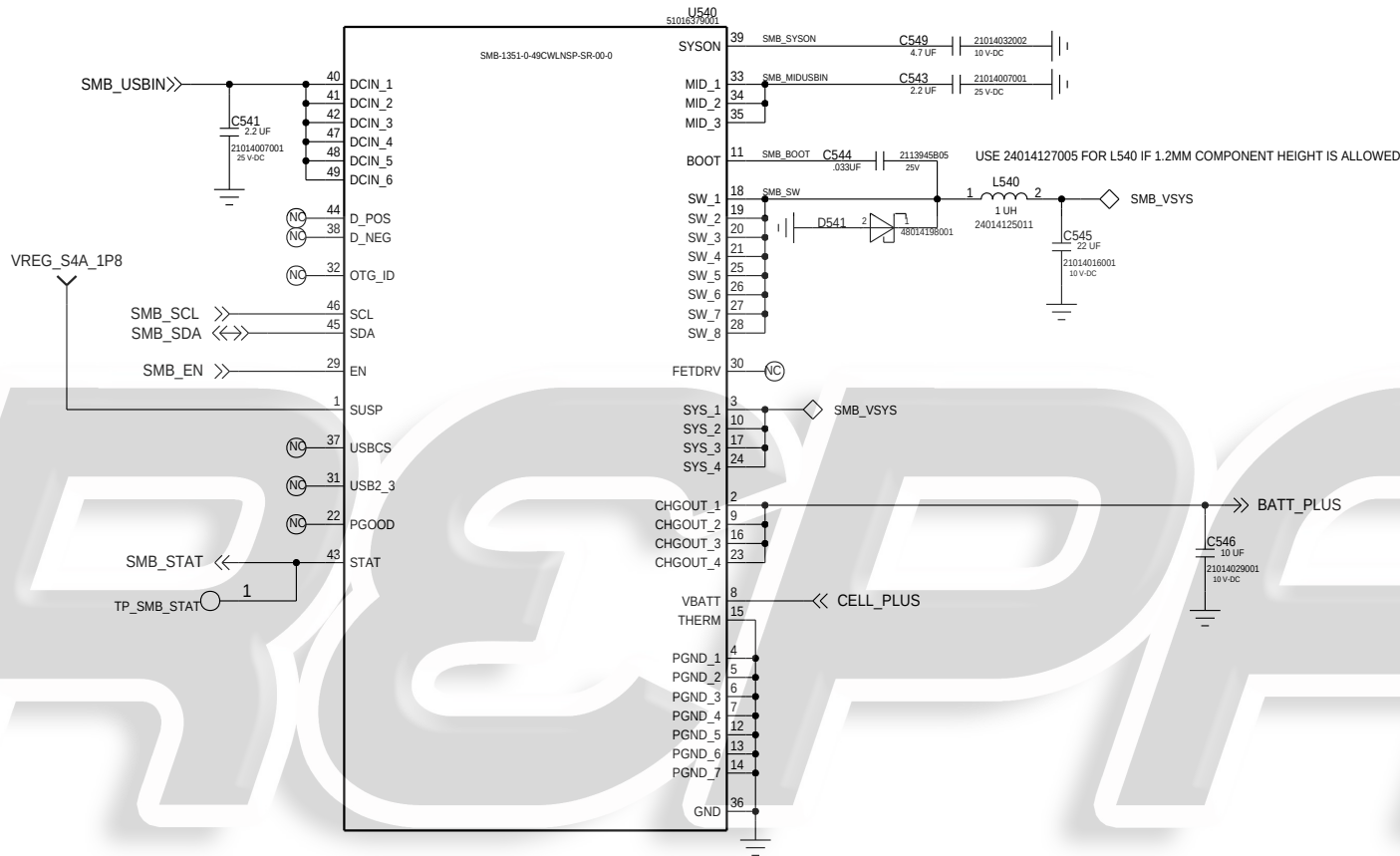
- R651 IS REQUIRED FROM EMU_ID TO VREG WITH FACTORY CABLE
- BATT_PLUS_CONN



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: UDI ELRAN Wed Jun 10 14:32:36 2015			
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		CAD FILE NAME:	PAGE 22 OF 78

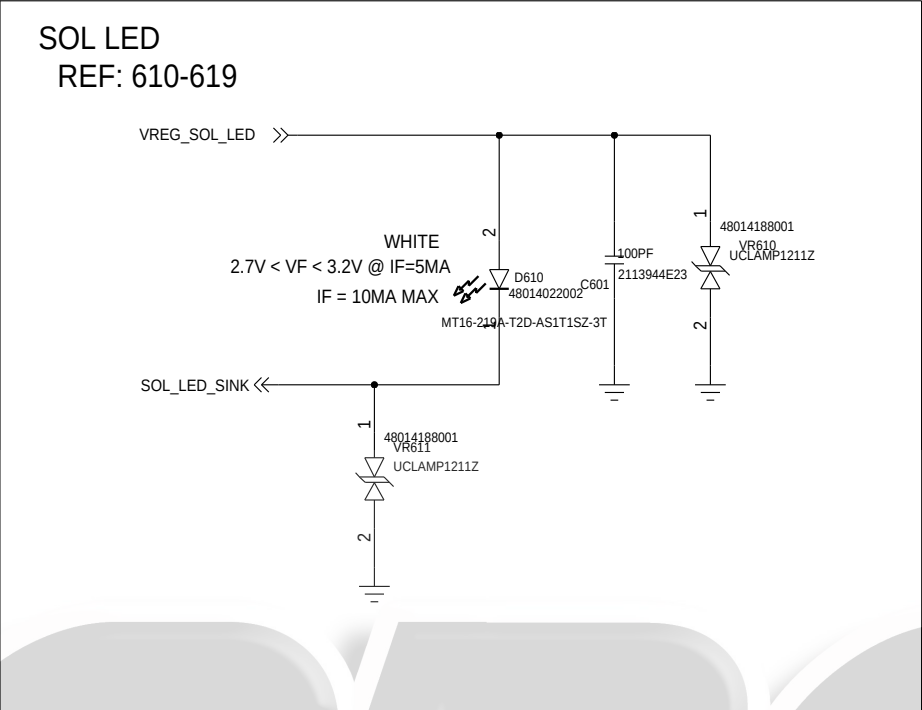
PWR: EXT CHG

REF: 540-559



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 23 OF 78	

PWR: LIGHTING

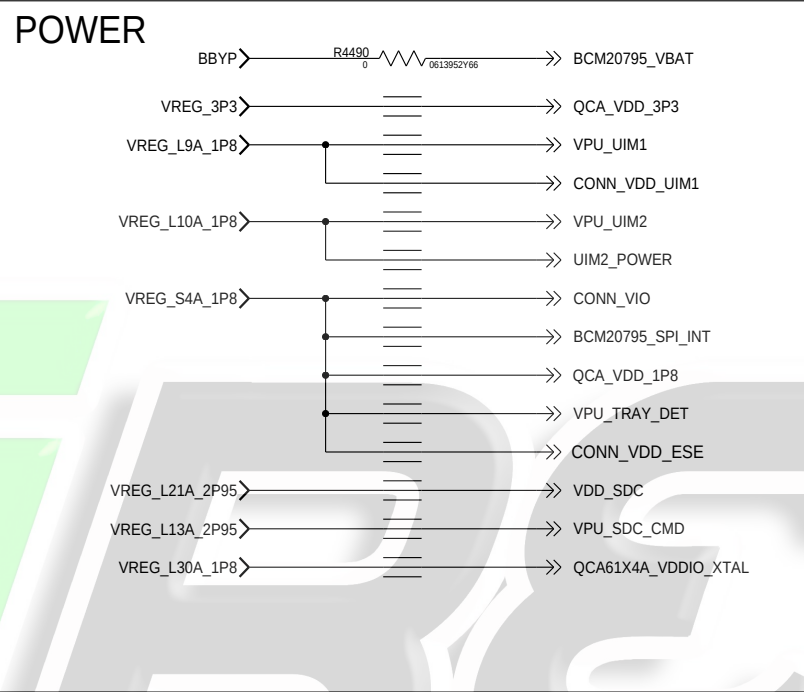


REPAIR

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LAST MODIFIED BY: TIM MILLER Wed Jun 10 14:32:38 2015		power	
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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 24 OF 78	

CONN: POWER
REF 4490-4499



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Wed Jun 10 14:32:26 2015		connectivity	
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		B 84016440001_b.cpm	01
		CAD FILE NAME:	PAGE 25 OF 78

CONN: PORTS

REF 4000-4499

BT/WLAN SYSTEM

BT_WLAN_32KHZ_CLK_IN [IN] QCA61X4_32KHZ_CLK_IN

WLAN

WLAN_ENABLE [IN] QCA61X4_WL_EN

PCIE

PCIE_CLK_REQ_OUT_N [OUT] QCA61X4_PCIE_CLKREQ_N
PCIE_CLKIN_DP [IN] QCA61X4_PCIE_REFCLKP
PCIE_CLKIN_DM [IN] QCA61X4_PCIE_REFCLKN
PCIE_RX_DP [IN] QCA61X4_PCIE_RXP
PCIE_RX_DM [IN] QCA61X4_PCIE_RXN
PCIE_TX_DP [OUT] QCA61X4_PCIE_TXP
PCIE_TX_DM [OUT] QCA61X4_PCIE_TXN
PCIE_RST_IN_N [IN] QCA61X4_PCIE_RST_N
PCIE_HOST_WAKE_OUT_N [OUT] QCA61X4_PCIE_WAKE_N

BT CONTROLS AND INTERFACES

BT_ENABLE [IN] QCA61X4_BT_EN

BT_PCM_IN [IN] QCA61X4_PCM_IN
BT_PCM_OUT [OUT] QCA61X4_PCM_OUT
BT_PCM_CLK [IN] QCA61X4_PCM_CLK
BT_PCM_SYNC [IO] QCA61X4_PCM_SYNC

BT_UART_TX [OUT] QCA61X4_UART_TXD
BT_UART_RX [IN] QCA61X4_UART_RXD
BT_UART_CTS_N [IN] QCA61X4_UART_CTS_N
BT_UART_RTS_N [OUT] QCA61X4_UART_RTS_N

CELLULAR MODEM/GPS COEXISTENCE

MODEM_LTE_SYNC_UART_RXD [IN] QCA61X4_LTE_SYNC
WLAN_LTE_PRI_UART_TXD [OUT] QCA61X4_LTE_PRI
WLAN_GPS_COEX [OUT] QCA61X4_GPS_COEX

NFC

NFC_CLK_IN [IN] BCM20795_EXT_CLK_IN
NFC_CLK_REQ_OUT [OUT] BCM20795_CLK_REQ_OUT
NFC_ENABLE_IN [IN] BCM20795_ENABLE_IN
NFC_IRQ_OUT [OUT] BCM20795_IRQ_OUT
NFC_SCL [IN] BCM20795_SCL
NFC_SDA [IO] BCM20795_SDA
NFC_POWER_ON [IN] BCM20795_REG_PU

SDCARD

SDC_CLK [IN] CLK_SDC
SDC_CMD [IN] CMD_SDC
SDC_DATA<3..0> [IO] DATA_SDC<3..0>

UIM1

UIM1_DATA [IO] UIM1_DATA
UIM1_CLK [IN] UIM1_CLK
UIM1_RST [IN] UIM1_RST

UIM2

UIM2_DATA [IO] UIM2_DATA
UIM2_CLK [IN] UIM2_CLK
UIM2_RST [IN] UIM2_RST

ANTENNA

STREAM0_BT [IO] BT_WLAN_ANT0
STREAM1 [IO] WLAN_ANT1

TRAY_DET [OUT] TRAY_DET

APPROVALS

ENGINEER: FROEHLING

LAST MODIFIED BY: FROEHLING
Mon Mar 30 17:43:19 2015

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MOBILE DEVICES

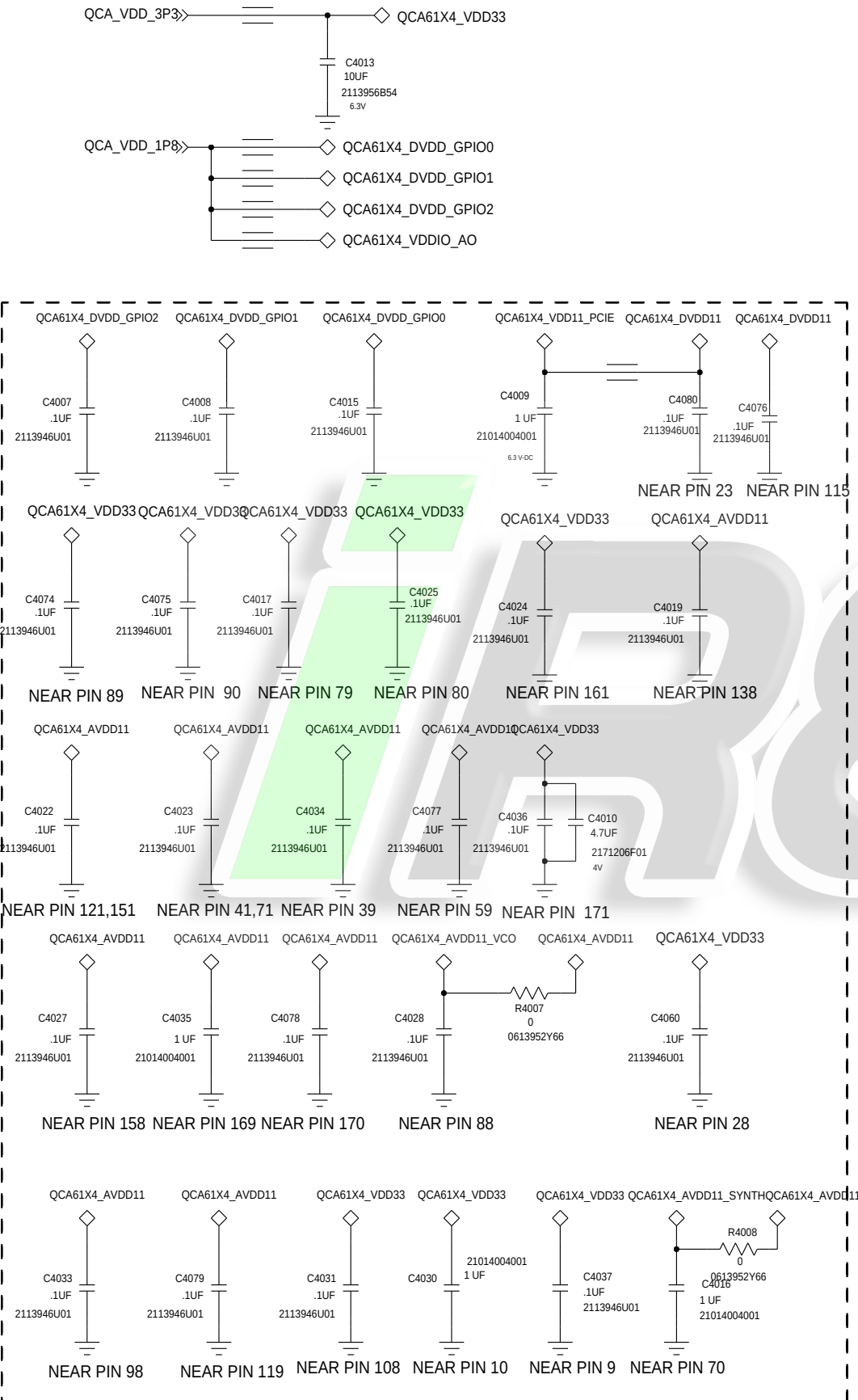
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connectivity

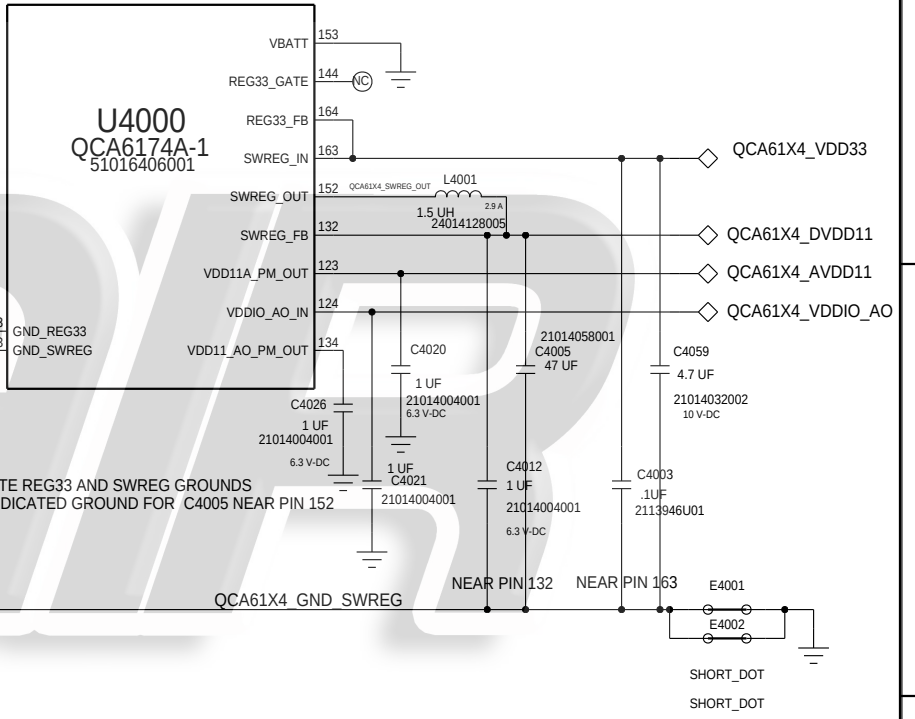
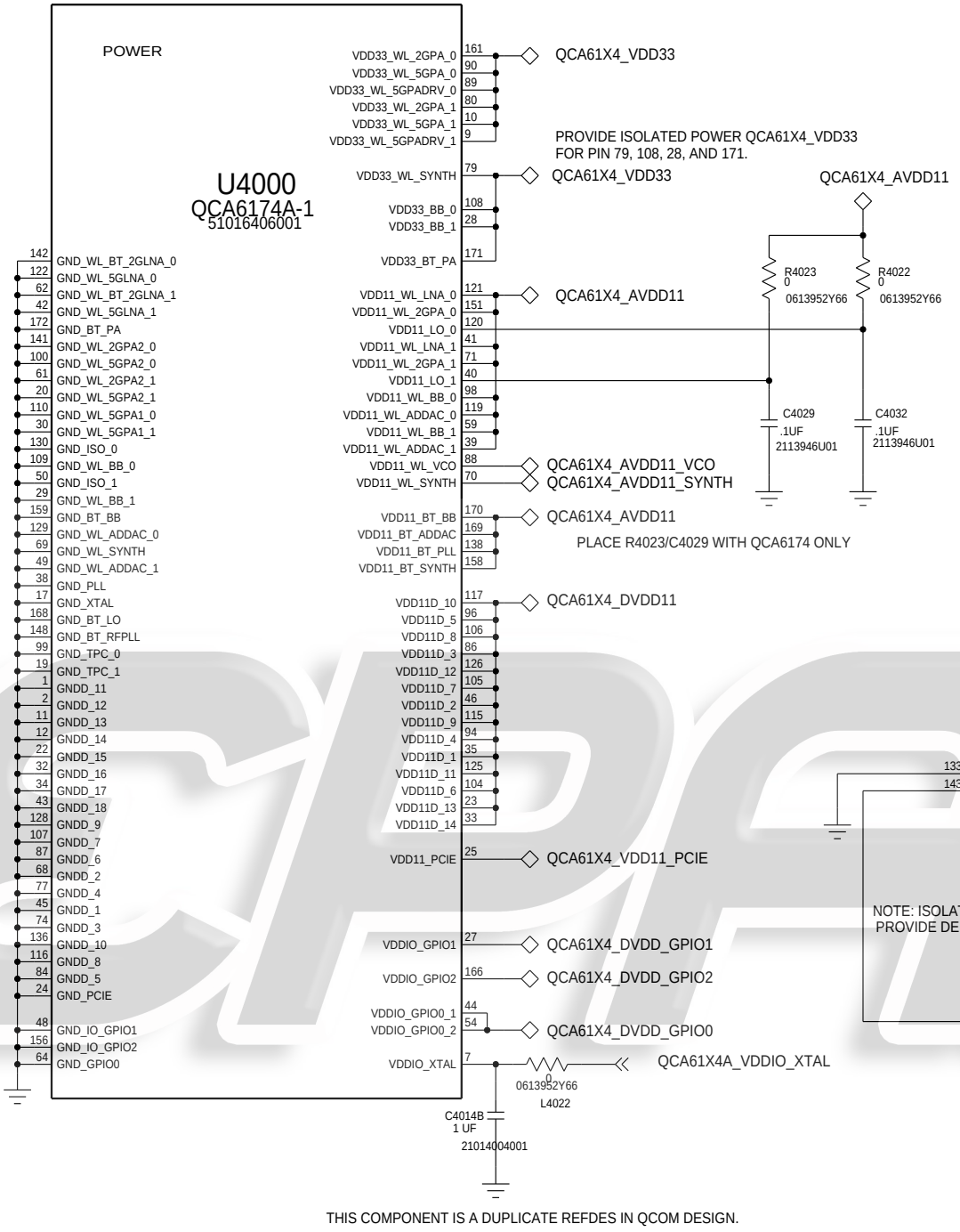
PROJECT NAME
84016440001_b.cpm
REV
01

CAD FILE NAME:
PAGE 26 OF 78

CONN: QCA61X4A POWER
REF 4000-4099



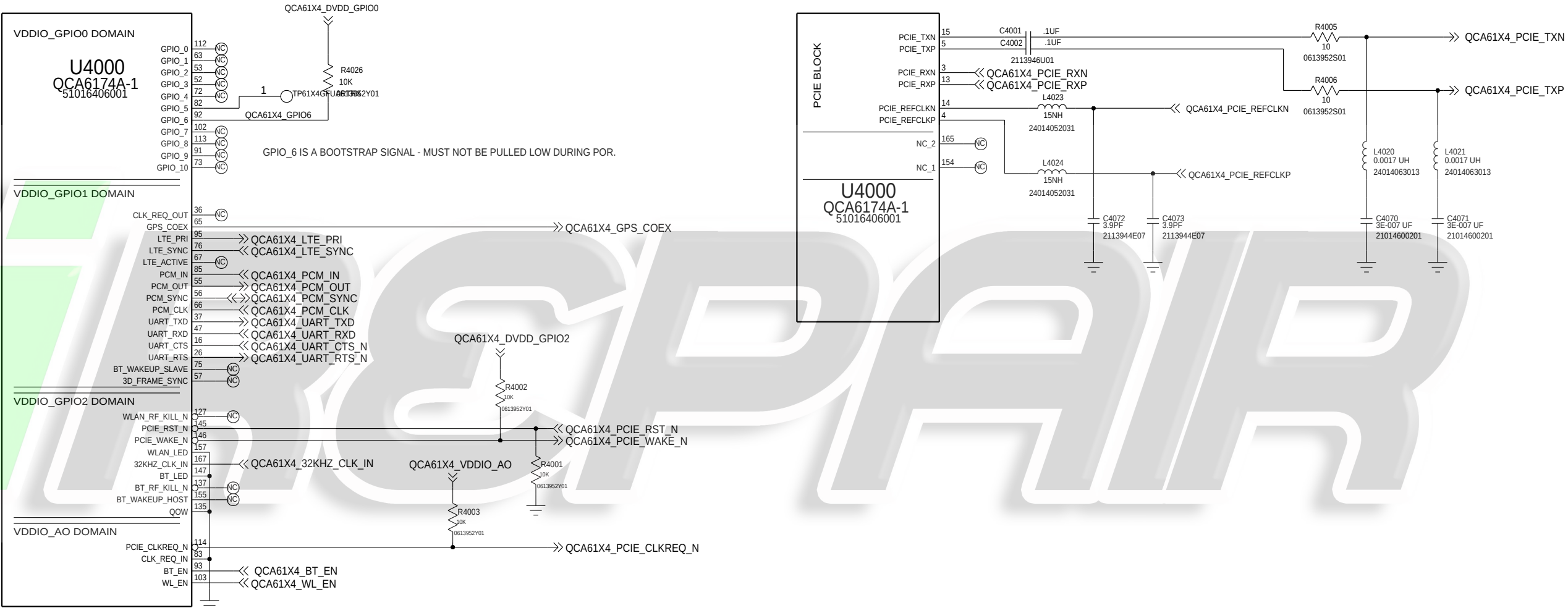
BASED ON 80-WL400-42 REV. B



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LAST MODIFIED BY: FROEHLING Wed Jun 10 14:32:29 2015		PROJECT NAME 84016440001_b.cpm	
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		PAGE 27 OF 78	

CONN: QCA61X4A HOST I/F

REF: 4000-4099

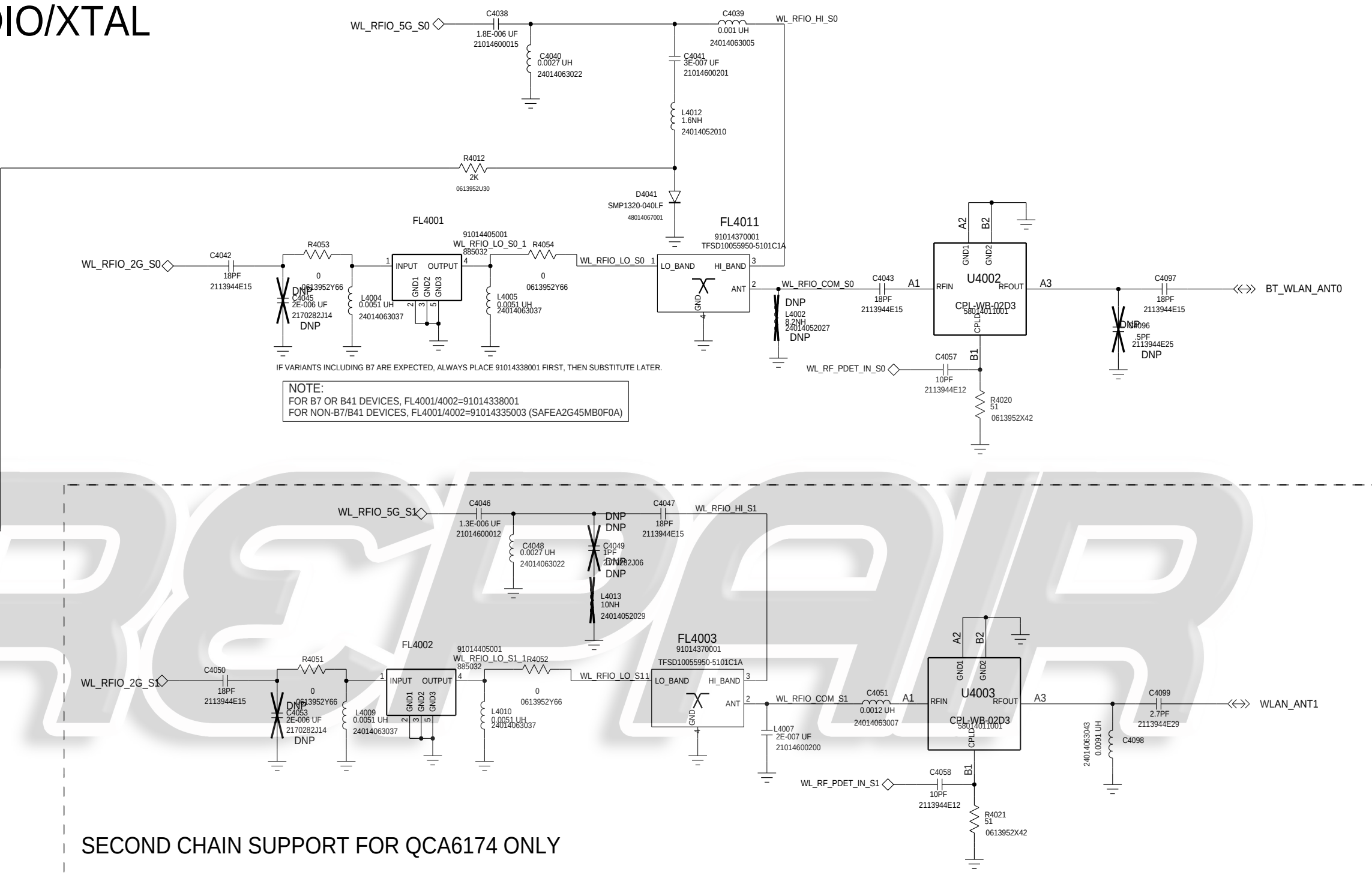
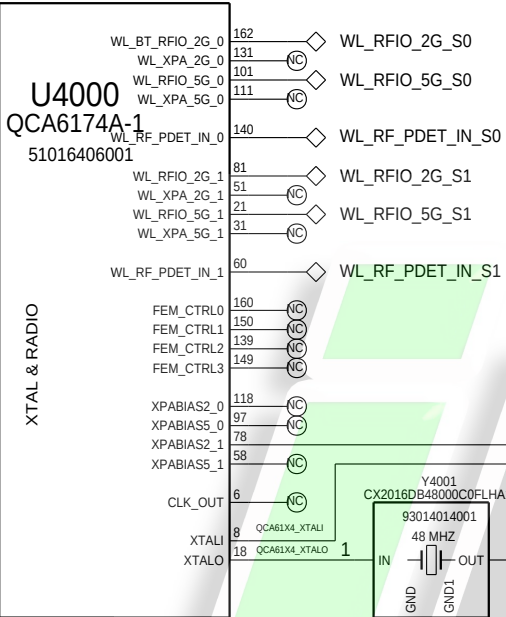


BASED ON 80-WL400/405-42 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

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		CAD FILE NAME: PAGE 28 OF 78	

CONN: QCA61X4A RADIO/XTAL
REF 4000-4099



SECOND CHAIN SUPPORT FOR QCA6174 ONLY

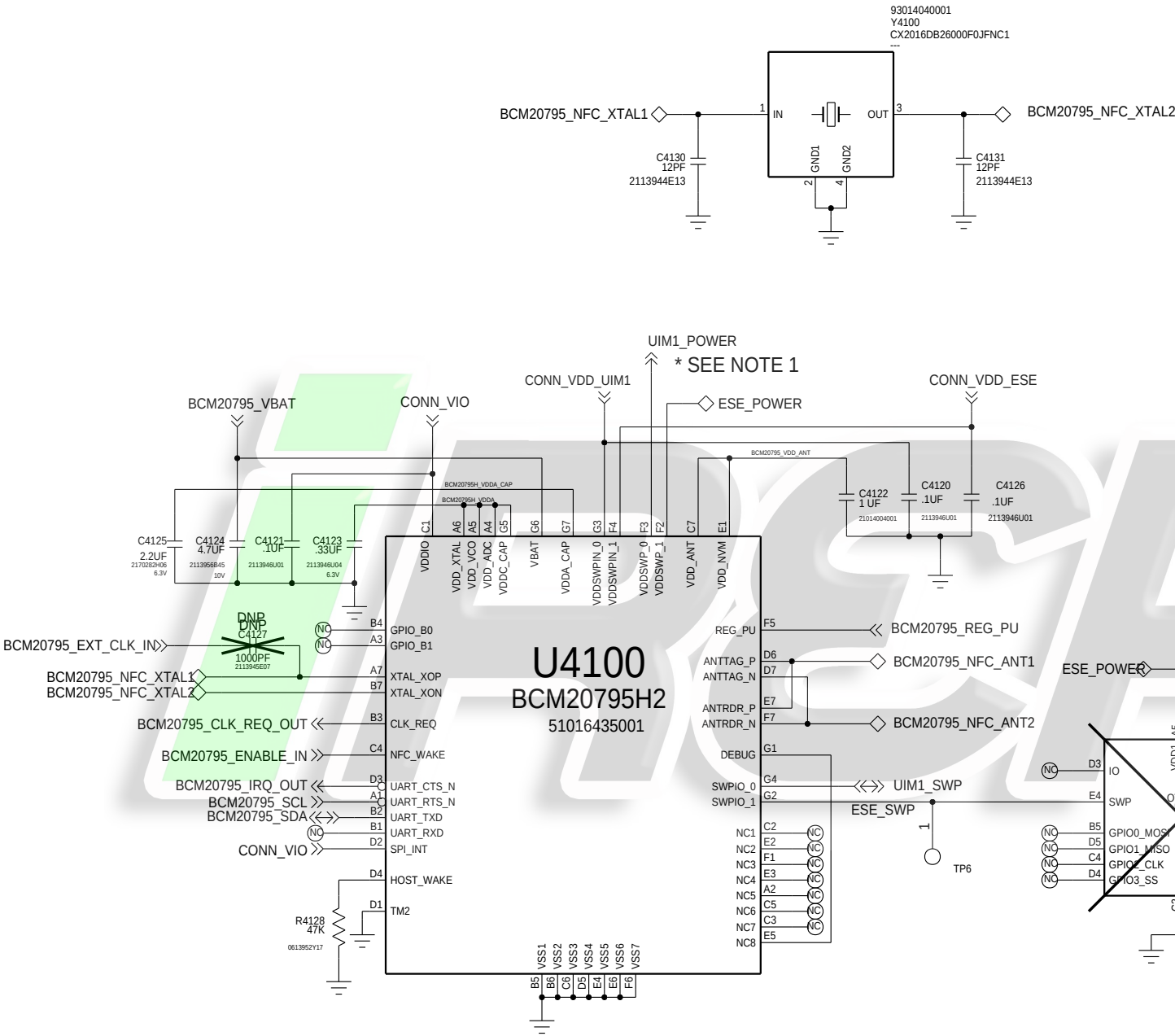
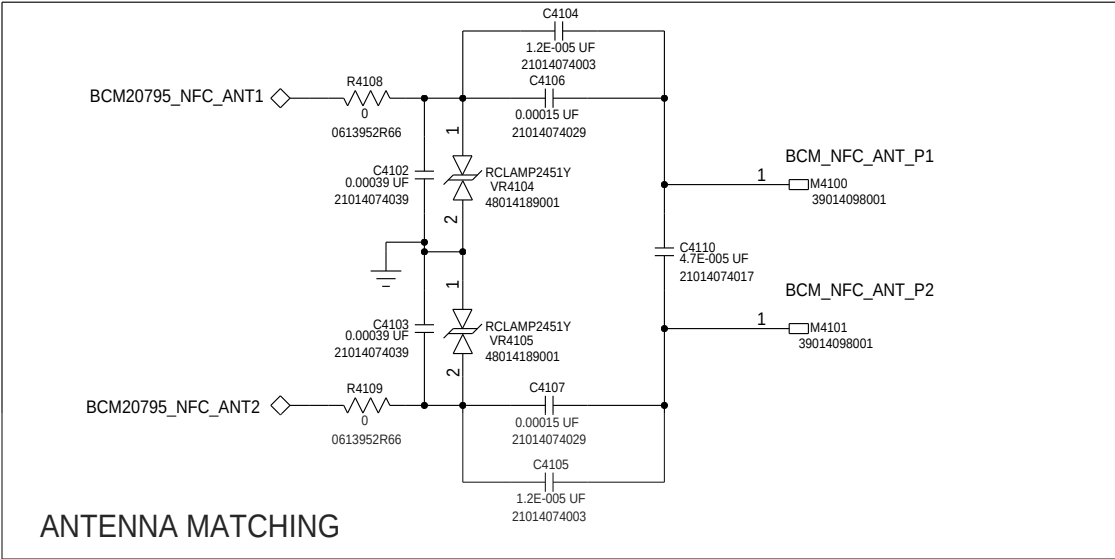
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LAST MODIFIED BY: FROEHLING Wed Jun 10 14:32:31 2015		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		CAD FILE NAME:	
		PAGE 29 OF 78	

BASED ON 80-WL400-42 REV. B

CONN: BCM20795 NFC

REF: 4100-4199

I2C ADDR: 0X77 (BOOT DEFAULT WHEN PIN 3 IS PULLED TO GROUND)
OR
I2C ADDR: 0X76 (BOOT DEFAULT WHEN PIN 3 IS FLOATED/PULLED HIGH)

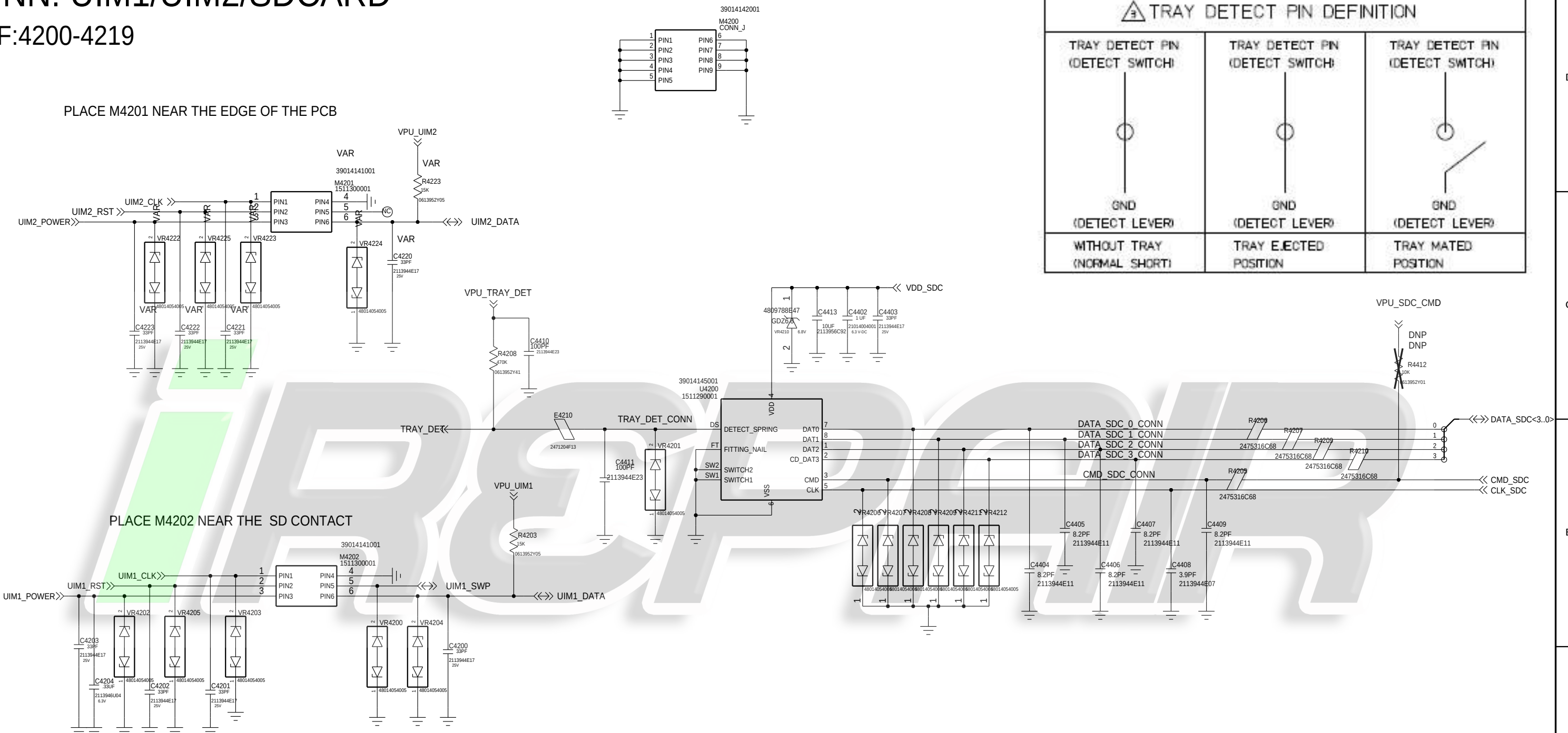


NOTES:
1. PER BROADCOM RECOMMENDATION, CAPACITANCE ON UIM1_POWER NET SHOULD BE 0.33UF. TOO LITTLE COMPROMISES REGULATOR STABILITY, TOO MUCH CAN CAUSE INADVERTENT POWER-ON RESETS AT VDD.
2. SECURE ELEMENT U4101 ONLY PLACED ON DSDA SKU (CHINA) REFER TO BELOW CHART:

	U4100	C4100
ATT/ROW	DNI	DNI
VZW/ SPRINT/ USC	DNI	DNI
TMO	DNI	DNI
CHINA	PLACE	PLACE

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Wed Jun 10 14:32:32 2015		connectivity	
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		B 84016440001_b.cpm	01
		CAD FILE NAME:	
		PAGE 30 OF 78	

CONN: UIM1/UIM2/SDCARD
REF:4200-4219



NOTE 1: USE 24.9K PULL-UP TO SUPPORT HOT SWAP WITH FREEZE-IO FEATURE.
NOTE 2: UIM_PWR BYPASS OF 0.33UF RECOMMENDED BY NFC IC SUPPLIER, PLACE NEAR CONNECTOR.

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKED AND BLOCKED			
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Wed Jun 10 14:32:33 2015			
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		CAD FILE NAME:	PAGE 31 OF 78

87654321

USR: PORTS

REF: 800-999

D

C

B

A

87654321

CAMERA_FLASH



SLICE INTERFACES



GPIO - INPUT

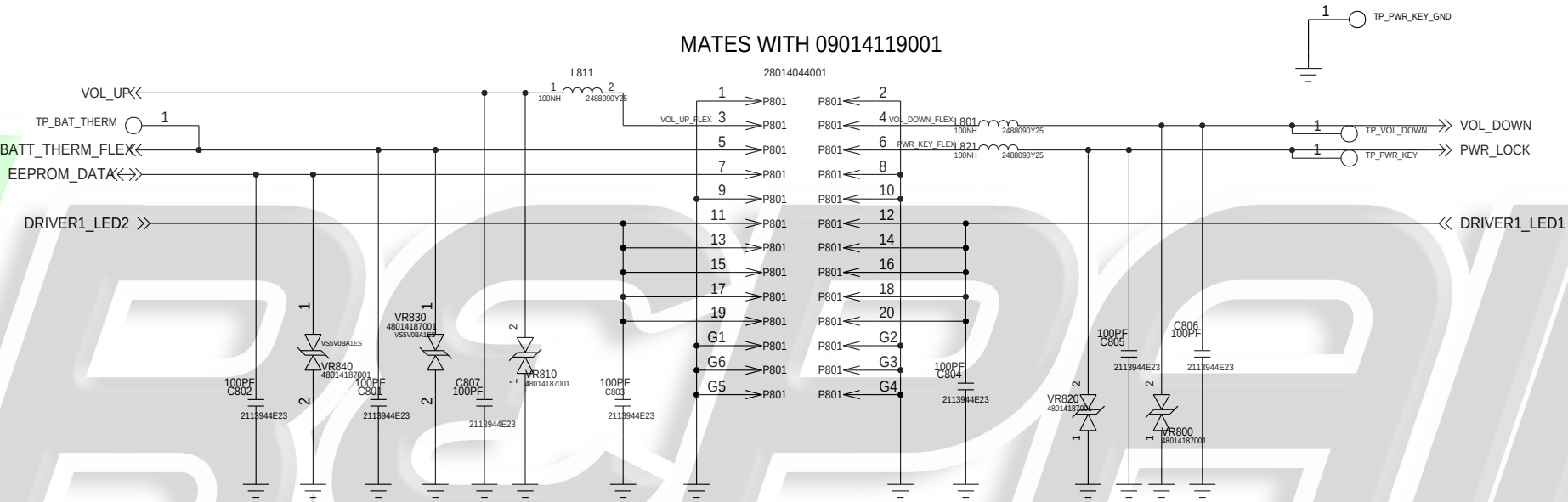
GPIO - OUTPUT



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: user_input	
LAST MODIFIED BY: NATE CONNELL Wed Nov 26 14:04:42 2014			
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		CAD FILE NAME: PAGE 40 OF 78	

USR: INPUT

REF: 800-899

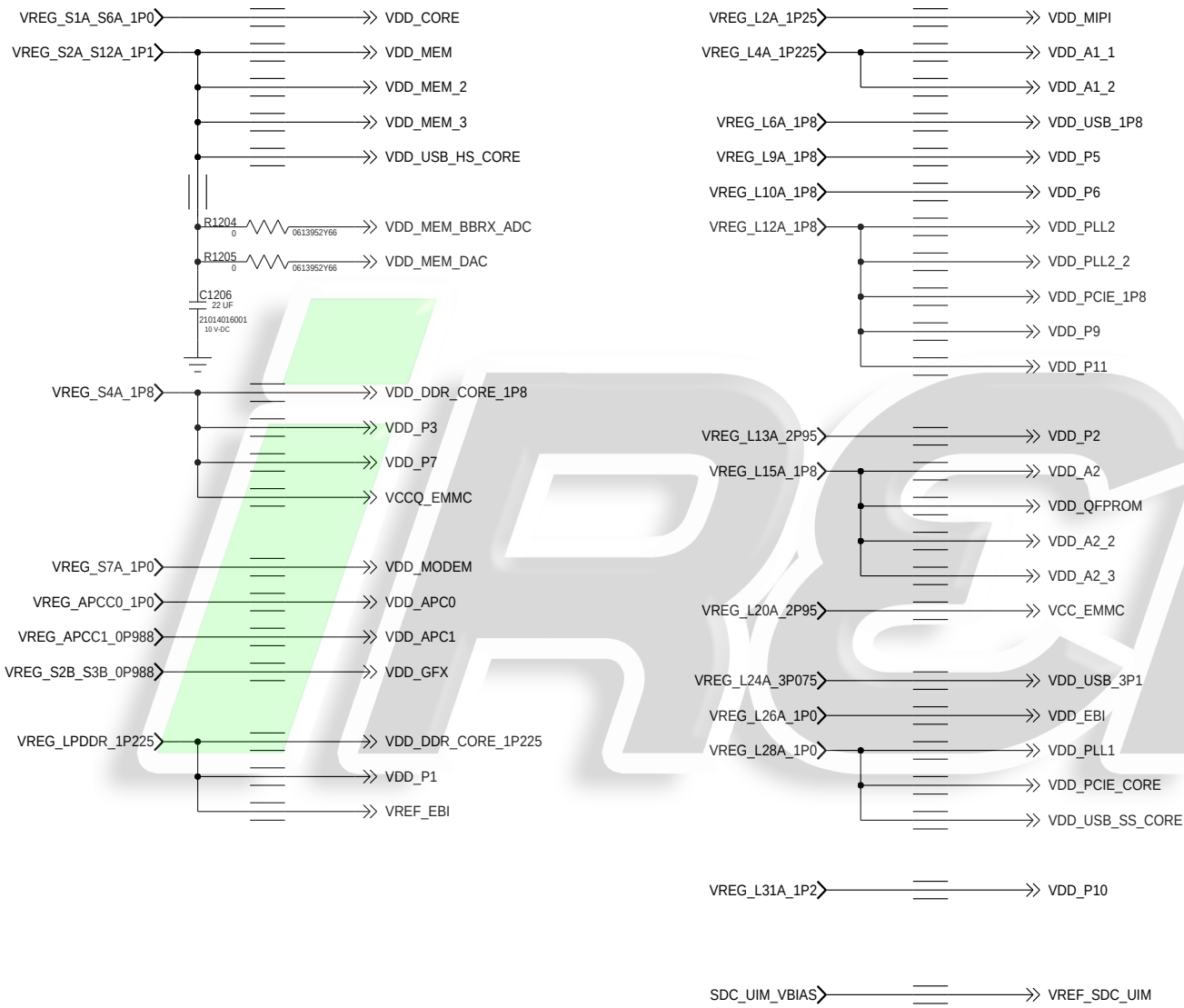


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		B 84016440001_b.cpm	
		CAD FILE NAME:	
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		PAGE 41 OF 78	

MSM8992: POWER

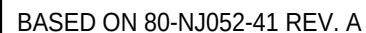
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SUPPLIES



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		CAD FILE NAME:	
		PAGE 42 OF 78	

REFDES: 1000-1099

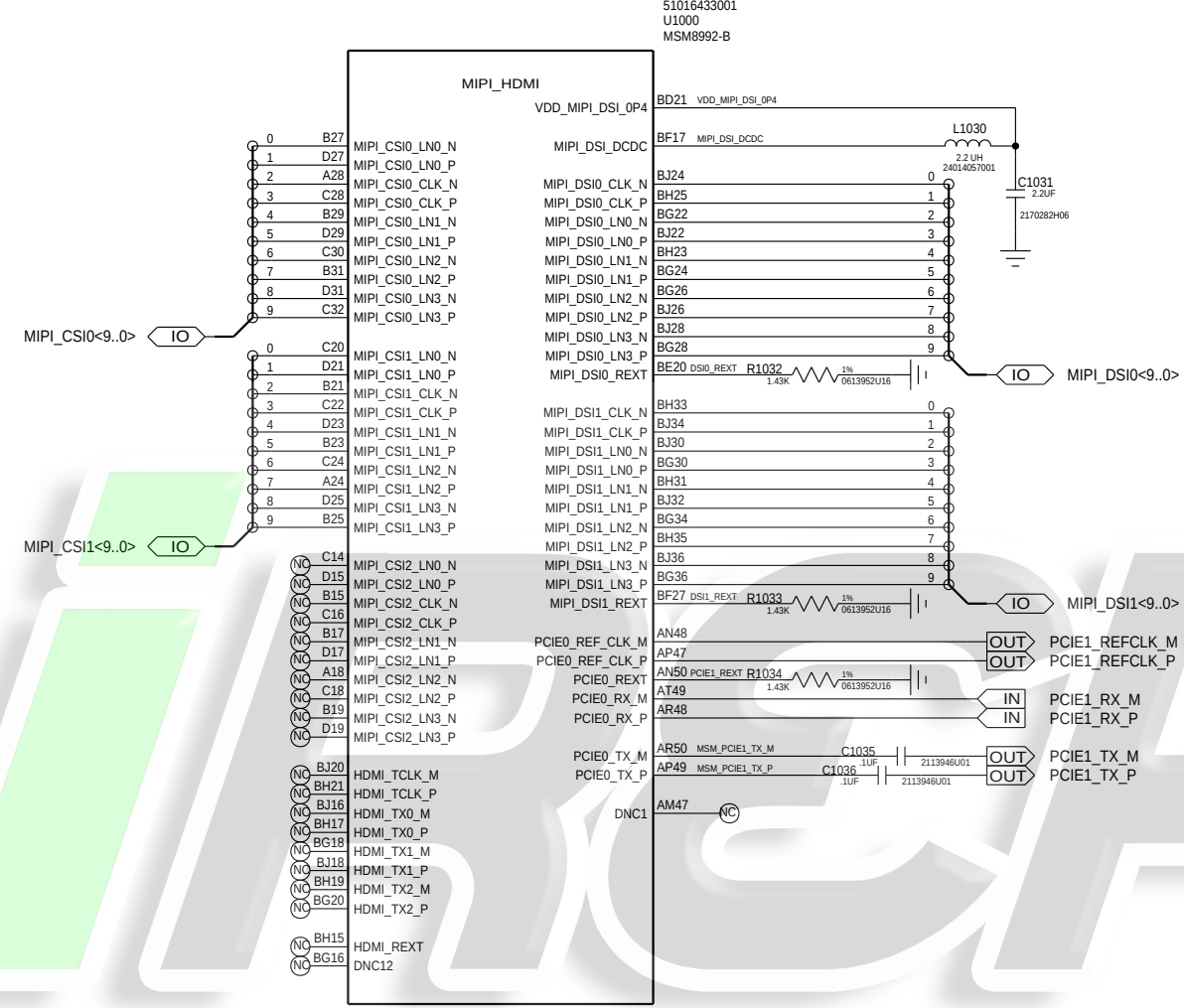


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		CAD FILE NAME:	PAGE 43 OF 78

MSM8992: MIPI/HDMI/PCIE

REFDES: 1030-1039

QCOM REFERENCE DESIGN HAS REAR CAMERA ON CSI0 AND FRONT CAMERA ON CSI2.



QCOM REFERENCE DESIGN HAS 1.43K 1% PULL-DOWN ON THE HDMI_REXT PIN.

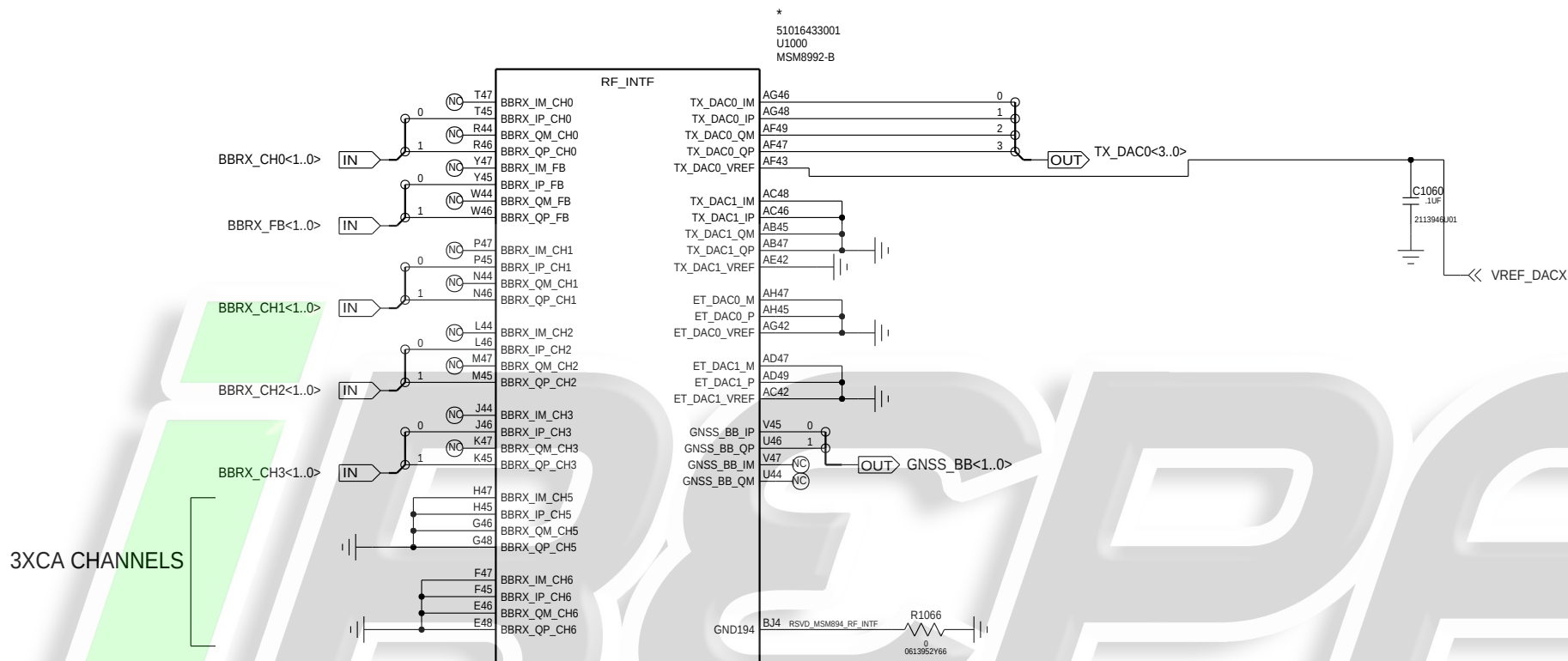
BASED ON 80-NJ052-41 REV. A

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		B 84016440001_b.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 44 OF 78	

MSM8992: RF INTERFACE

REFDES: 1040-1049

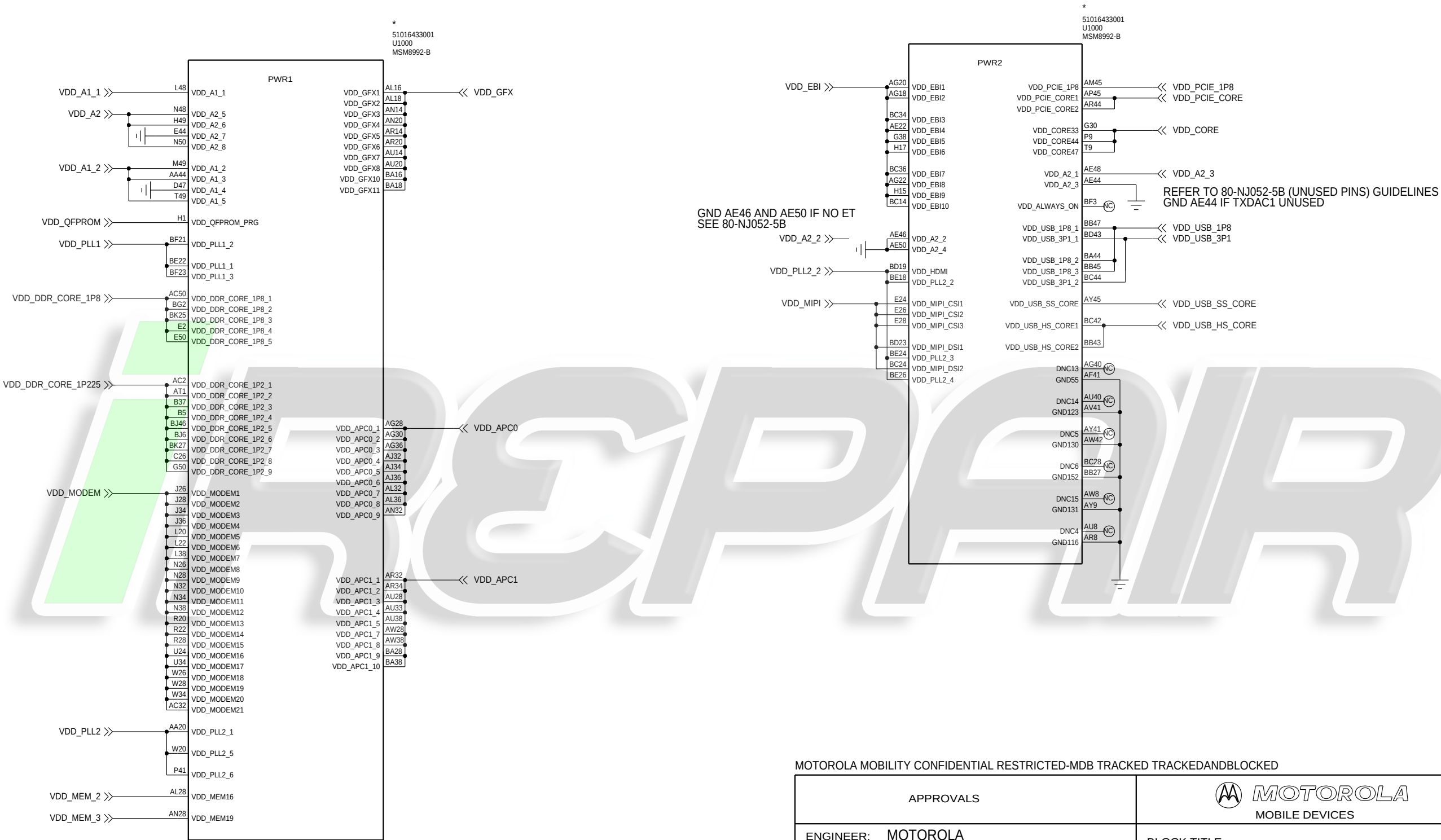


BASED ON 80-NJ051-41 REV. B

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		REV 01	
		CAD FILE NAME:	
		PAGE 45 OF 78	

MSM8992: POWER 1

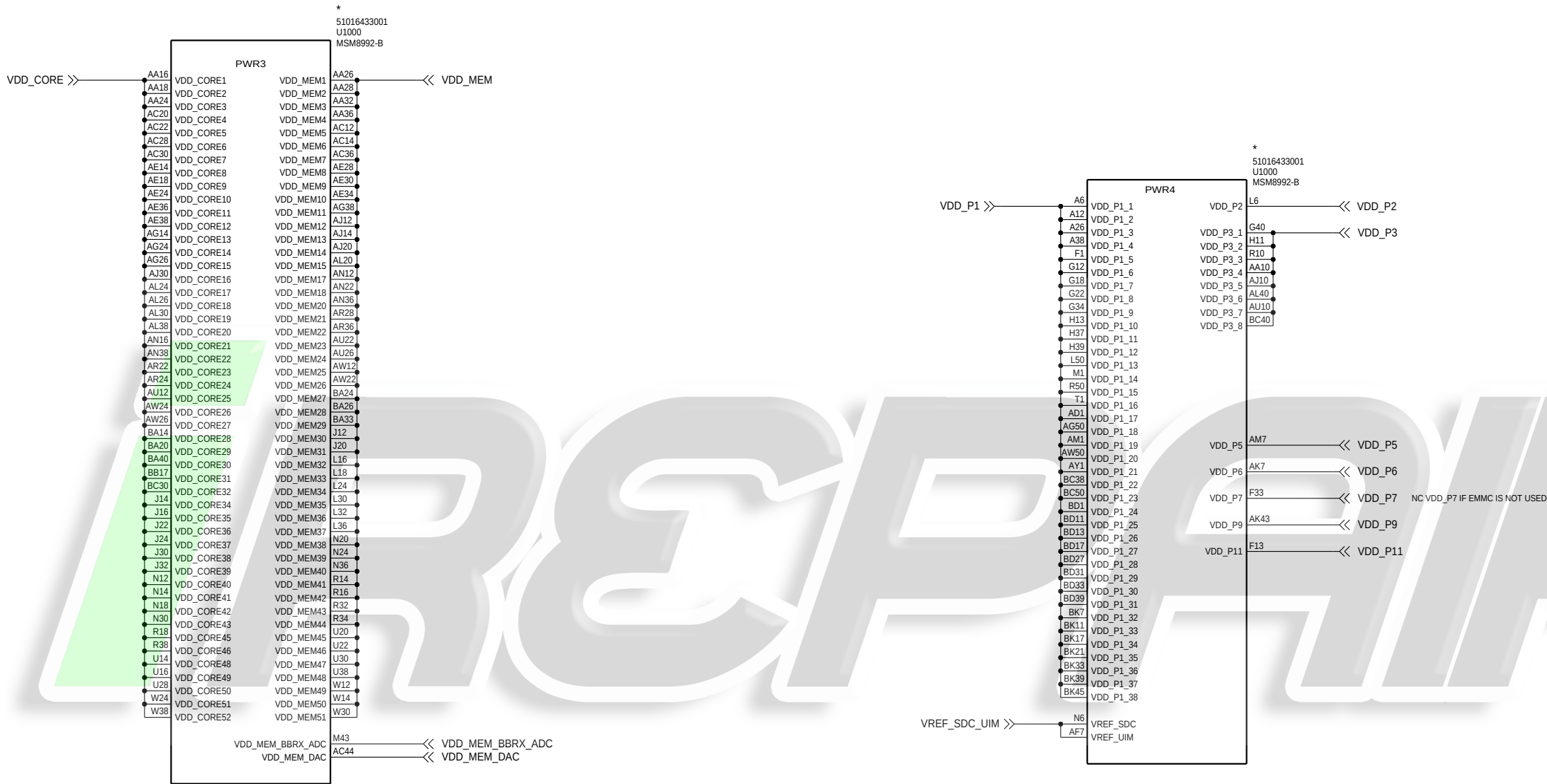


BASED ON 80-NJ052-41 REV. A

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
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		B 84016440001_b.cpm	01
CAD FILE NAME:			PAGE 46 OF 78

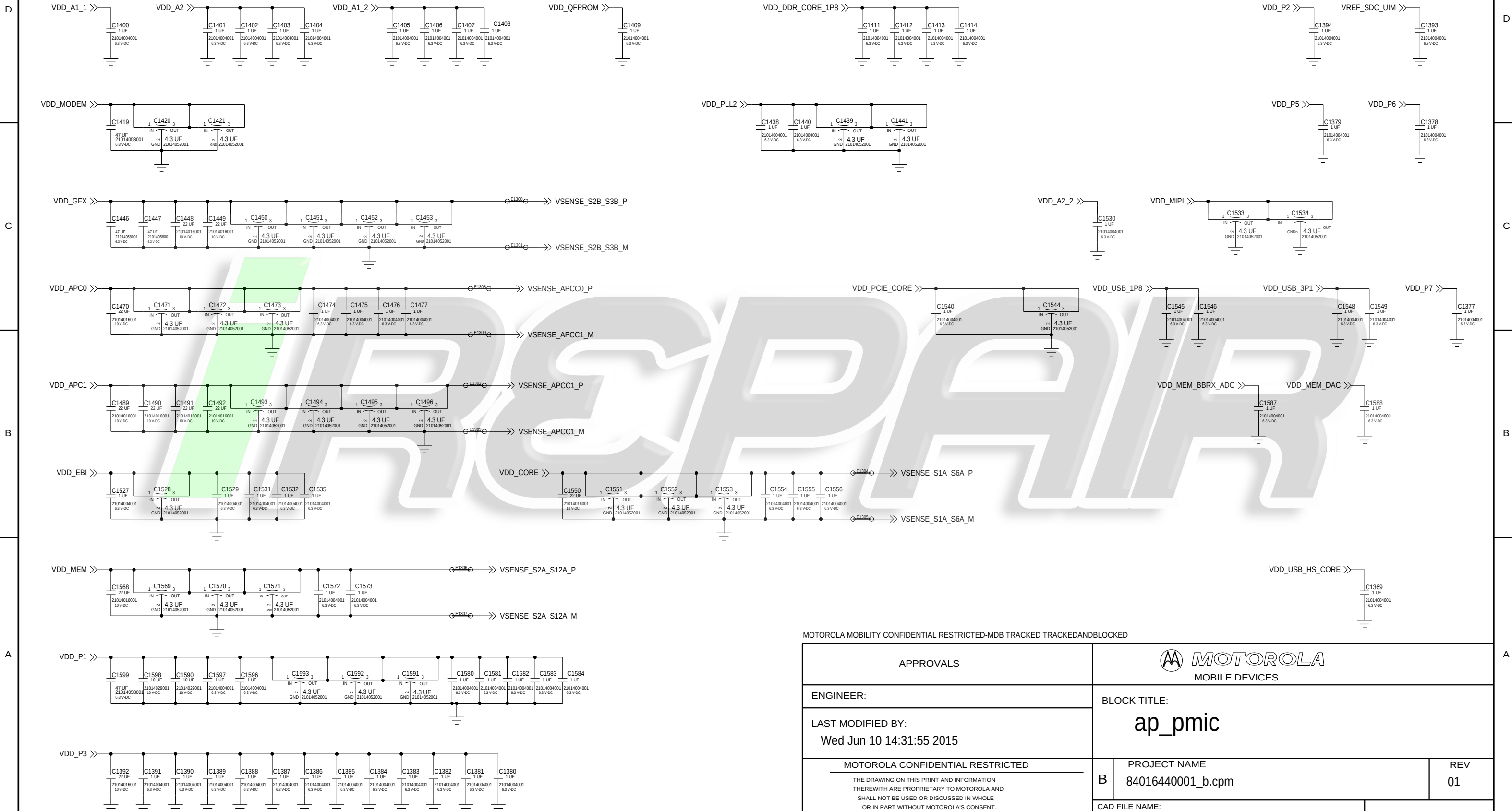
MSM8992: POWER 2



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Wed Jun 10 12:14:07 2015		ap_pmic	
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		B 84016440001_b.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 47 OF 78	

MSM8992: BYPASS CAPS

REFDES: 1300-1599

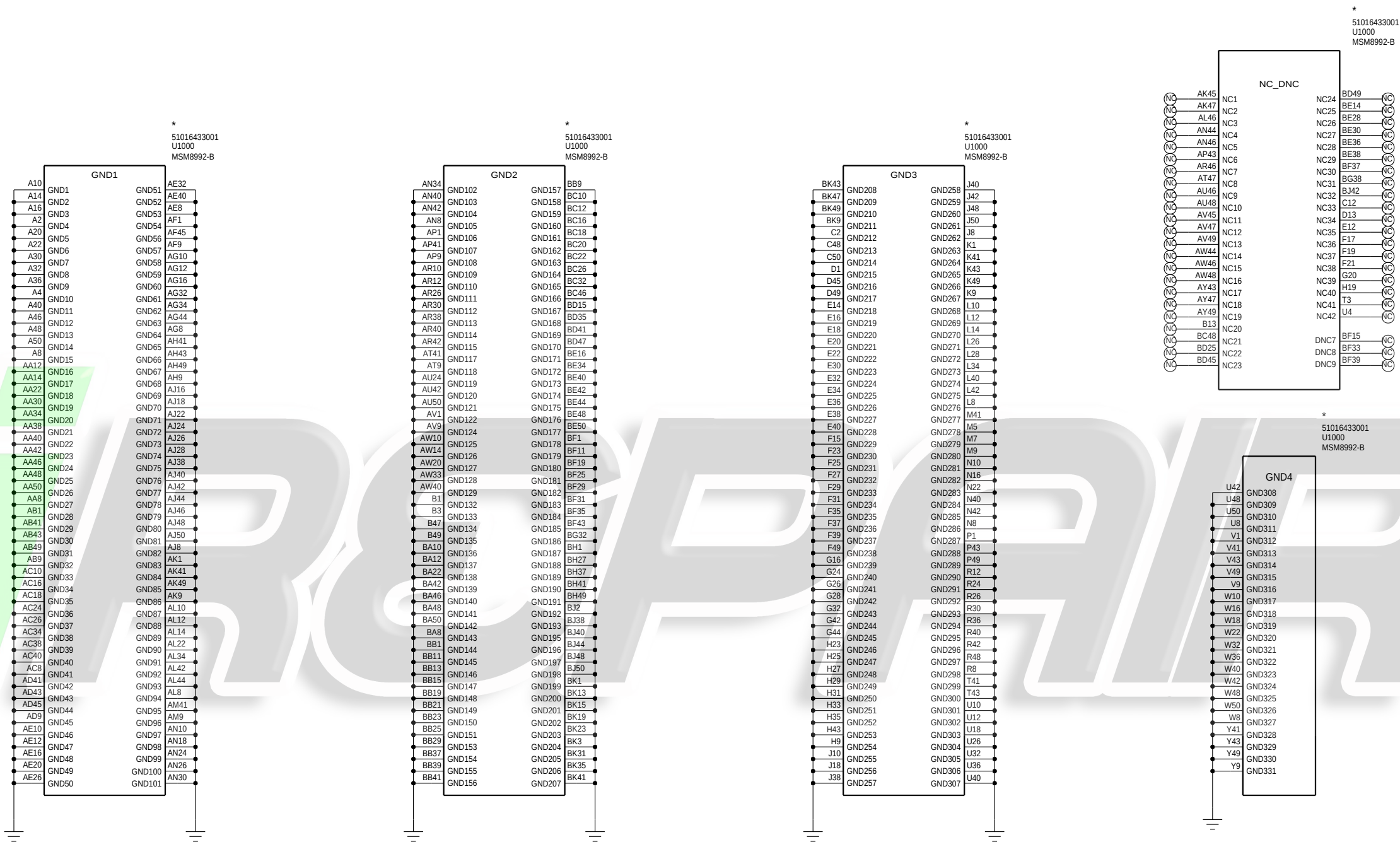


BASED ON 80-NJ051-41 REV. B

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		B 84016440001_b.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 48 OF 78	

MSM8992: GND/NC



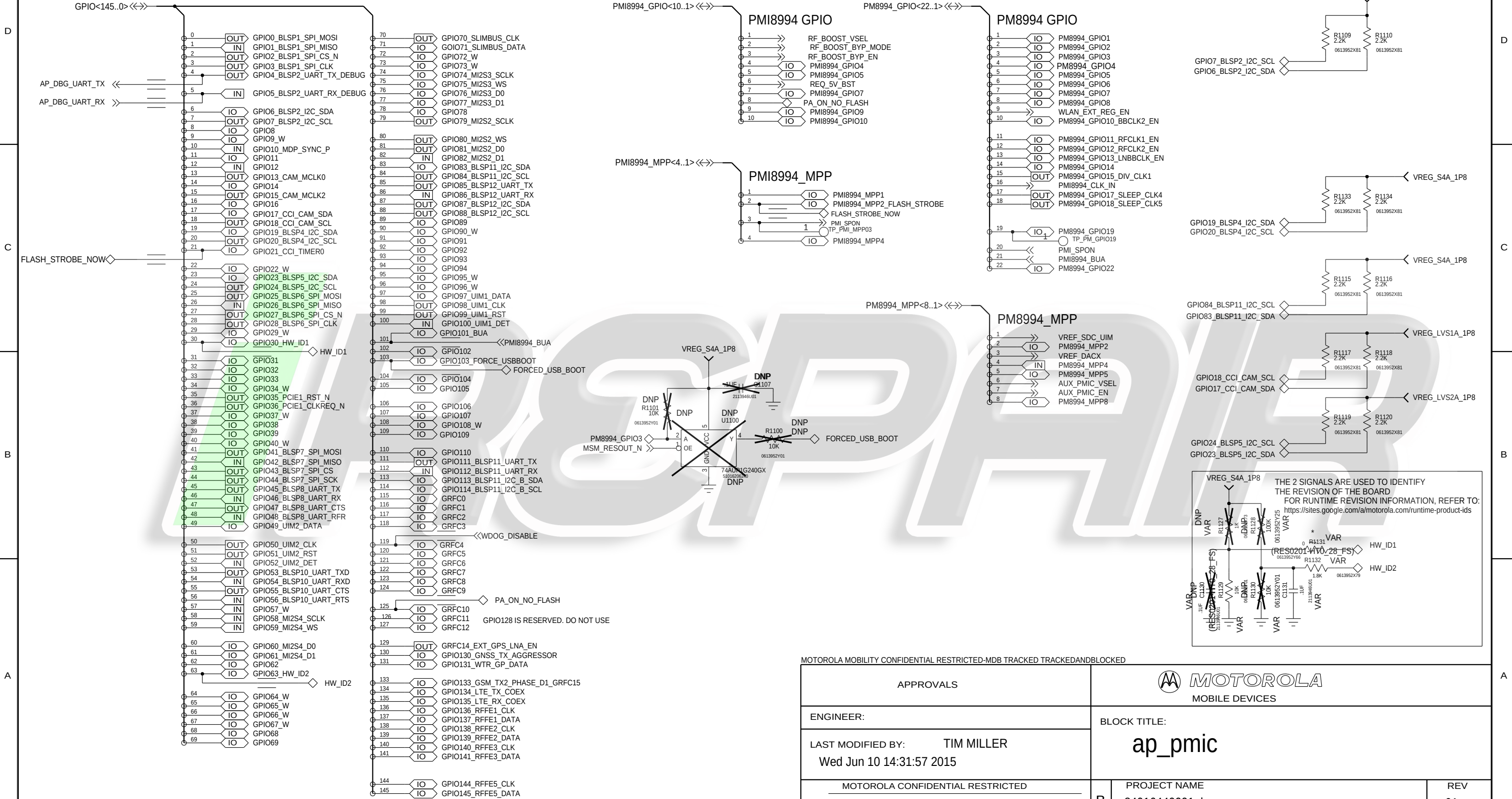
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B		REV 01	
CAD FILE NAME:		PAGE 49 OF 78	

MSM8992: GPIO MAP

REFDES: 1100-1199

PURPLE=MOT CUSTOM
CYAN=QCOM REFERENCE



BASED ON 80-NJ051-41 REV. B

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: TIM MILLER Wed Jun 10 14:31:57 2015		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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CAD FILE NAME:		PAGE 50 OF 78	

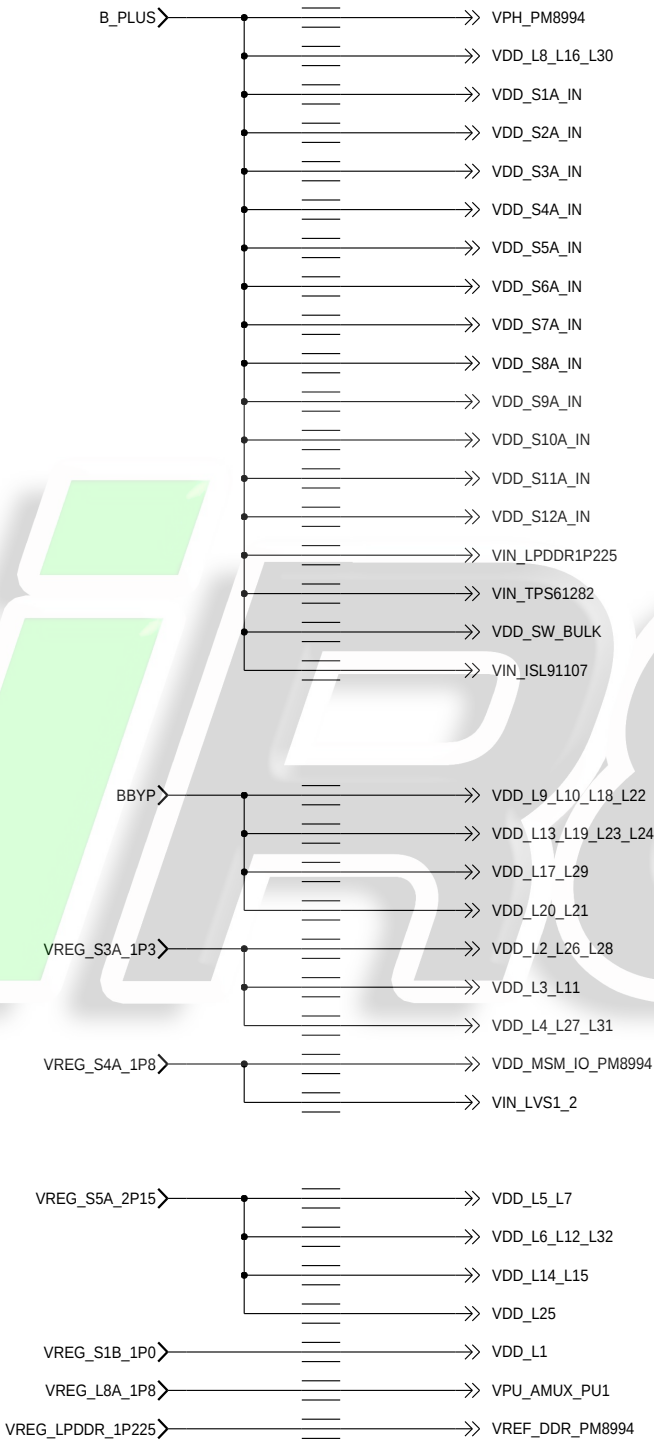
REFDES: 1600-1699

PAGE 51 OF 78

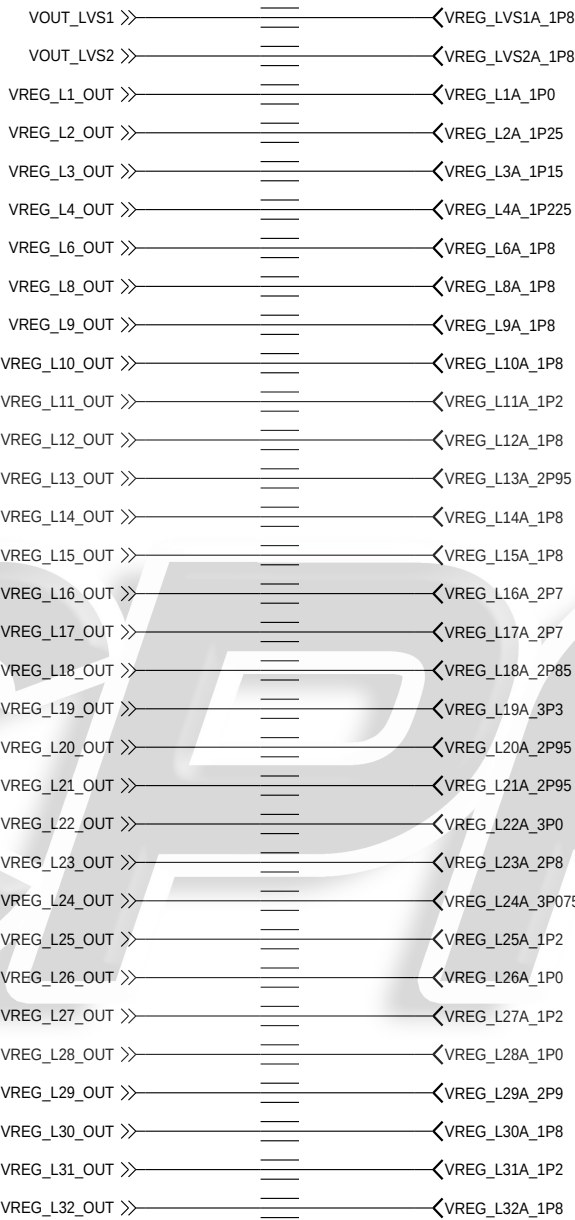
PM8994: POWER

REFDES: 2300-2399

INPUT POWER



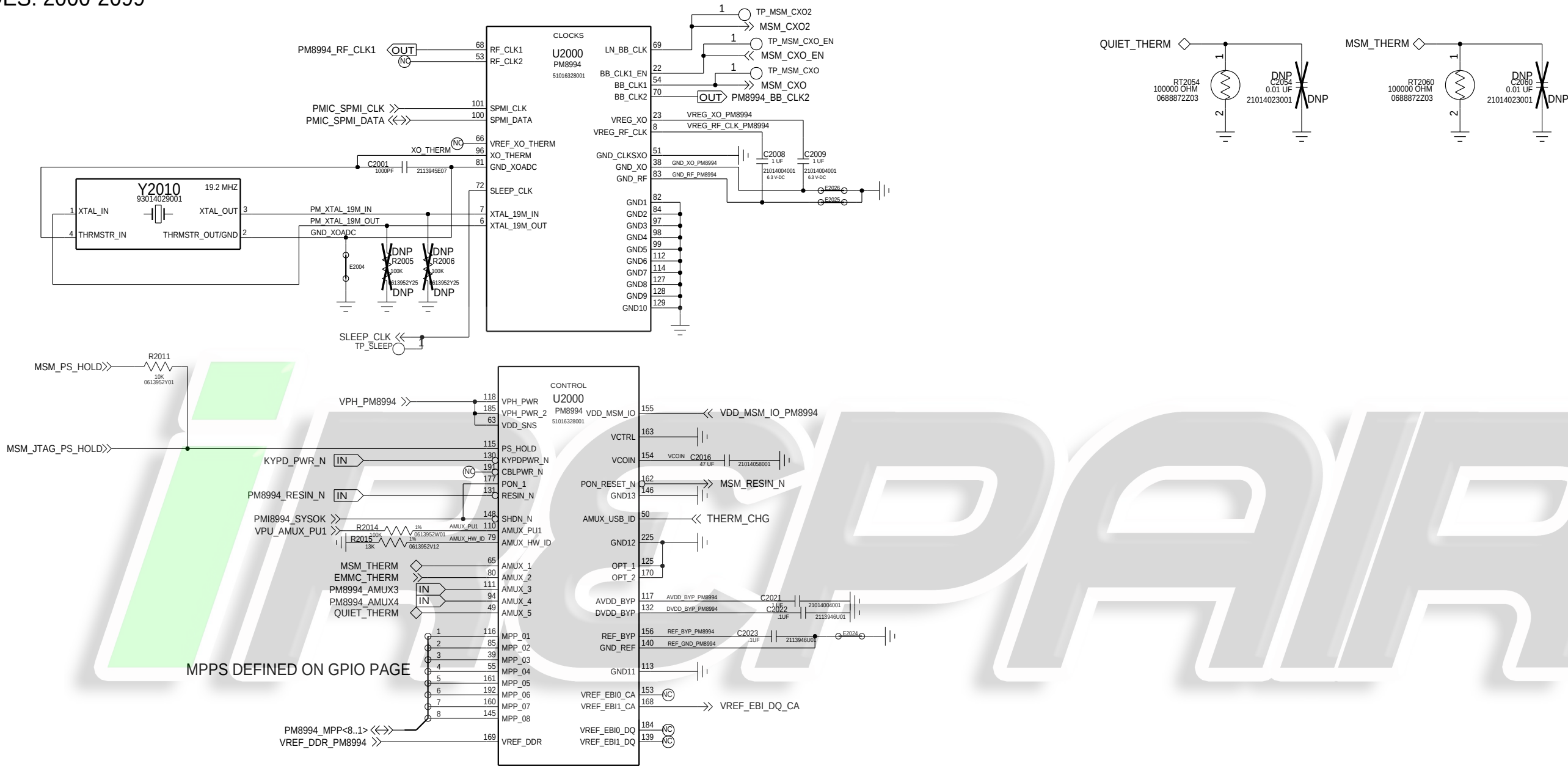
OUTPUT POWER



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:31:59 2015		ap_pmic	
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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 52 OF 78	

PM8994: CLK/CTRL/GPIO

REFDES: 2000-2099



FOLLOW CRYSTAL OSCILLATOR PLACEMENT/LAYOUT INSTRUCTIONS IN QCOM DOCUMENT 80-N5423-5 (P. 221 IN REV G)

*19.2MHZ CRYSTAL+THERMISTOR OPTIONS

PACKAGE PART NUMBER
2520 93014012001
2016 93014029001

PM GPIOs DEFINED ON MSM GPIO PAGE

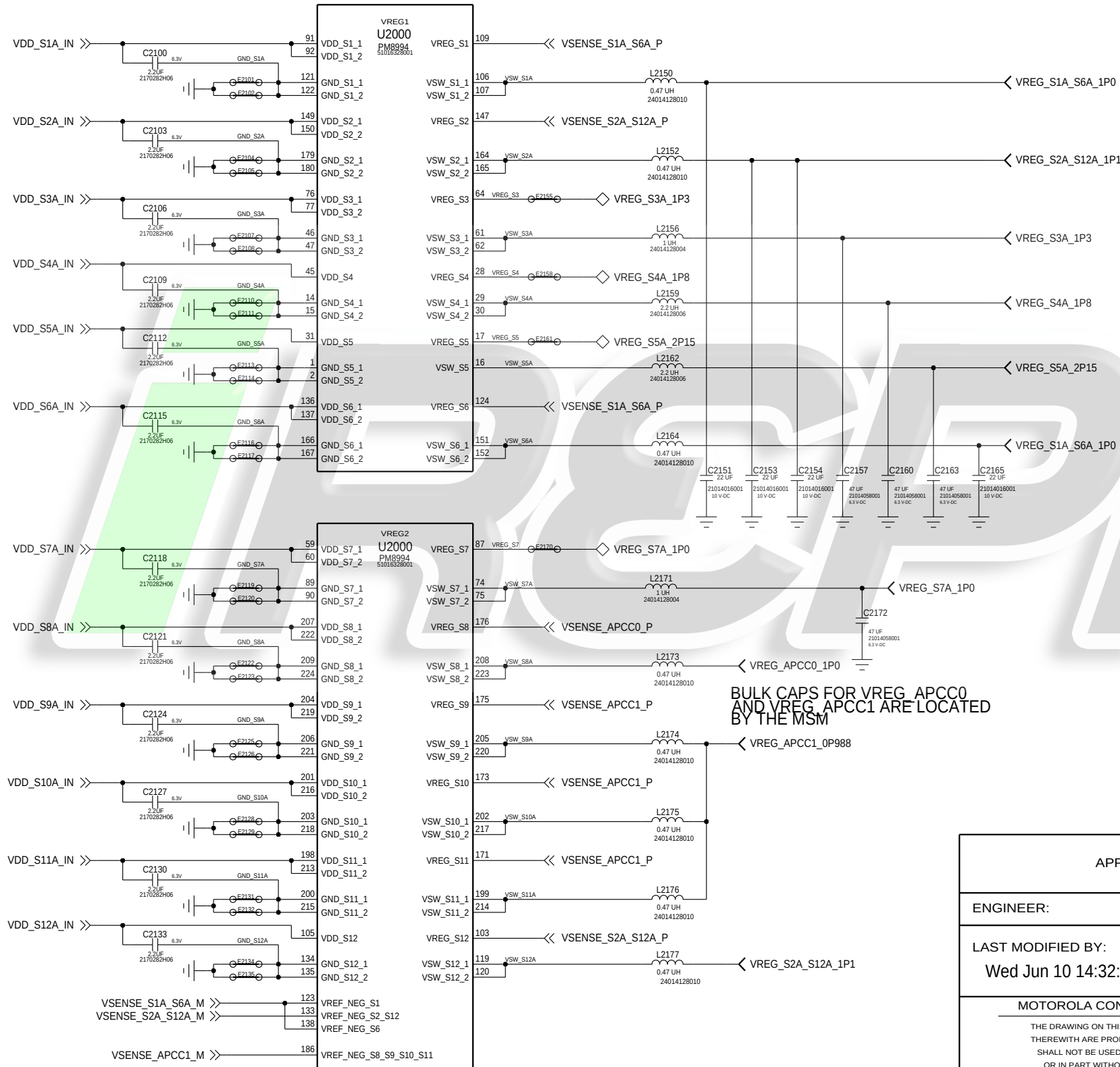
BASED ON 80-NJ051-41 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: Wed Jun 10 14:32:00 2015		PROJECT NAME B 84016440001_b.cpm	
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		CAD FILE NAME: PAGE 53 OF 78	

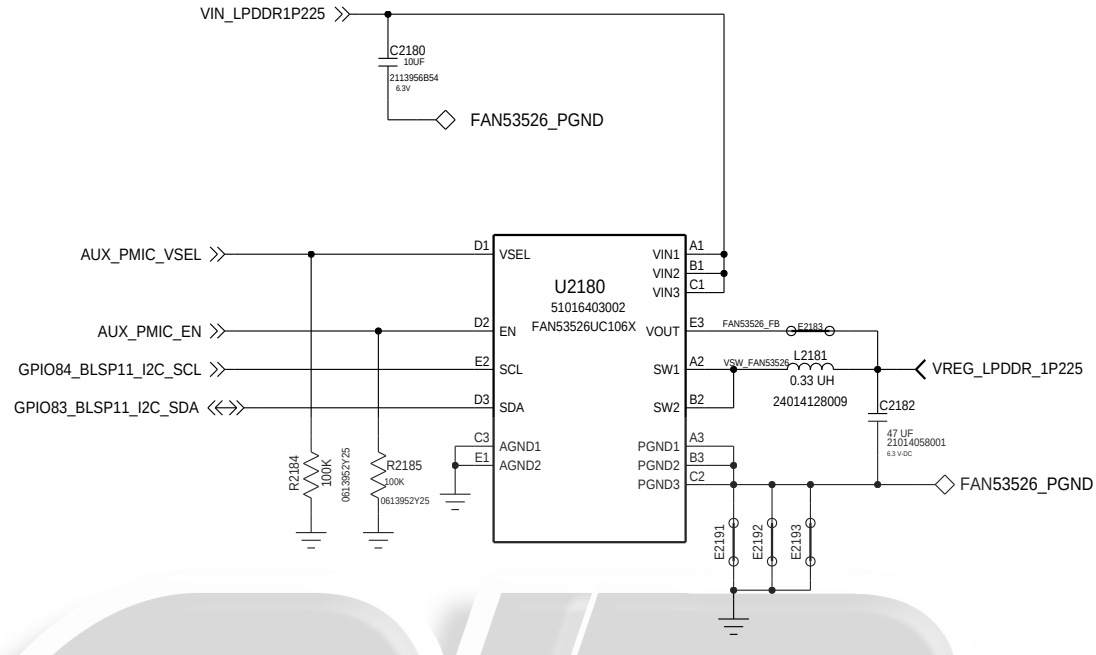
PM8994: SMPS

REFDES: 2100-2199

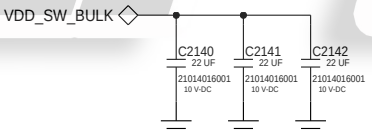


VREG_LPDDR3 EXTERNAL REGULATOR

*** EXT-REGULATOR FOR LPDDR3 I2C ADDR. 0X60



SWITCHER INPUT BULK CAPS



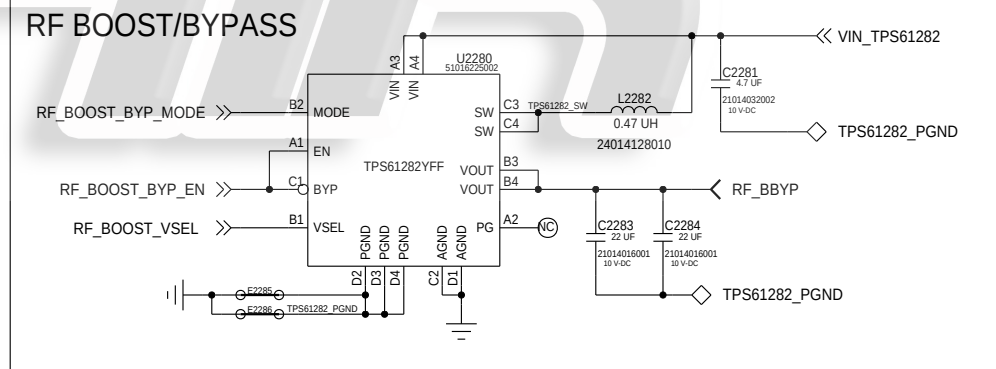
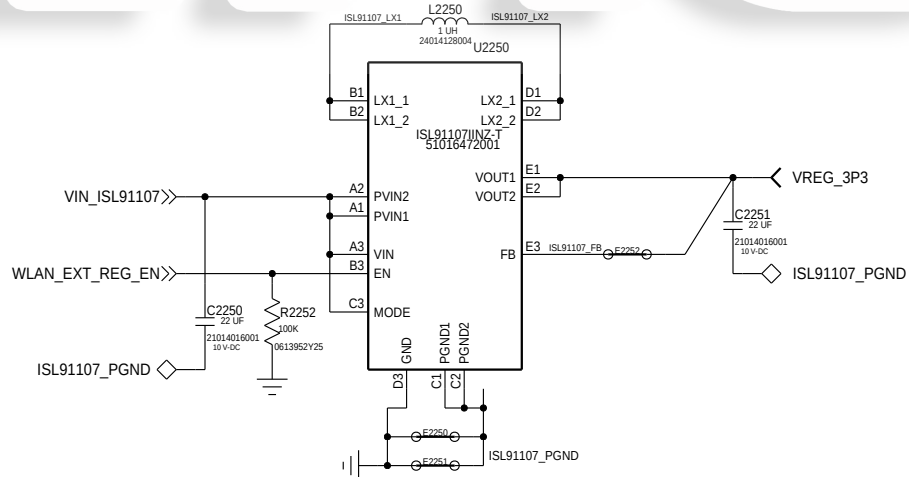
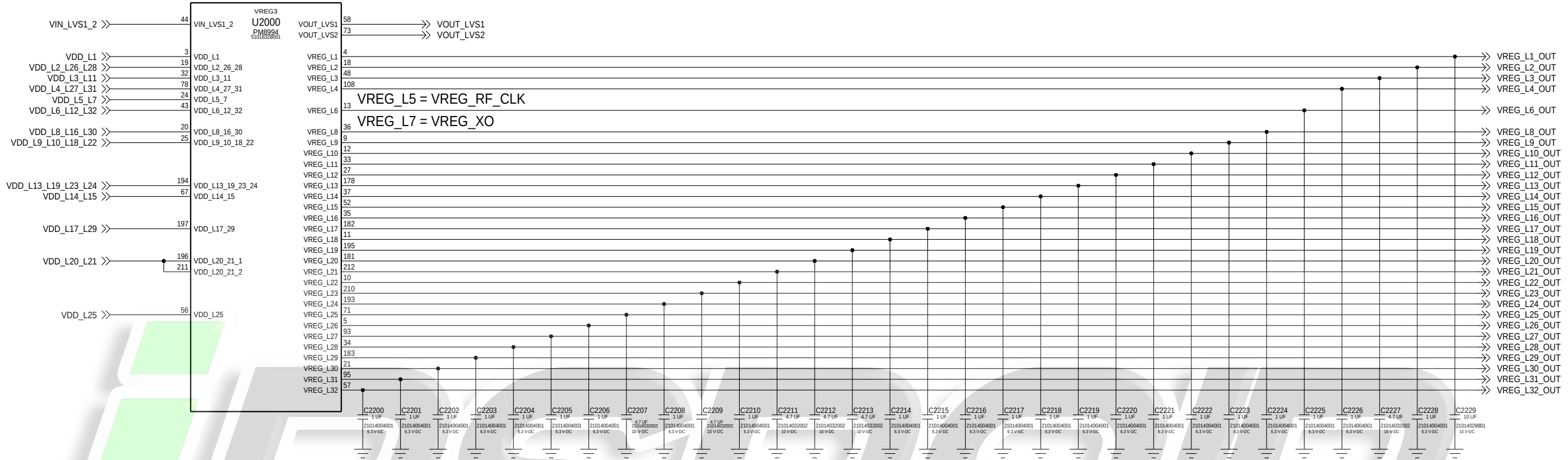
BULK CAPS FOR VREG_APCC0 AND VREG_APCC1 ARE LOCATED BY THE MSM

BASED ON 80-NJ051-41 REV. B

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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PM8994: VREGS

REFDES: 2200-2299



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APPROVALS		BLOCK TITLE:	
ENGINEER:		ap_pmic	
LAST MODIFIED BY:		PROJECT NAME	
Wed Jun 10 14:32:04 2015		84016440001_b.cpm	
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87654321

D

C

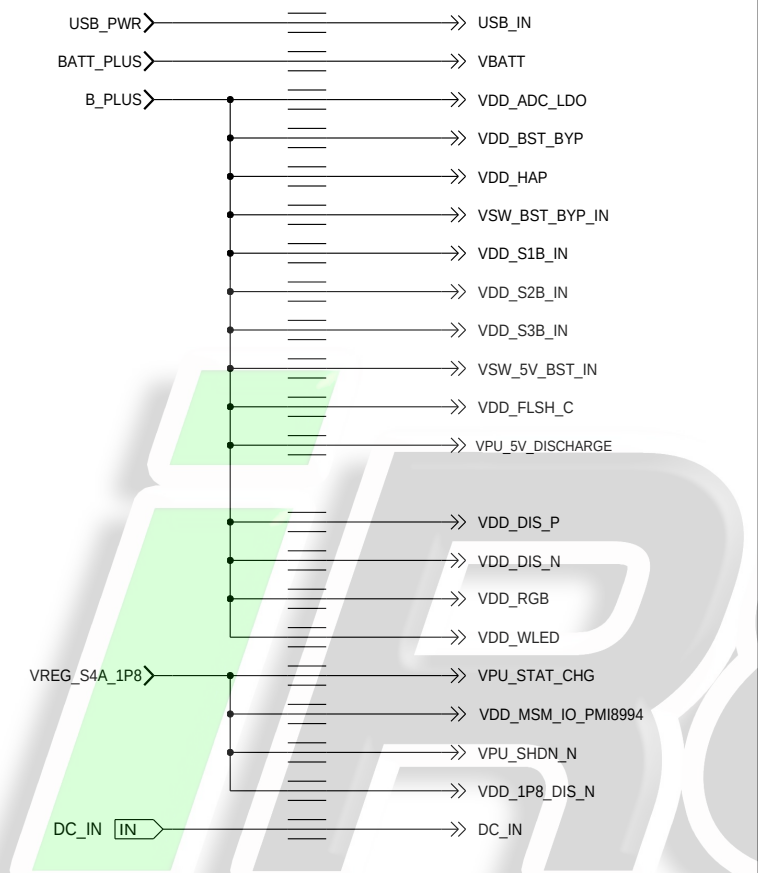
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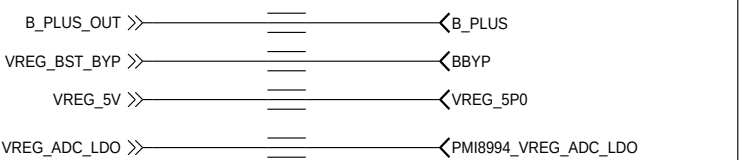
PMI8994: POWER

REFDES: 2400-2499

INPUT POWER

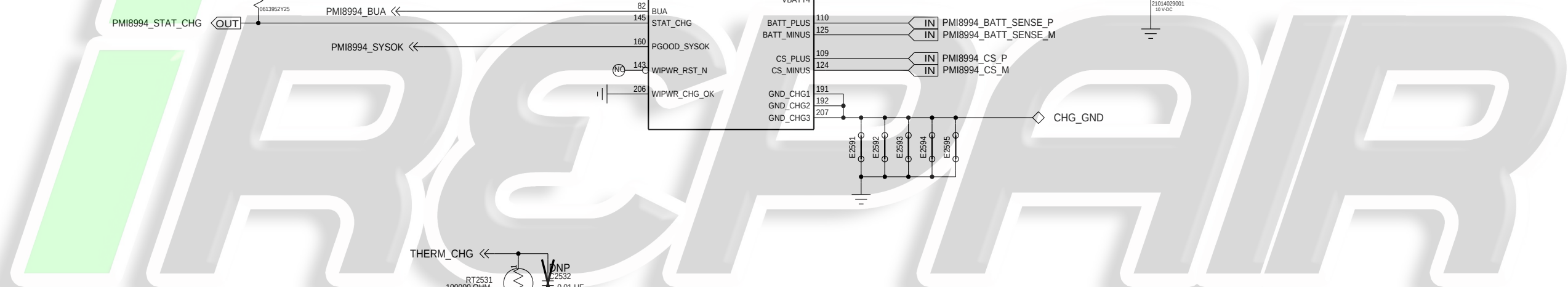


OUTPUT POWER



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LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:05 2015			
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		CAD FILE NAME: PAGE 56 OF 78	

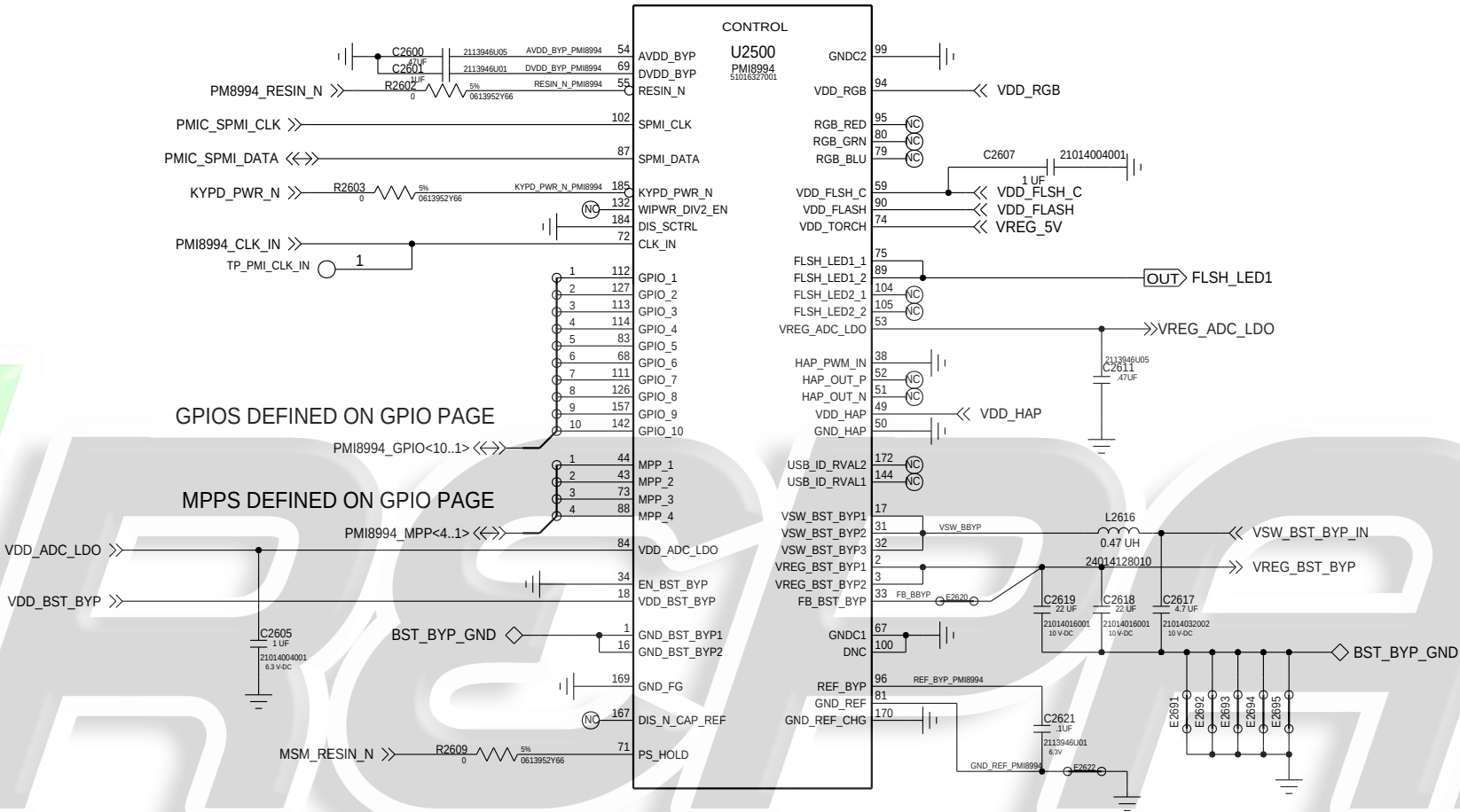
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	PAGE 57 OF 78

PMI8994: GPIO/LED/RGB

REFDES: 2600-2699

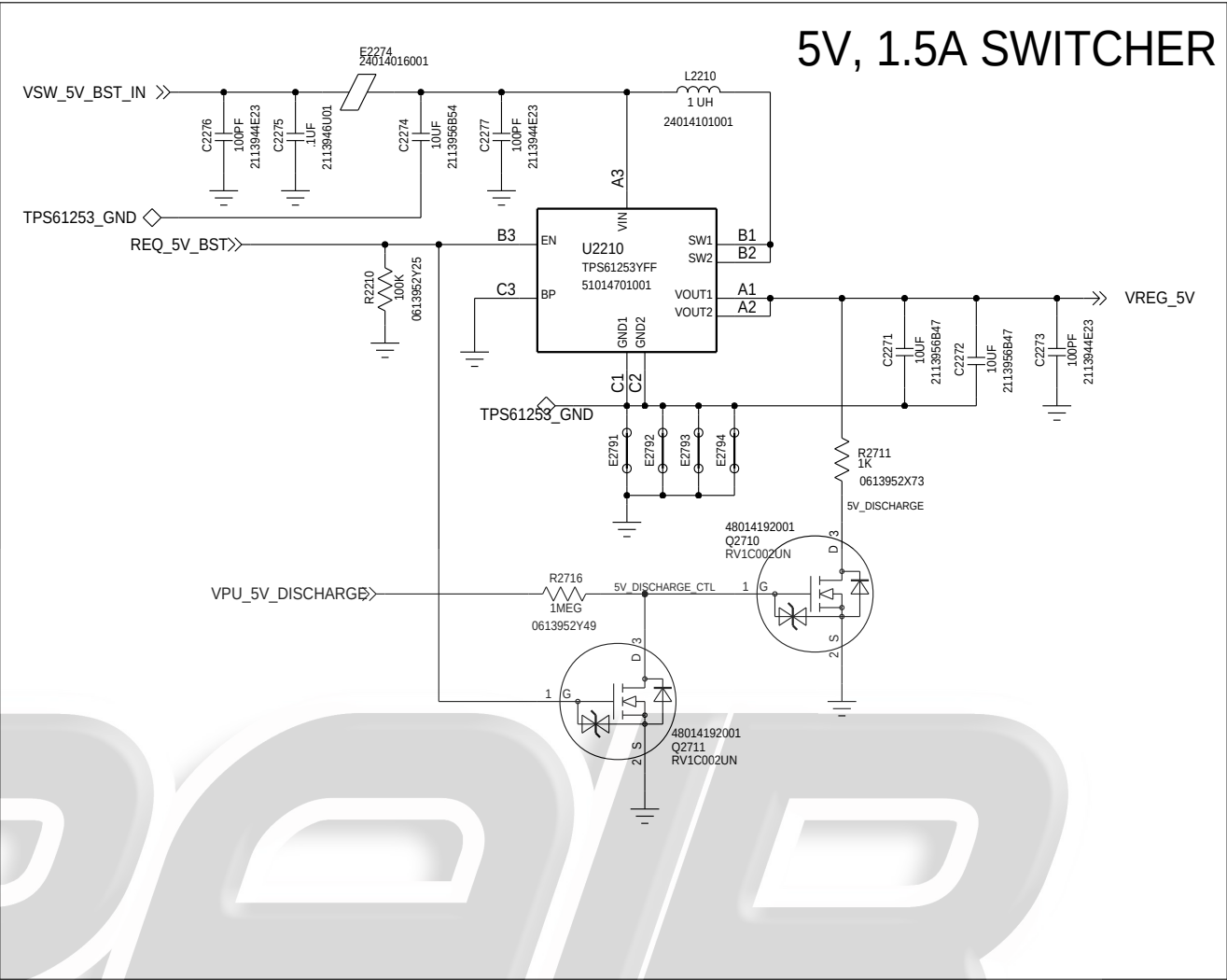
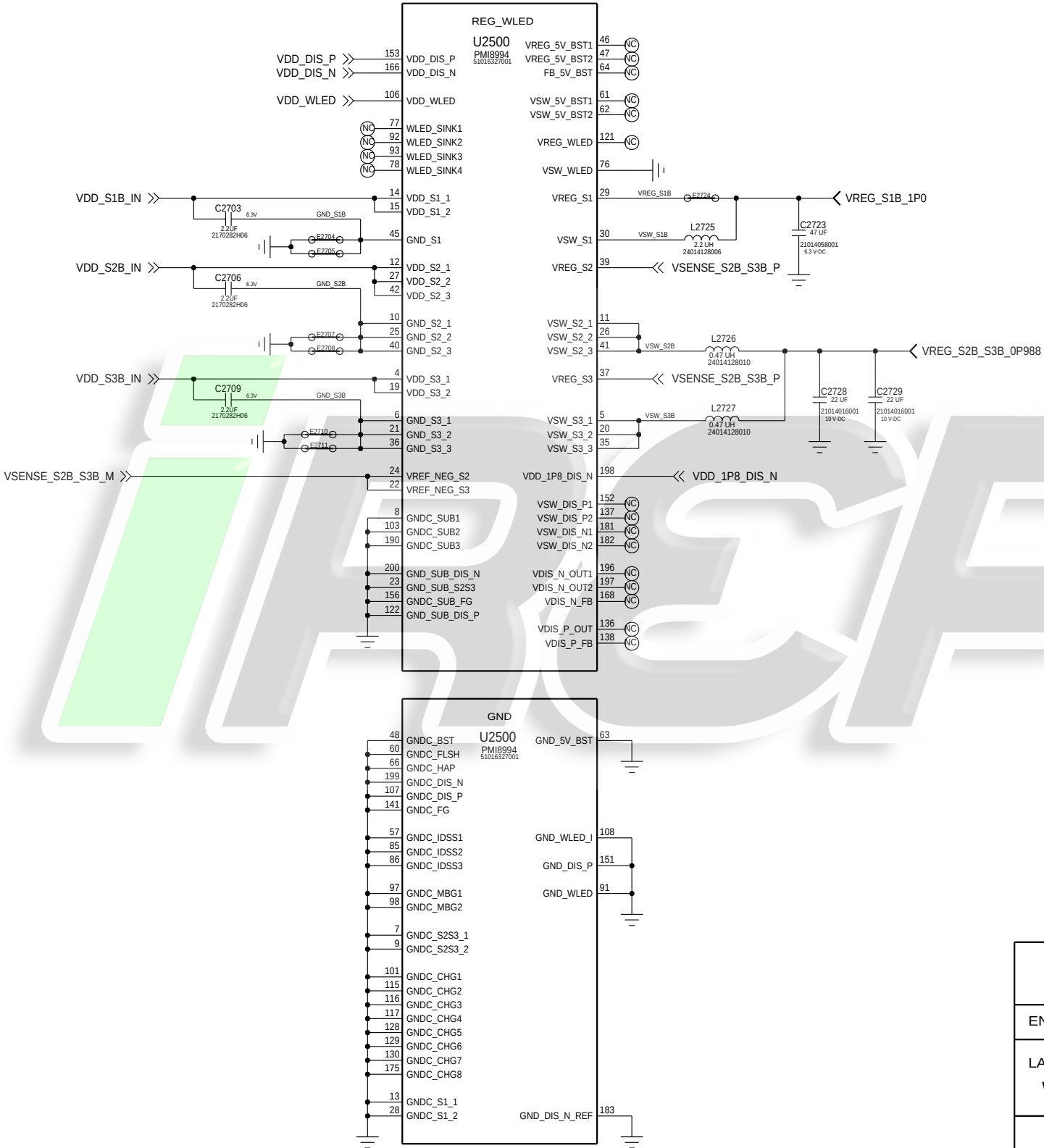


BASED ON 80-NJ051-41 REV. B

APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:08 2015		ap_pmic	
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		CAD FILE NAME:	
		REV 01	
		PAGE 58 OF 78	

PMI8994: REF/WLED/GND

REFDES: 2700-2799



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: NATE CONNELL Wed Jun 10 14:32:09 2015		PROJECT NAME 84016440001_b.cpm	
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		PAGE 59 OF 78	

REF 3120-3199

APPROVALS



MOTOROLA

MOBILE DEVICES

ENGINEER:

BLOCK TITLE:

LAST MODIFIED BY:

Wed Jun 10 14:31:48 2015

block_wtr3925

PROJECT NAME

B	89016440001_b.cpm
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REV

01

CAD FILE NAME:

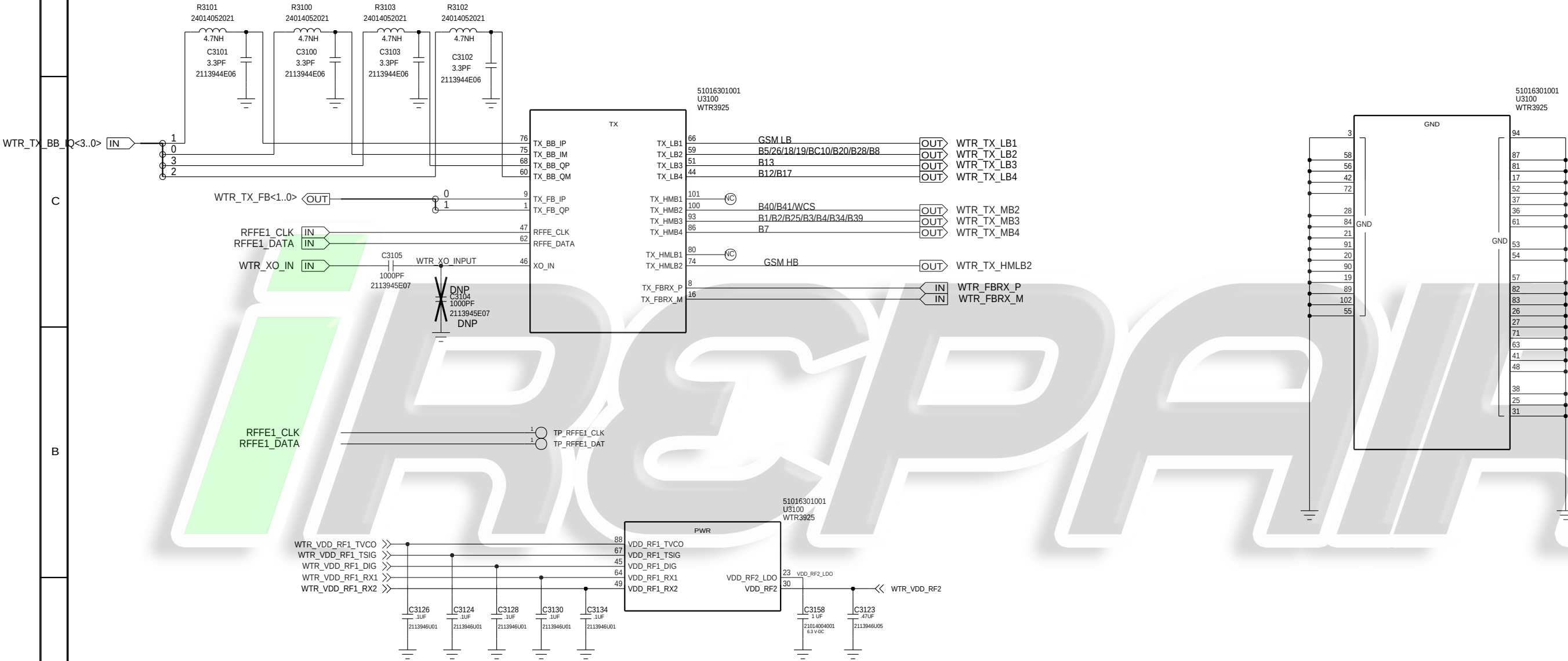
PAGE 60 OF 78

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WTR3925: TX/PWR/GROUND

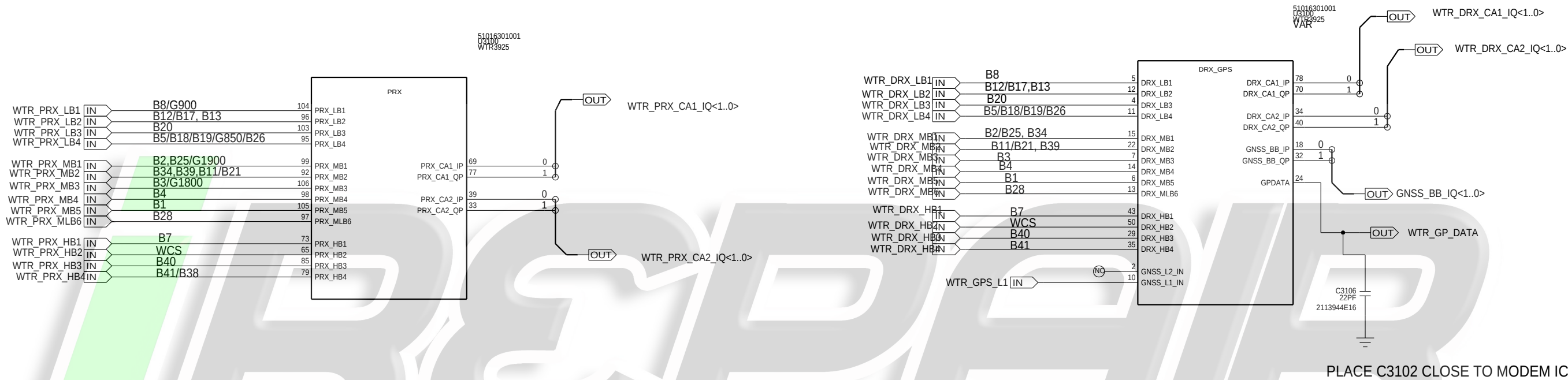
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		REV 01	
		PAGE 61 OF 78	

WTR3925: PRX/DRX/GPS

REF 3110-3119



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		block_wtr3925	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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SLICE: I/F SLICE DELETED

SL_VBUS<<>> SL_VBUS

SL_I2C_SDA <<>> SL_I2C_SDA
SL_I2C_SCL << SL_I2C_SCL
SL_BPLUS>> SL_BPLUS
SL_DET_N<< SL_DET_N

SL_INT_N>> SL_INT_N
SL_BPLUS_EN<< SL_BPLUS_EN
SL_VBUS_EN<< SL_VBUS_EN



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDELOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Wed Mar 11 13:17:37 2015		slice_if	
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		84016440001_b.cpm	
		REV 01	
		PAGE 63 OF 78	

RFFE: BLOCK DIAGRAM

REFDES: 7000-8999

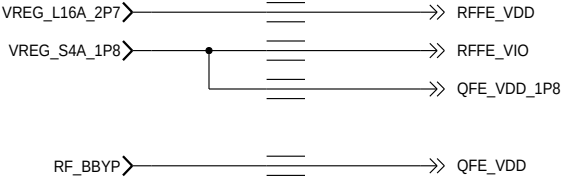


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		CAD FILE NAME: PAGE 64 OF 78	

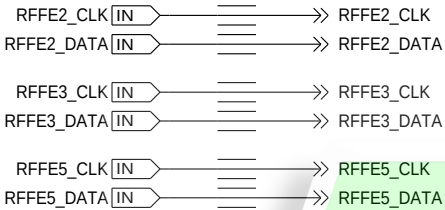
RFFE: PORTS

REFDES: 7000-8999

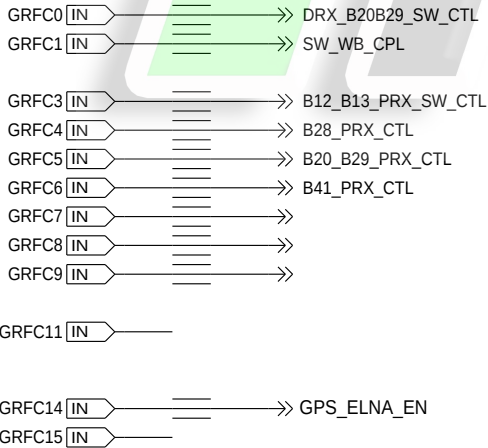
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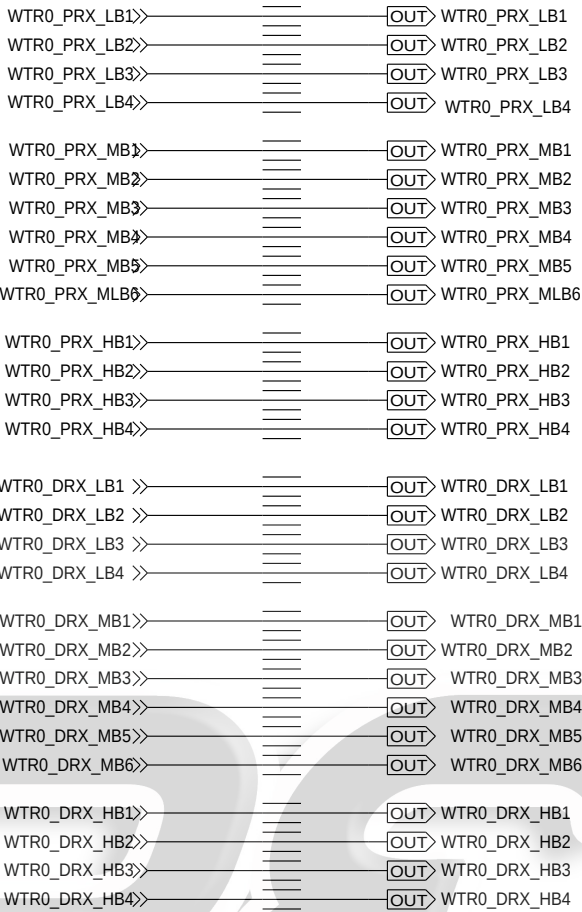
INTERFACES - RFFE MIPI



INTERFACES - RFFE GPIO/GRFC



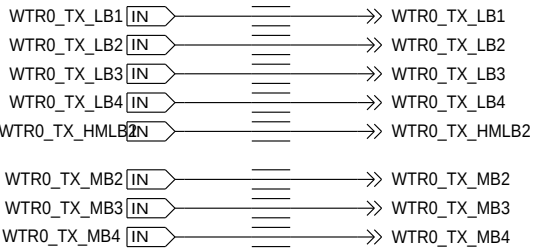
OUTPUTS - RF RECEIVER - WTR0



OUTPUTS - RF RECEIVER - 3DL

OUTPUTS - RF RECEIVER - DSDA

INPUTS - RF DRIVER - WTR0



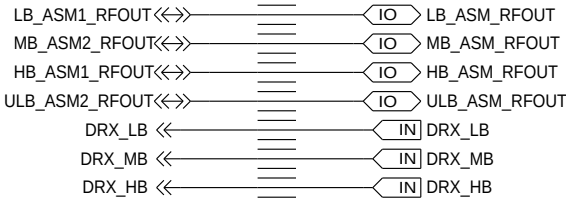
INPUTS - RF DRIVER - 3DL

INPUTS - RF DRIVER - DSDA

INPUTS /OUTPUTS - DETECTORS



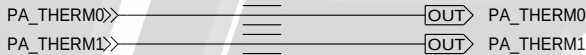
OUTPUTS - RF ANTENNA CONNECTORS



GPS



THERMISTORS

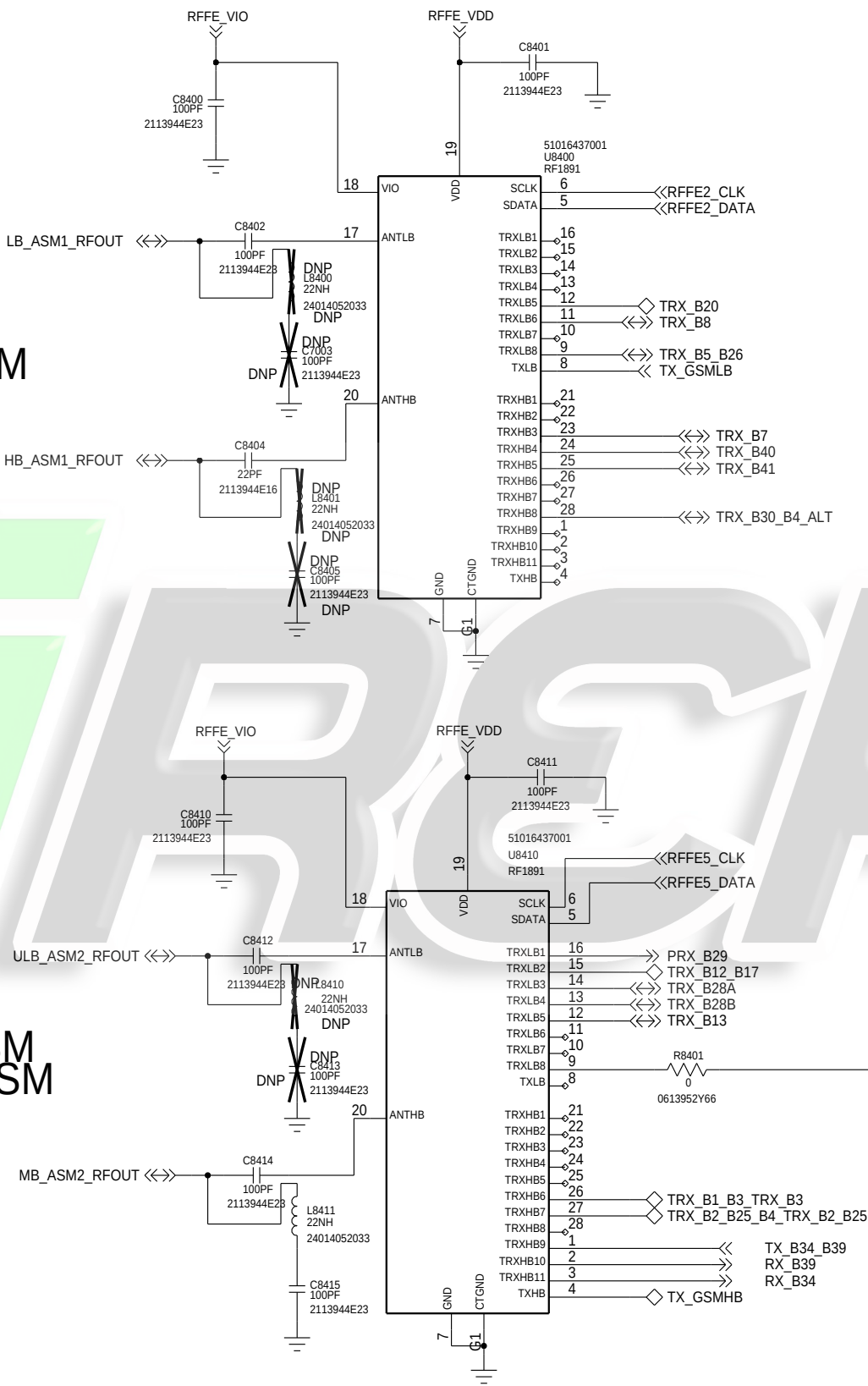


APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Wed Jun 10 14:31:26 2015			
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		CAD FILE NAME: PAGE 65 OF 78	

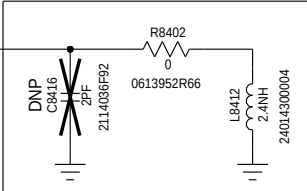
RFFE: ASM
REF 8400-8499

ASM1
CA: LB+HB ASM
DSDA: DATA ASM

ASM2
CA: ULB+MB ASM
DSDA: VOICE ASM

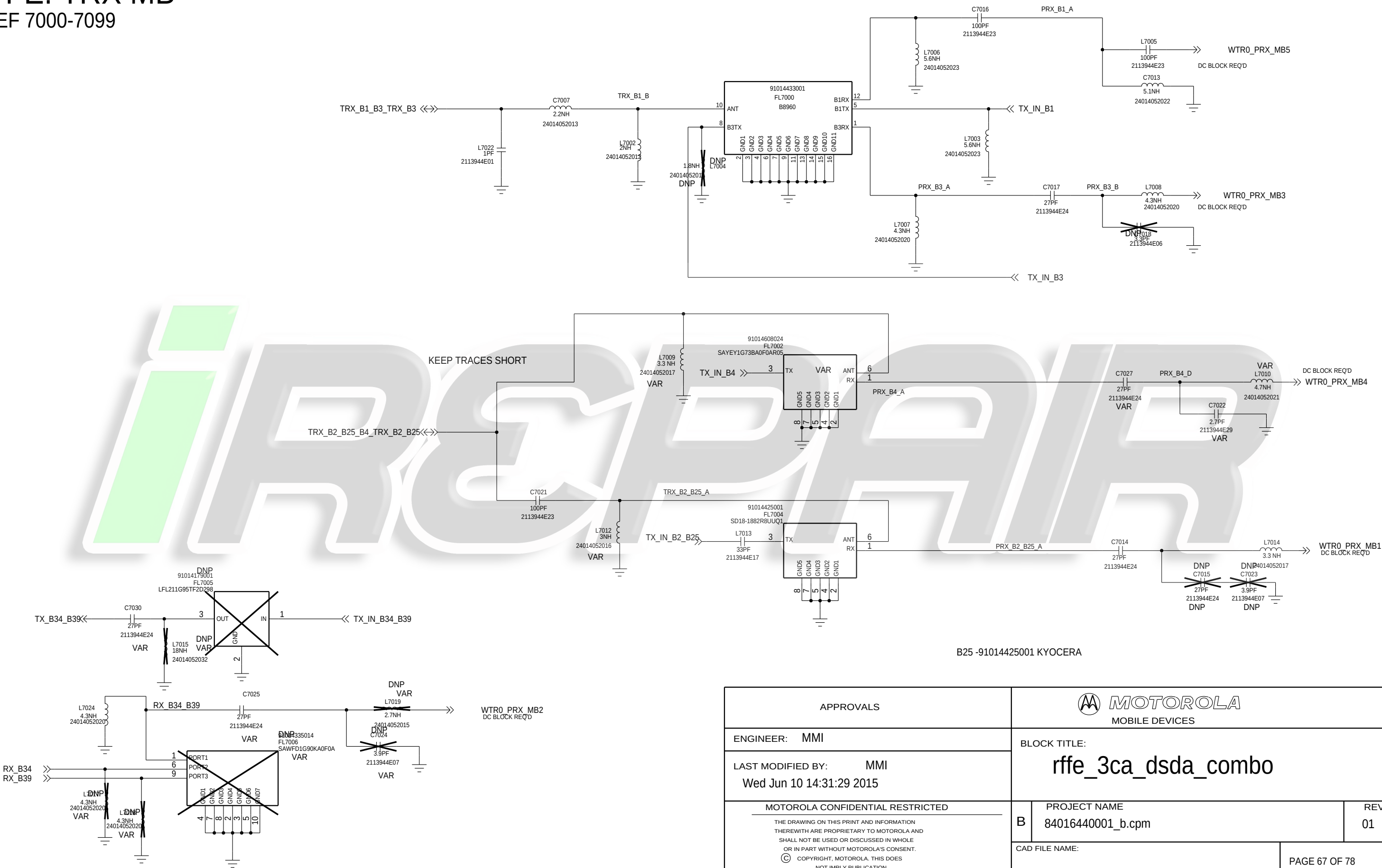


PLACE PI OUTSIDE OF SHIELD CAN



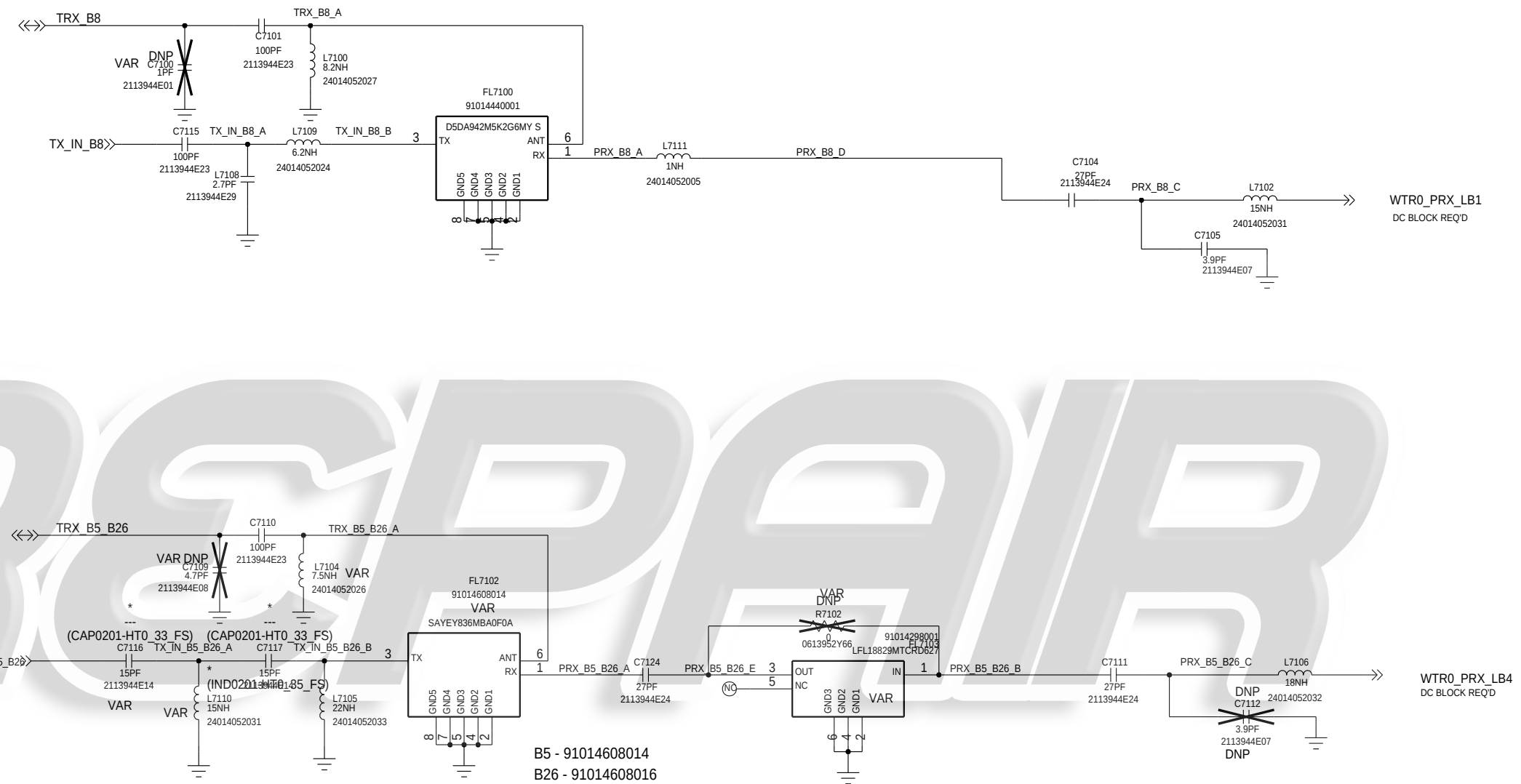
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		PAGE 66 OF 78	

RFFE: TRX MB
REF 7000-7099



APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Wed Jun 10 14:31:29 2015		PROJECT NAME 84016440001_b.cpm	
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		PAGE 67 OF 78	

RFFE: TRX LB
REF 7100-7199



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY:		rffe_3ca_dsda_combo	
Wed Jun 10 14:31:30 2015			
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		PAGE 68 OF 78	

RFFE: TRX ULB
REF 7200-7299

D

C

B

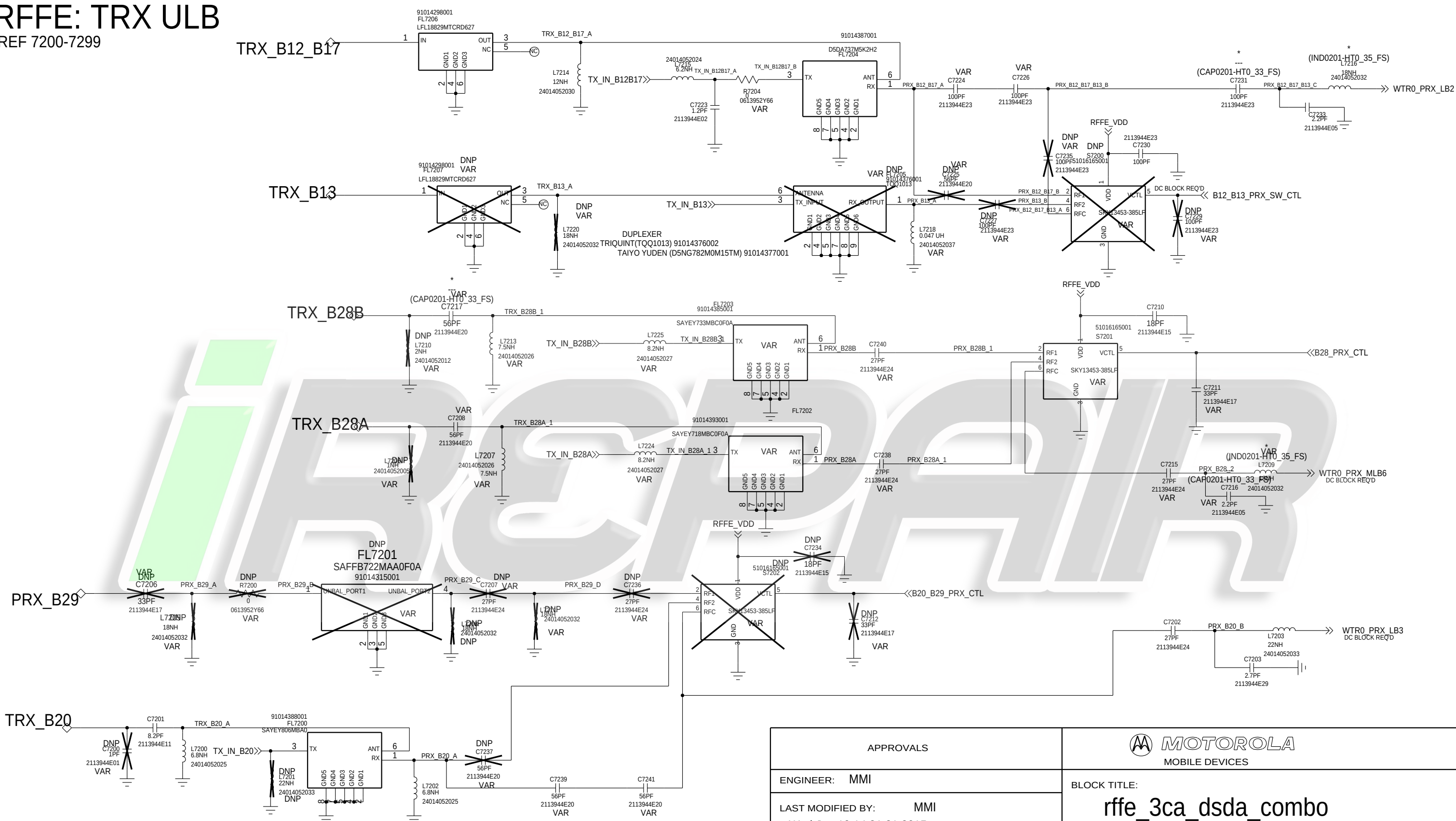
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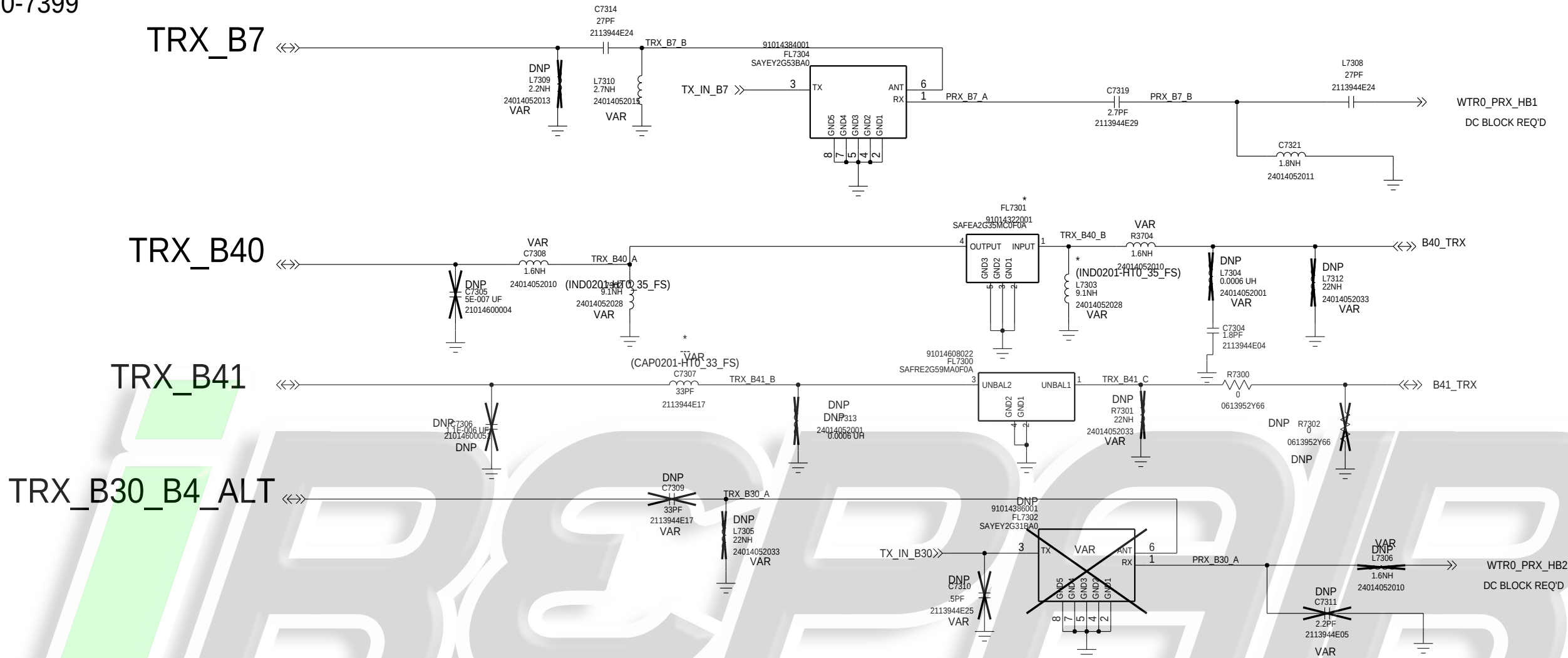
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		PAGE 69 OF 78	

RFFE: TRX HB
REF 7300-7399

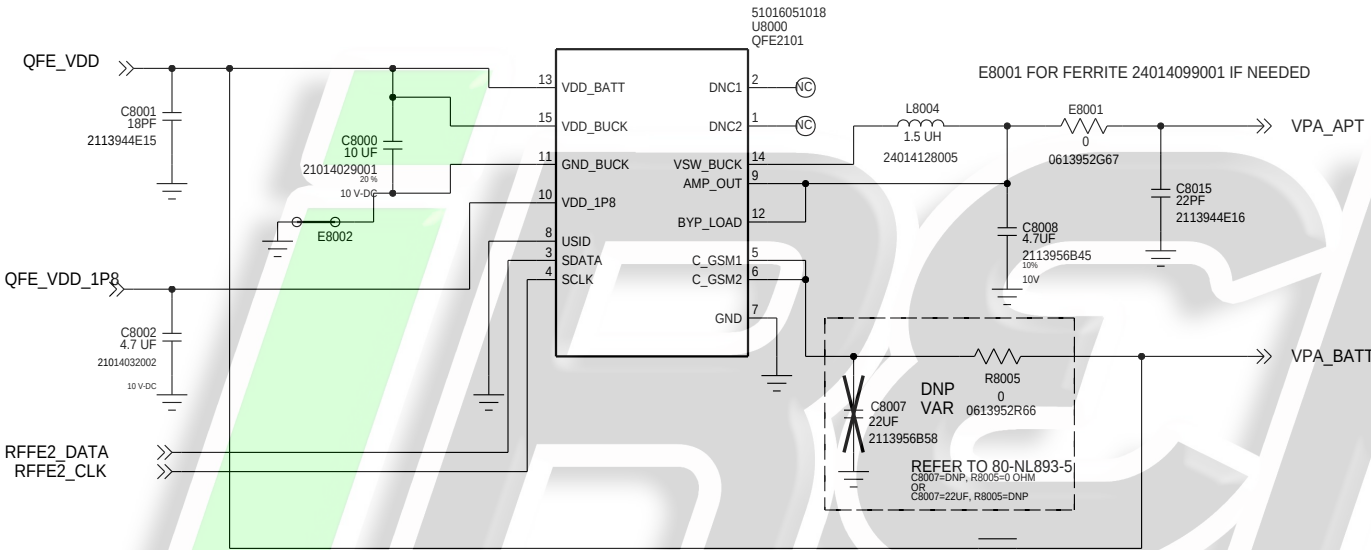


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		B 84016440001_b.cpm	REV 01
		CAD FILE NAME:	
		PAGE 70 OF 78	

RFFE: APT

REF 8000-8099

MAIN APT



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 71 OF 78	

RFFE: TX MMB PA

REF 7400-7499

WTR0_TX_MB3

D

WTR0_TX_LB2

C

WTR0_TX_LB4

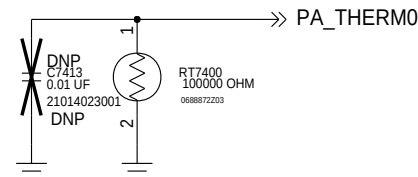
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WTR0_TX_LB3

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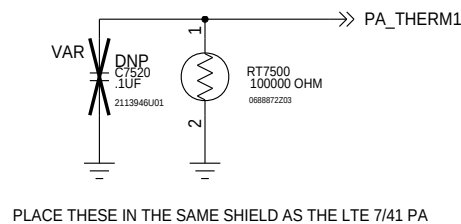
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WTR0_TX_LB1



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		PAGE 72 OF 78	

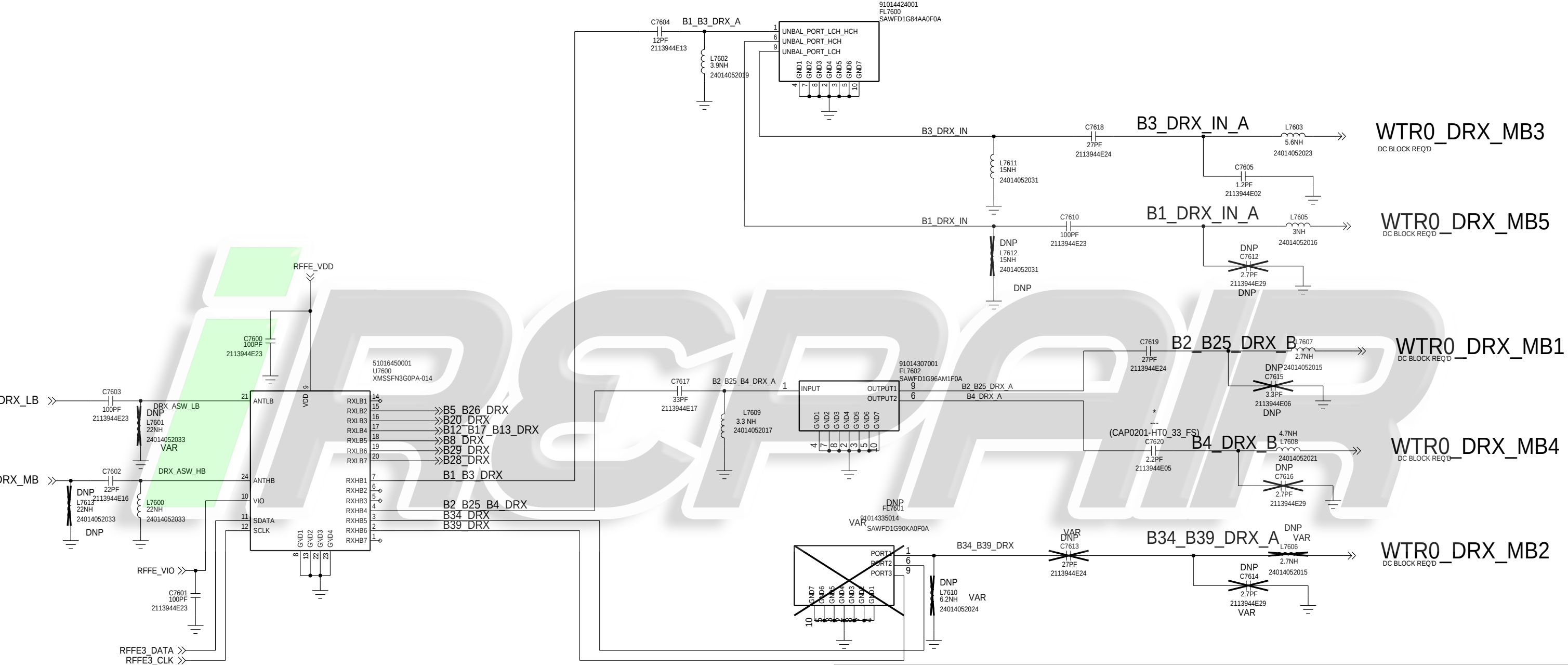
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	PAGE 73 OF 78

RFFE: DRX ANTENNA SWITCH

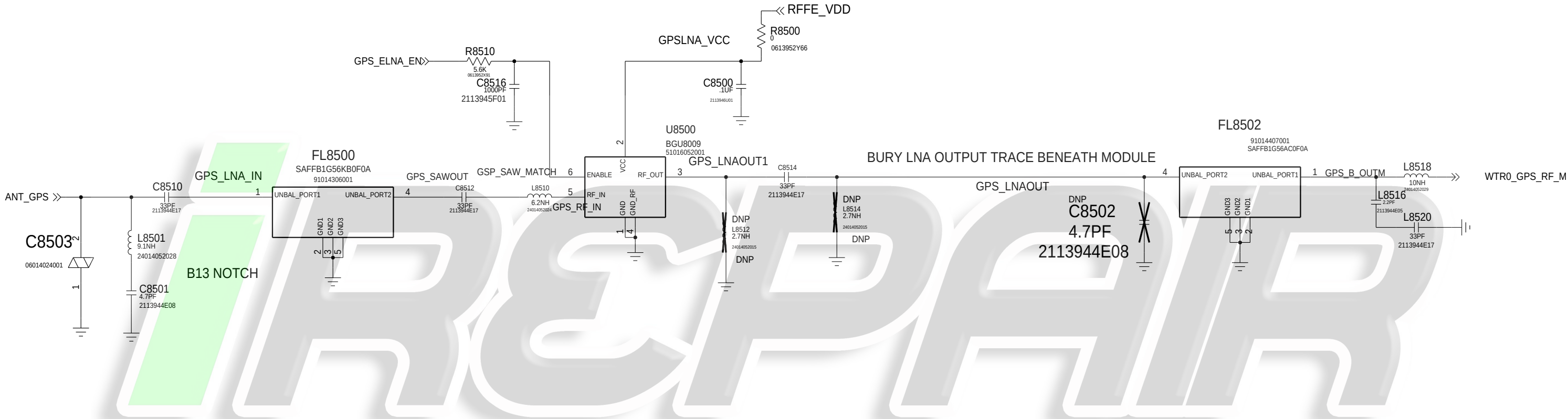
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APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Wed Jun 10 14:31:38 2015		rffe_3ca_dsda_combo	
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		CAD FILE NAME:	
		PAGE 74 OF 78	

RFFE: GPS

REF : 8500-8549



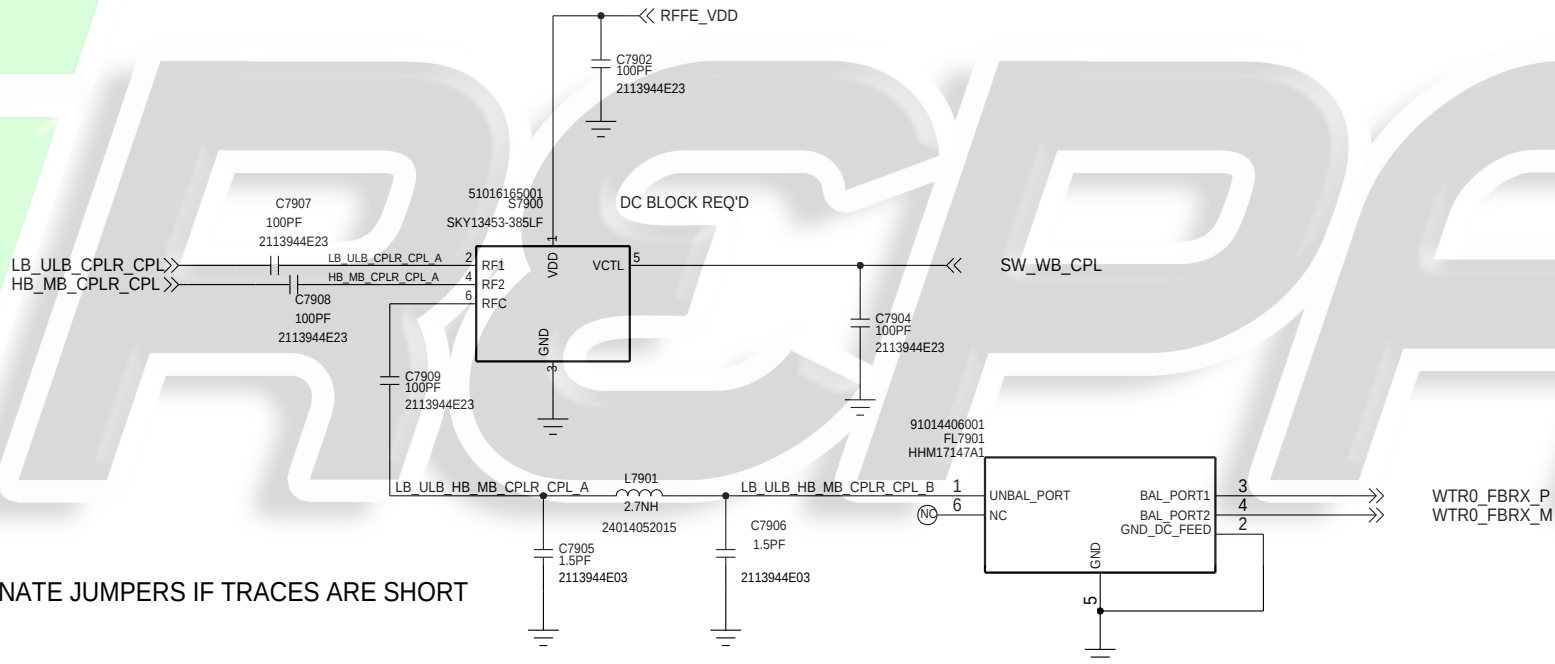
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RFFE: PDET

REF : 7900-7949

COULD ELIMINATE JUMPERS IF TRACES ARE SHORT

DEVICE LOCATION - TBD



APPROVALS		 MOTOROLA MOBILE DEVICES	
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